POLY PMNA | REVISION 2026 | COURSE 1021 | PRACTICUM

Basic Mechanical Engineering

A complete machine-and-energy handbook for mechanical fundamentals, thermodynamics, fluid power, transmission, manufacturing and industrial safety.

Course code	**1021**
Semester	**I**
Type	**Practicum**
Scheme	**3:0:3:0**
Contact hours	**90**
Credits	**4.5**
Summary CIE	**40**
Summary ESE	**100**

Programmes: Fire Technology and Safety, Mechatronics, Mechanical Engineering, Tool & Die Engineering, and Wood and Paper Technology.

OFFICIAL SOURCE DECLARATION

Official Source Declaration and Verified Inconsistencies

This handbook is an original educational expansion of the supplied SITTTR Kerala Diploma Curriculum Revision 2026 PDF for **Course 1021 — Basic Mechanical Engineering**. Official topics, hours, outcomes, activities, assessment values and references are preserved.

Contact-hour inconsistency: the course-description table states **90 contact hours**. The module table allocates 45 theory hours and 39 practical hours, totalling **84 instructional hours**. Both official values are retained; the six-hour difference is not silently reassigned.

Detailed-practical-hour inconsistency: the major-topic table assigns Module II practical 9 h and Module IV practical 11 h. The detailed rows additionally show Series Lab Exam I (3 h) and Series Lab Exam II (3 h); summing those examination rows with the listed identification rows would give 12 h and 14 h respectively. The handbook treats the series exams as assessment periods and preserves the major-topic practical allocations.

Assessment inconsistency: the course-description summary lists CIE 40 and ESE 100. The teaching-and-assessment table separately lists theory CIE 20 + ESE 60 = 80 and practical CIE 20 + ESE 40 = 60, producing a combined total of 140. The detailed practicum methodology also converts internal components to 40 and retains end-semester written 60 plus lab 40. Each value is shown in its official context.

Source limitations: no official web resources and no software requirements are listed. The prerequisite row states Basic Science at High School level but does not specify a course code or course title.

COURSE MAP

Course identity, objectives and outcomes

45 h

Theory allocation

39 h

Practical allocation

90 h

Self-learning

4 COs

Understand-level focus

Introduction

To get an idea about mechanical engineering and develop enthusiasm in mechanical engineering and related programmes.

Course prerequisite

Basic Science at High School level.

Official objectives

1. Introduce fundamental concepts of mechanical engineering.
2. Provide basic knowledge of machines, manufacturing and energy systems.
3. Familiarise students with power transmission systems, thermodynamics and fluid power.
4. Provide awareness of energy sources, maintenance and industrial safety.

Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the scope of Mechanical Engineering, basic engineering concepts, units and basic principles of thermodynamics.	Understand
CO2	Explain basic fluid mechanics, fluid power systems and power generation.	Understand
CO3	Explain working principles of mechanical power transmission systems.	Understand
CO4	Describe basic manufacturing and machining processes used in industry.	Understand

Official CO–PO mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	–	–	1	1	–	–	1	–	2
CO2	2	2	2	1	2	2	–	–	1	–	2
CO3	2	2	2	1	2	1	–	–	1	1	2
CO4	2	2	2	2	2	2	1	–	1	1	2

NAVIGATION

How to use this handbook

1. Learn

Open every topic, read the principle, inspect its labelled visual and explain the Malayalam support sentence aloud.

2. Observe safely

Use the practical record centre only in the institution laboratory or workshop under faculty supervision.

3. Prepare

Revise formulas, worked examples, diagrams, one-/three-/seven-mark questions, model paper and viva cues.

SAFETY FIRST

Workshop and laboratory safety centre

Never operate an engine, pump, turbine, lathe, drill, mill, grinder, welding set or compressed-air rig without faculty authorisation. Isolate electrical, pressure, thermal and mechanical energy before touching a component.

Rotating machinery

Remove loose clothing and jewellery, tie hair, keep guards fitted and never measure or clean a moving part.

Pressure systems

Depressurise before disconnecting a hose. Never direct compressed air at skin, eyes or another person.

Heat and fire

Keep fuel and hot surfaces controlled. Know alarm, extinguisher, emergency-stop and evacuation locations.

PPE

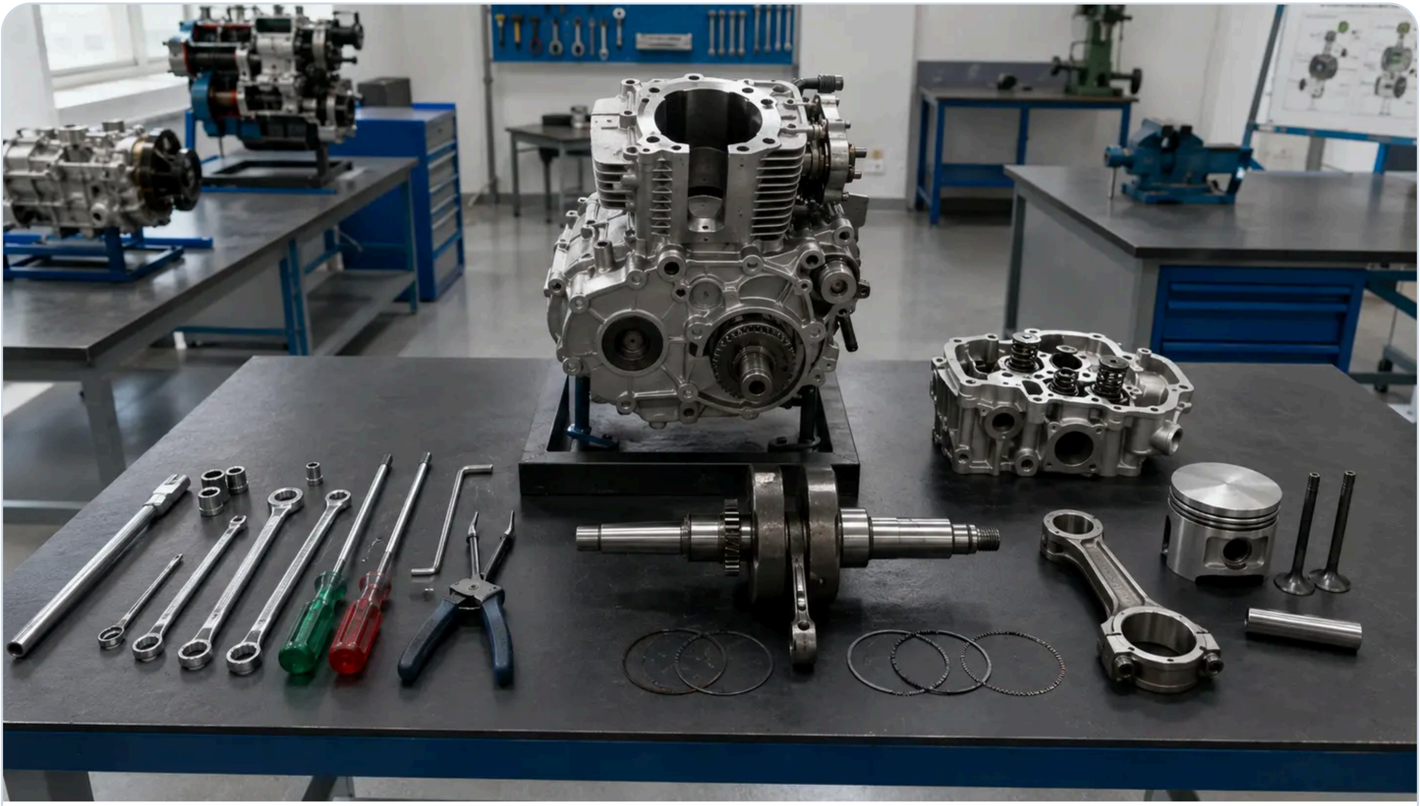
Select goggles, face shield, gloves, footwear, hearing protection and welding PPE for the actual hazard.

Malayalam safety: Machine start □□□□□□□□□□□□□ □□□□□□ guard, emergency stop, loose clothing, pressure release, PPE □□□□□□ □□□□□□□□□□□□. Rotating part-□□ □□□□□□□□ □□□□□□□.

ORIGINAL COURSE PHOTOGRAPHS

Mechanical engineering laboratory visual studio

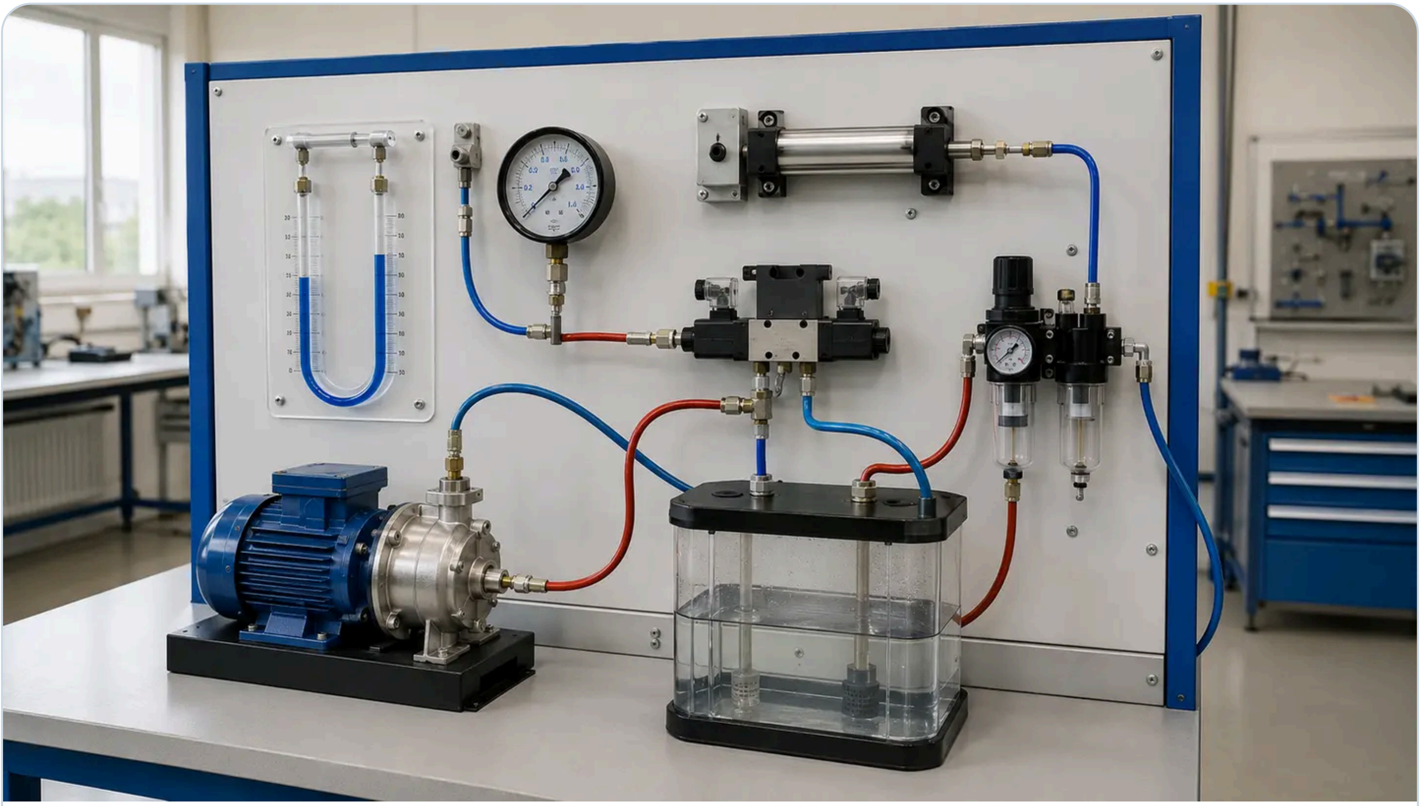
Use photographs for component recognition and the labelled SVG diagrams for working principle, motion, energy flow and examination drawing practice.



Module I — IC-engine identification

Recognise cylinder, piston, rings, connecting rod, crankshaft, cylinder head, valves and safe dismantling tools.

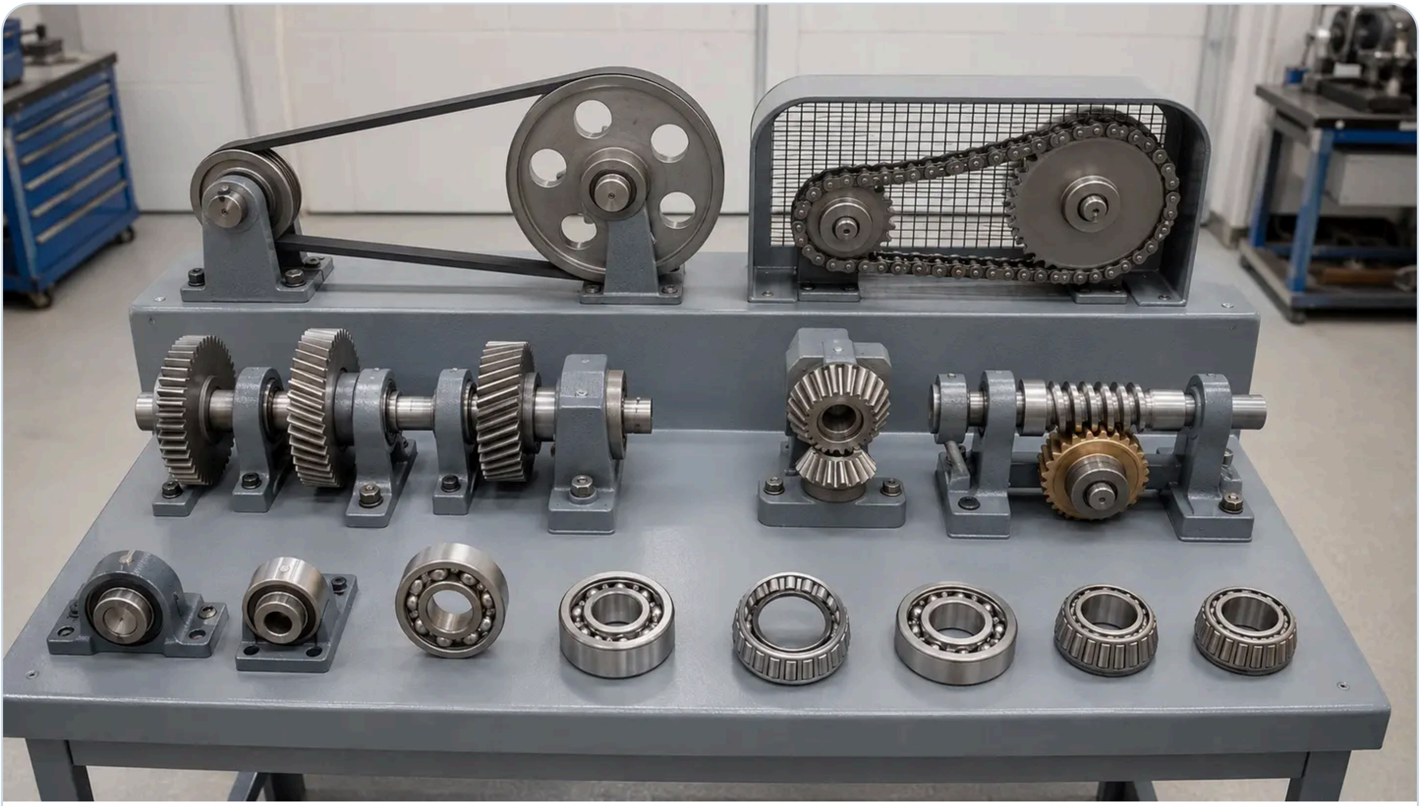
Parts-□□□□ □□□□□□, motion, function □□□□□□ dismantling □□□□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□.



Module II — fluid-power equipment

Trace pressure measurement, pump delivery and the air path through preparation, control and actuator components.

Pressure line depressurise □□□□□□□□□ □□□□□□ hose/fitting disconnect □□□□□□□.



Module III — transmission and bearing identification

Compare positive and friction drives, shaft orientation, tooth form and bearing rolling elements

Driver, driven member, rotation direction, speed change □□□□□□ □□□ drive-□□□ □□□□□□□□□□.

Module IV — manufacturing and safety

Identify machine zones, foundry pattern/mould, welding area, soldering station, PPE and safe walkways

Process □□□□□□□□□□□□□□□□ □□□□□□ hazard, guard, PPE, ventilation, housekeeping □□□□□□□□□□□□.

MODULE I · CO1

Mechanical Engineering, Thermodynamics and IC Engines

Theory 10 h | Practical 6 h | CO1 | Understand level

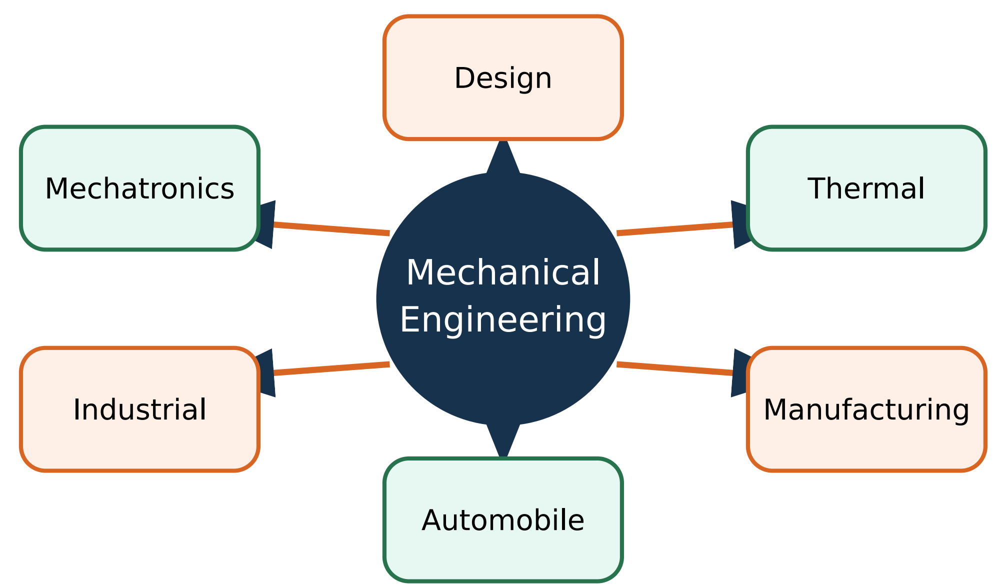
1.1 Scope, importance and branches of Mechanical Engineering

1 h · Understand

Mechanical Engineering applies force, motion, energy, materials and manufacturing knowledge to design, produce, operate and maintain machines and systems. Mechanical engineers work in transport, power, manufacturing, building services, maintenance, safety, automation and sustainable development, often alongside electrical, electronics, civil, computer and chemical specialists.

- Design converts a need into specifications, calculations, drawings and a safe product
- Production plans processes, machines, tools, quality and cost
- Maintenance preserves reliability through inspection, lubrication, repair and condition monitoring
- Thermal Engineering deals with energy conversion, engines, refrigeration and power plants
- Manufacturing, design, industrial, automobile, mechatronics and materials are major branches

Malayalam support: Force, motion, energy, material, manufacturing □□□□□□ □□□□□□□□□□ machine/system design, production, operation, maintenance □□□□□□□□□□□□□ Mechanical Engineering.



Mechanical Engineering branches. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

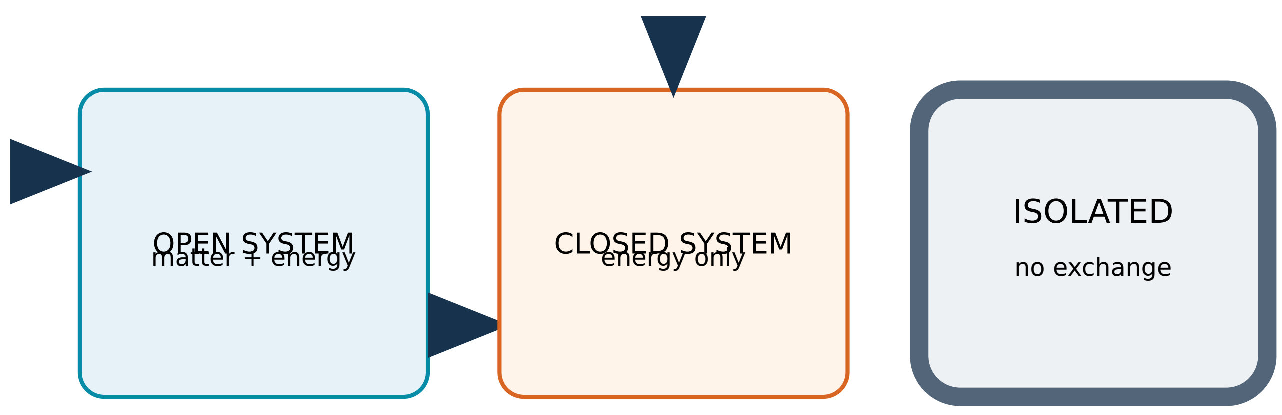
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

1.2 Thermodynamic systems, properties and processes 3 h · Understand −

A thermodynamic system is a chosen quantity of matter or region in space; everything outside it is the surroundings, separated by a real or imaginary boundary. Open systems exchange matter and energy, closed systems exchange energy but not matter, and isolated systems ideally exchange neither. A state is described by properties; a path connects states and a cycle returns to the initial state.

- Isothermal process: temperature remains constant
- Isobaric process: pressure remains constant
- Isochoric process: volume remains constant
- Adiabatic process: no heat crosses the boundary
- Polytropic process follows PV^n = constant as a useful general relation
- Intensive properties do not depend on mass; extensive properties do

Malayalam support: System □□□□□□□□□□□ □□□□□□□□□□□ region □□□. Open system matter-□□ energy-□□□ exchange □□□□□□□□; closed system energy □□□□□□□; isolated system □□□□□□ exchange □□□□□□□□□□.



Thermodynamic systems and P-V processes. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

1.3 Energy, work, heat and laws of thermodynamics 1 h · Understand –

Energy is the capacity to produce change. Work is ordered energy transfer through a force acting over a distance or another generalised displacement; heat is energy transfer caused by a temperature difference. The Zeroth law establishes thermal equilibrium and temperature measurement, the First law expresses conservation of energy, and the Second law gives direction and limits to energy conversion.

Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other

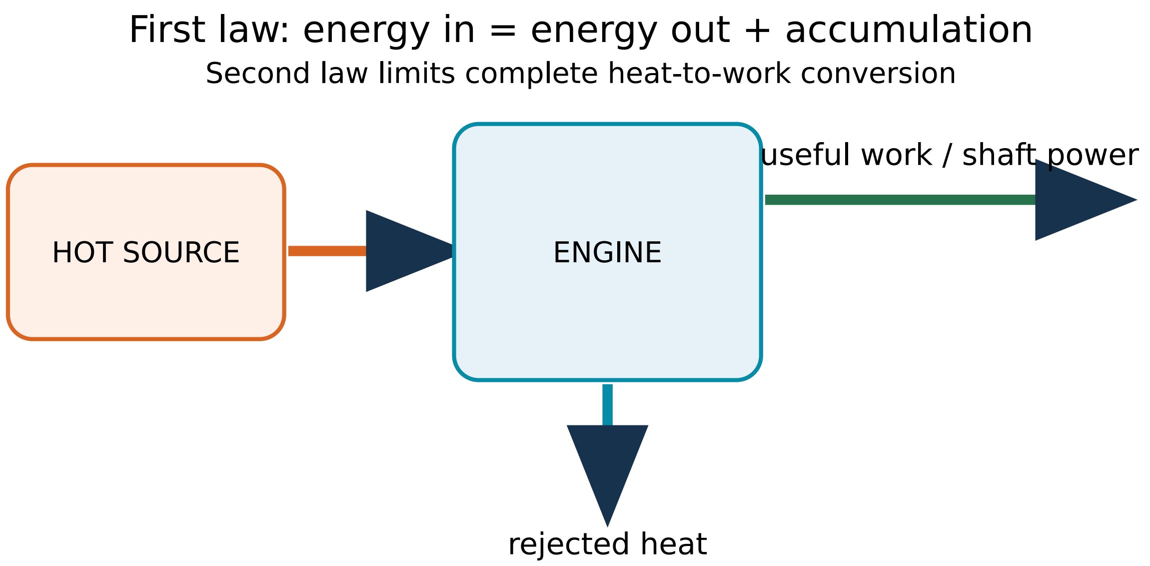
First law: energy is conserved; it changes form or crosses a boundary

Second law: heat does not spontaneously flow from cold to hot and no heat engine converts all supplied heat into work

Heat and work are boundary interactions, not stored properties

Use consistent SI units: joule for energy/work/heat and watt for power

Malayalam support: Zeroth law temperature measurement-□□□□ □□□□□□□□□□□□. First law energy conservation □□□□□□□□; Second law energy conversion-□□□□ direction-□□ limitation-□□ □□□□□□□□□□□□□□□□□.



Thermodynamic laws and energy balance. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

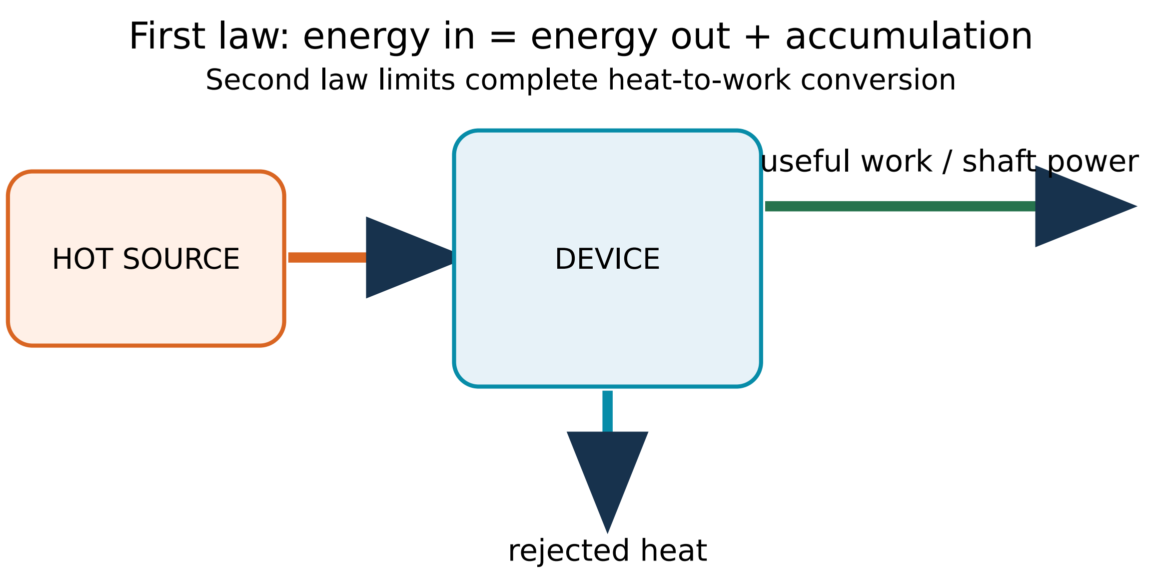
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

1.4 Heat engines, refrigerators, steam boilers and turbines 1 h · Understand –

A heat engine receives heat from a high-temperature source, converts part to work and rejects the remainder to a low-temperature sink. A refrigerator uses work input to remove heat from a low-temperature region and reject it at a higher temperature. A boiler converts water to steam using heat, while a steam turbine expands steam to produce shaft work.

- Heat-engine output is always less than heat supplied
- Refrigerator components form a closed circulation system; it does not create cold
- Boilers require pressure control, water-level control and safety devices
- A turbine converts fluid energy into rotating mechanical energy
- Power-plant chain: fuel/heat source → boiler → steam turbine → generator → condenser/feed system

Malayalam support: Heat engine heat-□□□□ □□□ □□□□ work □□□□□□. Refrigerator work input □□□□□□□□□□ low-temperature space-□ □□□□□□ heat □□□□□□□□□□□ □□□□□□□. Boiler steam □□□□□□□□□□; turbine shaft work □□□□□.



Heat engine refrigerator boiler and turbine. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

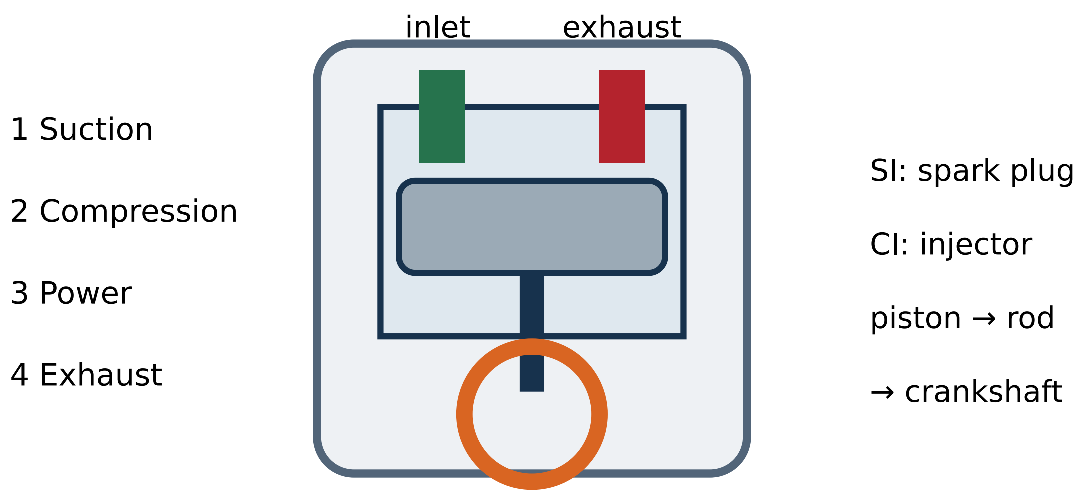
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

1.5 Internal combustion engines: types and working 4 h · Understand –

In an internal combustion engine, fuel releases energy inside the cylinder and gas pressure acts on a piston to rotate the crankshaft. Spark-ignition petrol engines ignite a prepared air-fuel mixture using a spark plug; compression-ignition diesel engines compress air strongly and inject fuel near the end of compression. Four-stroke engines complete a cycle in two crankshaft revolutions, while two-stroke engines complete a cycle in one.

- Four-stroke sequence: suction, compression, power and exhaust
- Piston reciprocation becomes crankshaft rotation through connecting rod and crank
- Inlet and exhaust valves control gas exchange in a four-stroke engine
- Petrol/SI engine uses a spark plug; diesel/CI engine uses high compression and fuel injection
- Two-stroke engines use ports and combine events, giving higher power-to-weight but poorer scavenging/emissions in simple designs
- Applications include automobiles, pumps, generators and industrial drives

Malayalam support: Cylinder-□ combustion pressure piston-□□ □□□□□□□; connecting rod, crank □□□□□□ reciprocating motion-□□ rotary motion □□□□□□. Four-stroke cycle: suction, compression, power, exhaust.



Four-stroke IC engine. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

MODULE II · CO2

Fluid Mechanics, Fluid Power and Energy Sources

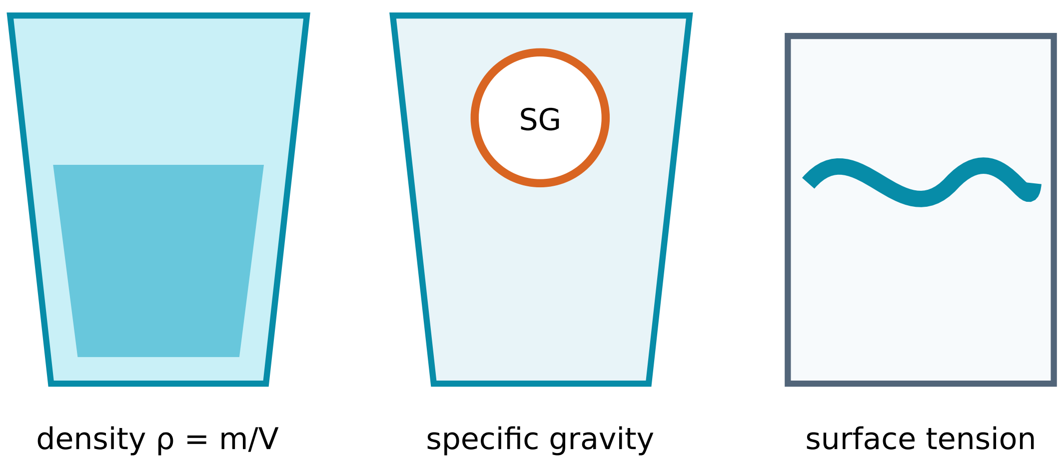
Theory 12 h | Practical 9 h | CO2 | Understand level

2.1 Properties of fluids 1 h · Understand –

A fluid deforms continuously under shear and takes the shape of its container. Density relates mass to volume; specific weight relates weight to volume; specific volume is volume per unit mass; specific gravity compares density with a reference, normally water for liquids. Surface tension arises from molecular attraction at a liquid surface.

- Density $\rho = m/V$, unit kg/m^3
- Specific weight $\gamma = \rho g$, unit N/m^3
- Specific volume $v = 1/\rho$, unit m^3/kg
- Specific gravity is dimensionless
- Surface tension unit is N/m and influences droplets, capillary action and wetting

Malayalam support: Fluid property-□□□□ density, specific weight, specific volume, specific gravity, surface tension □□□□□□□□□□□□. Definition-□□□□□□ SI unit □□□□□□□.



Fluid properties. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

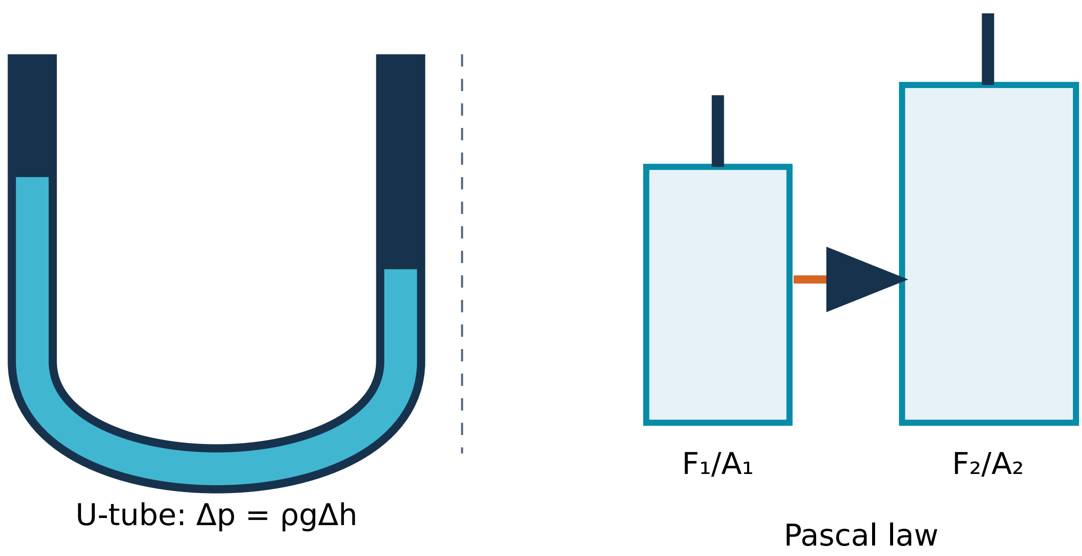
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.2 Pressure, manometers, gauges and Pascal's law 2 h · Understand –

Pressure is normal force per unit area. A piezometer uses liquid-column height for positive liquid pressure; a U-tube manometer balances pressure against a known liquid column; mechanical pressure gauges convert pressure-induced deformation into pointer movement. Pascal's law states that pressure applied to a confined fluid is transmitted undiminished in all directions.

- Absolute pressure is measured from perfect vacuum
- Gauge pressure is measured relative to local atmospheric pressure
- Vacuum pressure is below atmospheric pressure
- Hydrostatic pressure increases with density, gravity and depth
- Pascal's law enables hydraulic jack, lift, press and brake force multiplication
- Read a manometer vertically and use the correct level difference

Malayalam support: Pressure = normal force/area. Manometer liquid-column difference □□□□□□□□□□ pressure □□□□□□□□□□□□□. Pascal's law hydraulic jack, lift, brake □□□□□□□□□□ □□□□□□□□□□□□.



U-tube manometer and hydraulic press. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

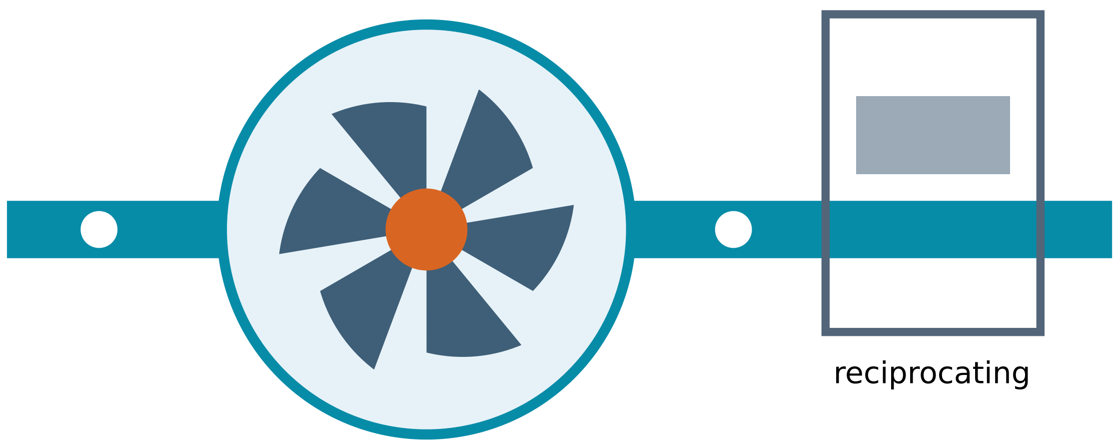
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.3 Hydraulic pumps: centrifugal and reciprocating 2 h · Understand −

A pump adds mechanical energy to a liquid so it can move or rise in pressure. A centrifugal pump uses a rotating impeller and volute casing for continuous flow; a reciprocating pump uses a piston or plunger and non-return valves for positive displacement. Pump choice depends on flow, head, liquid, efficiency, priming and maintenance.

- Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover
- Priming removes air and fills the suction/casing with liquid before starting
- Reciprocating pump: cylinder, piston/plunger, suction valve and delivery valve
- Centrifugal pumps suit moderate head and large continuous flow
- Reciprocating pumps suit high pressure and lower pulsating flow
- Never run a pump dry unless the manufacturer specifically permits it

Malayalam support: Pump liquid-□□ mechanical energy □□□□□□□□. Centrifugal pump impeller □□□□□□□□□□ continuous flow □□□□□; reciprocating pump piston, non-return valves □□□□□□□□□□□.



Centrifugal and reciprocating pumps. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

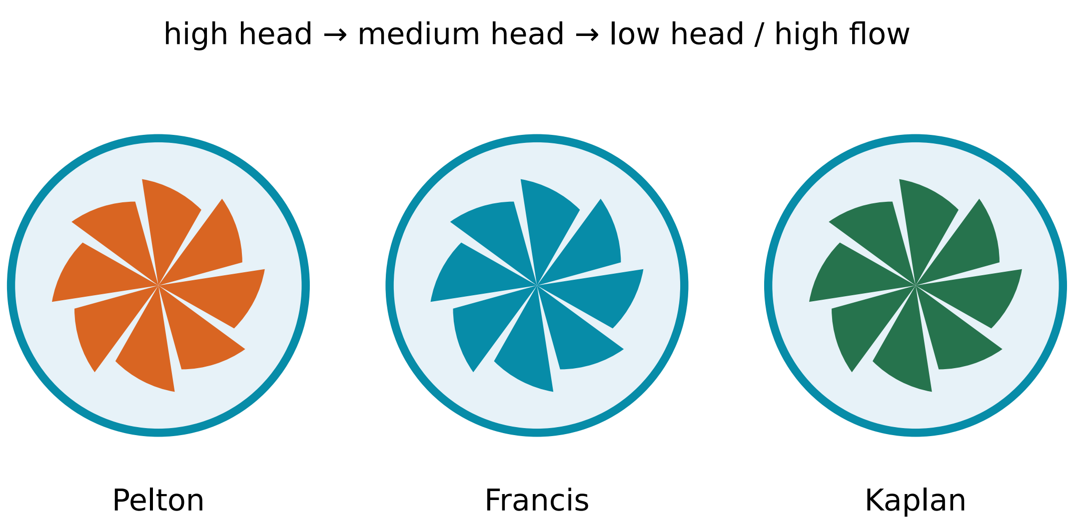
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.4 Hydraulic turbines: Pelton, Francis and Kaplan 2 h · Understand –

A hydraulic turbine removes energy from flowing or falling water and delivers rotating shaft power. Pelton is an impulse turbine for high head and low flow; Francis is a mixed-flow reaction turbine for medium head; Kaplan is an axial-flow reaction turbine for low head and high flow. A casing, runner and draft tube or nozzle guide the energy conversion according to type.

- Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure
- Francis: water enters radially/mixed and leaves axially through a draft tube
- Kaplan: propeller-like adjustable blades suit large flow
- Turbine converts hydraulic energy to mechanical energy; pump performs the reverse direction of energy transfer
- Speed control and safe water isolation are essential

Malayalam support: Turbine water energy-□□ shaft power □□□□□□. Pelton high head, Francis medium head, Kaplan low head/high flow applications-□□□□ □□□□□□□□□.



Pelton Francis and Kaplan turbines. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and

assumptions.

Self-check

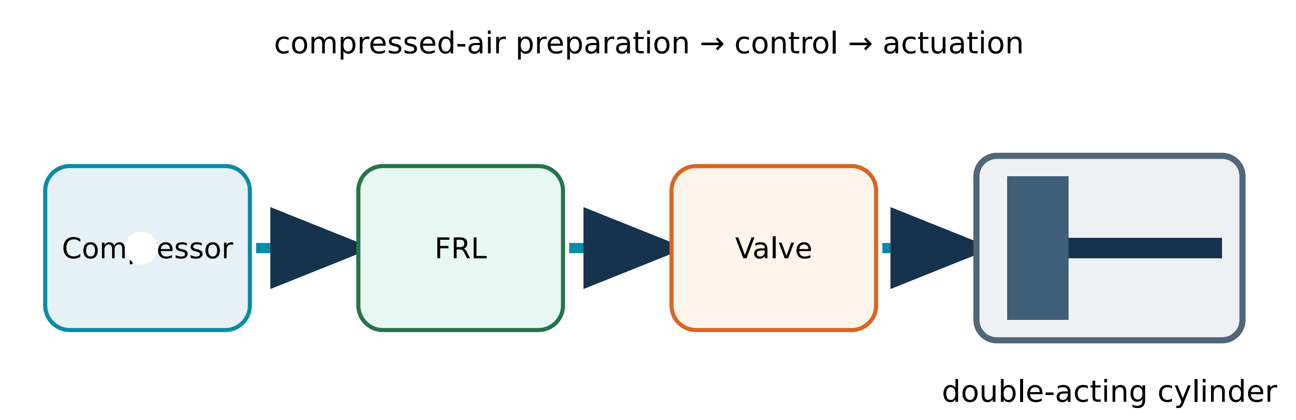
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.5 Pneumatic systems and compressed-air applications 1 h · Understand –

Pneumatics uses compressed air to transmit and control power. A typical system has compressor, receiver, dryer/filter, regulator, lubricator where required, directional valve, flow control, tubing and cylinder or air motor. Pneumatic systems are clean, fast and overload-safe, but air compressibility reduces stiffness and energy efficiency.

FRL means filter, regulator and lubricator	Directional control valves choose actuator motion
Flow-control valves adjust cylinder speed	Single-acting cylinder uses air in one direction and spring/load return
Double-acting cylinder uses air for extension and retraction	Common applications: clamping, pick-and-place, packaging, doors and light automation

Malayalam support: Compressed air-□□□□ path: compressor → receiver → FRL → control valve → actuator. Hose disconnect □□□□□□□□□□□□□ □□□□□□ pressure release □□□□□□□.



Pneumatic circuit. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

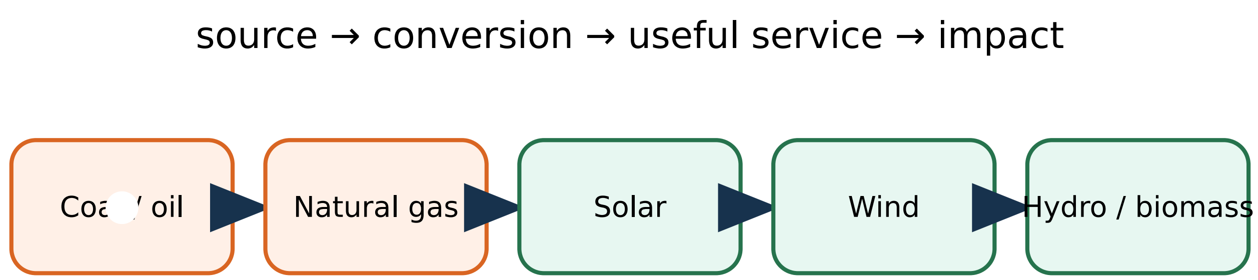
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.6 Conventional energy sources 1 h · Understand −

Coal, petroleum and natural gas are concentrated fossil energy sources formed over geological time. They support electricity generation, transport, process heating and chemical industries, but extraction, combustion and leakage create pollution, greenhouse-gas emissions, health impacts and resource-security concerns.

- Coal is widely used for thermal power and industrial heat
- Petroleum refining produces fuels, lubricants and petrochemical feedstocks
- Natural gas burns more cleanly than coal per unit energy but methane leakage matters
- Fossil fuels are finite and geographically uneven
- Efficiency, emission control and safe handling reduce but do not eliminate impacts

Malayalam support: Coal, petroleum, natural gas conventional energy sources □□□. Energy density □□□□□□□□□□□□□□□□□ pollution, greenhouse emission, finite resource □□□□□ limitations □□□□□.



Conventional energy chain. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and

assumptions.

Self-check

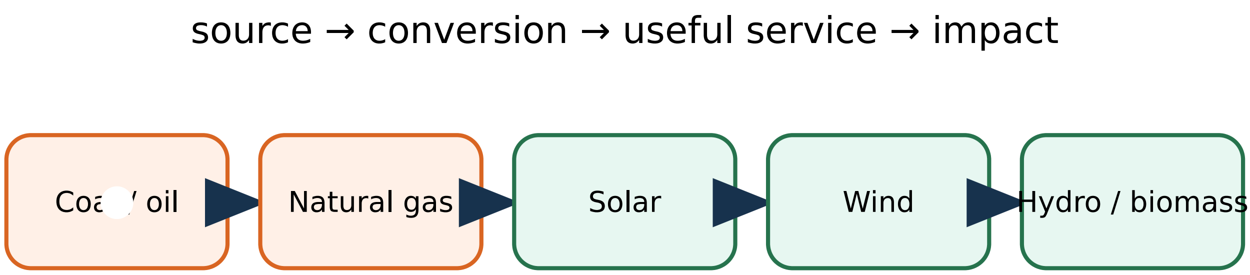
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.7 Renewable energy: solar, wind, hydro and biomass **2 h · Understand** –

Renewable sources are replenished by natural flows. Solar devices convert radiation to heat or electricity; wind turbines convert air motion; hydropower uses water head/flow; biomass stores solar energy chemically. Benefits include lower operating emissions and energy diversification, while variability, location, storage, land, ecological and initial-cost constraints must be managed.

- Solar PV output varies with irradiance, temperature, orientation and shading
- Wind power varies strongly with wind speed and requires suitable sites
- Hydro can be controllable but may affect river ecology and communities
- Biomass sustainability depends on feedstock, land, combustion and lifecycle management
- Hybrid systems, storage, grid management and efficiency improve reliability
- Renewable does not mean impact-free

Malayalam support: Solar, wind, hydro, biomass renewable sources □□□. Low operating emission advantage □□□□□□□□□□□□ variability, storage, land/ecology, initial cost □□□□□□ □□□□□□□□□□□□.



Renewable energy sources. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

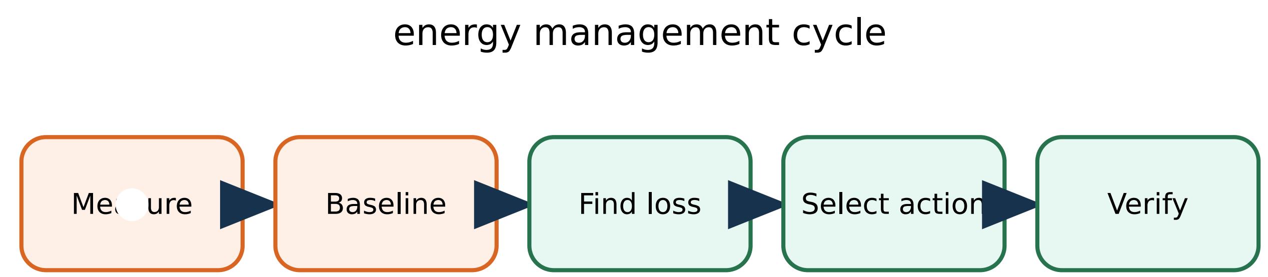
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

2.8 Energy conservation and management 1 h · Understand –

Energy conservation reduces avoidable consumption while maintaining required service, safety and quality. Energy management measures, records and improves performance through policy, metering, baseline comparison, audits, maintenance, efficient equipment and staff participation. The best first step is to identify where, when and why energy is used.

- Avoid idle running, leakage, throttling and unnecessary heating/cooling
- Maintain alignment, bearings, lubrication, filters and insulation
- Match motor/pump/compressor capacity to the load
- Record kWh, fuel, production and operating hours to create a baseline
- Prioritise measures by safety, saving, cost, payback and lifecycle benefit
- Verify savings after implementation

Malayalam support: Energy audit-▯ usage measure ▯▯▯▯▯▯ waste ▯▯▯▯▯▯▯▯▯ action plan ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯. Leakage, idle running, poor maintenance, wrong sizing ▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯.



Energy audit flow. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

MODULE III · CO3

Mechanical Power Transmission, Bearings and Lubrication

Theory 10 h | Practical 13 h | CO3 | Understand level

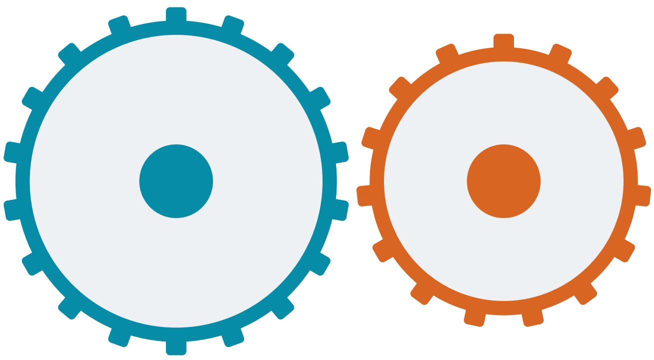
3.1 Motion and mechanical power transmission 1 h · Understand –

Machines transmit power from a prime mover to a driven member while changing speed, torque, direction, shaft position or motion type. Motion may be rotary, reciprocating, oscillating or linear. Selection considers power, speed ratio, centre distance, slip, shock load, noise, maintenance, alignment and safety guarding.

- Shafts transmit torque; couplings join shafts
- Belt and rope drives use friction and suit longer centre distances
- Chain and gear drives are positive drives with no normal slip
- Gears provide precise ratios and compact transmission
- Guards prevent access to nip points, rotating shafts and thrown parts

> **Malayalam support:** Prime mover-□ □□□□□□ driven machine-□□□□□□ power transfer □□□□□□□□□□□□□ transmission. Speed, torque, direction, centre distance, slip, maintenance □□□□□□ drive □□□□□□□□□□□□□□.

meshing teeth give a definite speed ratio



spur/helical: parallel · bevel: intersecting · worm: skew shafts

Mechanical power transmission. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

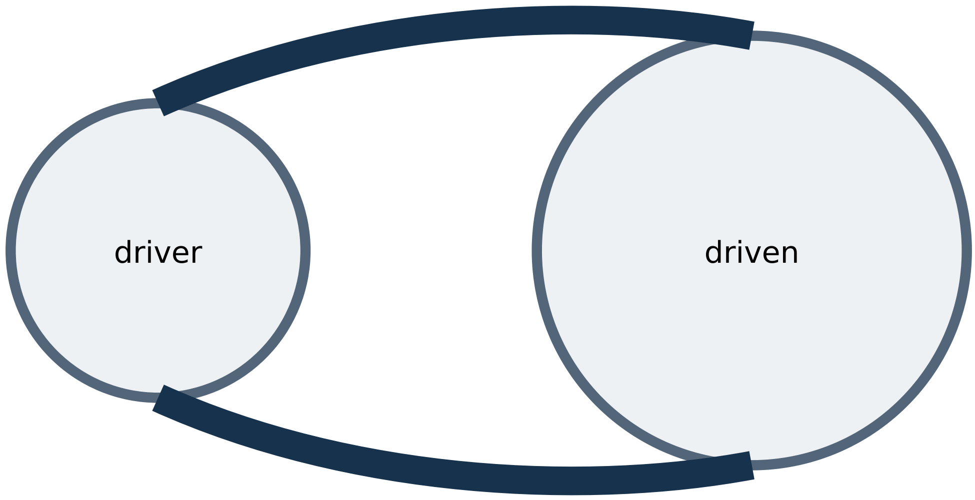
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

3.2 Open, crossed and V-belt drives 2 h · Understand –

A belt drive transmits power by friction between a flexible belt and pulleys. An open belt drive rotates parallel shafts in the same direction; a crossed belt reverses the driven direction and increases wrap but adds rubbing; a V-belt wedges into a grooved pulley for compact higher-capacity transmission.

- Driver is connected to the power source; driven pulley receives power
- Tight side carries greater tension than slack side
- Slip and creep cause actual velocity ratio to differ from the ideal
- Correct tension and alignment prevent heat, wear and bearing overload
- Advantages: simple, quiet, shock-absorbing and inexpensive
- Applications: fans, blowers, conveyors and machine tools

Malayalam support: Open belt-□ shafts same direction; crossed belt-□ opposite direction. V-belt groove-□ wedge action □□□□□□□□□□□□□□. Tension, alignment, guard □□□□□□□□□□□□.



open belt: same direction · V-belt uses wedge action

Open crossed and V-belt drives. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

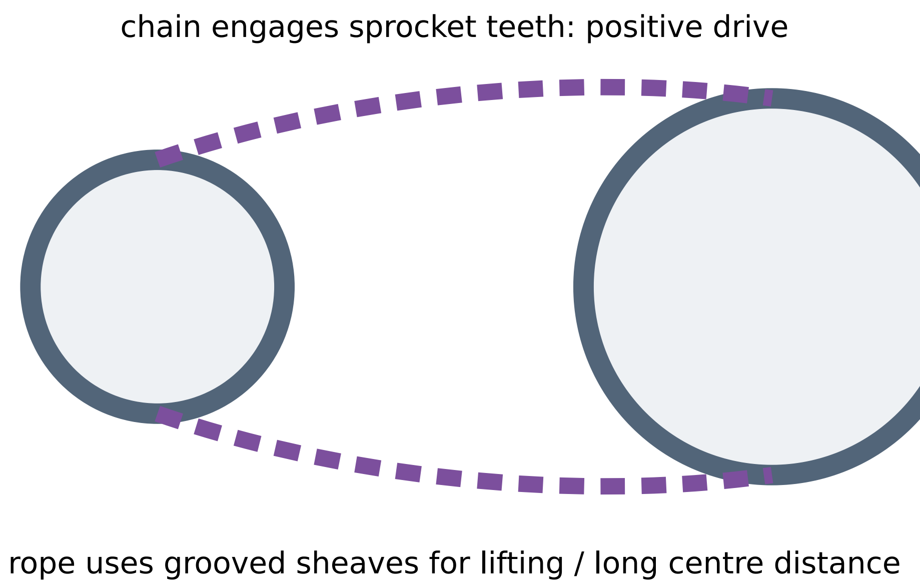
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

3.3 Rope drives and chain drives 1 h · Understand –

Rope drives use fibrous or wire ropes with grooved pulleys for long-distance or lifting duties. Chain drives use an articulated metal chain engaging sprocket teeth, giving a positive velocity ratio without slip. Chain drives need lubrication, alignment and controlled slack; ropes require inspection for wear, broken strands and correct groove contact.

- Rope drives suit large centre distance and heavy duty according to rope type
- Chain drive has driver sprocket, driven sprocket and chain links
- Chain polygonal action creates small speed fluctuation
- Chain drives suit bicycles, motorcycles, conveyors and timing systems
- Never operate an exposed chain or rope drive; inspect under isolation

Malayalam support: Chain sprocket teeth-□□□□ engage □□□□□□□□□□□□□□ normal slip □□□□. Rope/chain wear, alignment, lubrication, guard □□□□□□□□□□□□□.



Rope and chain drives. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

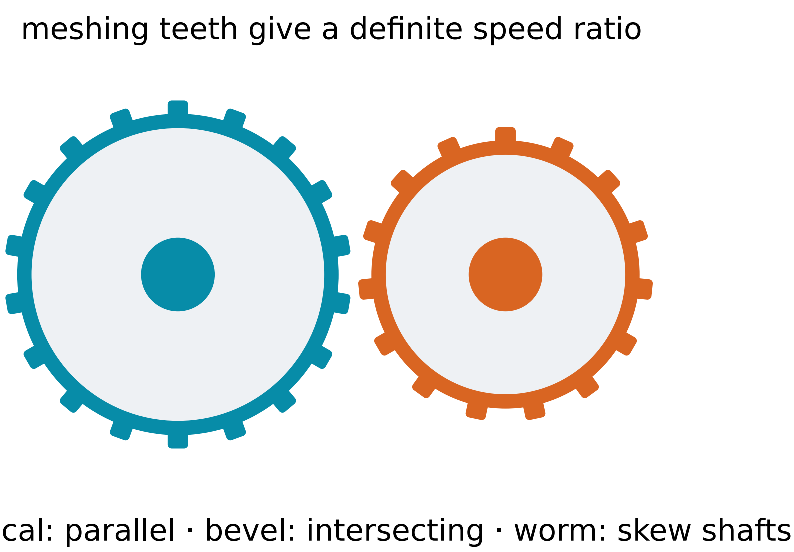
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

3.4 Spur, helical, bevel and worm gears 2 h · Understand –

Gears transmit power through meshing teeth and provide a definite speed ratio. Spur gears join parallel shafts with straight teeth; helical gears use inclined teeth for smoother engagement; bevel gears intersect shaft axes; worm gears connect skew shafts, commonly near 90°, and can provide a large reduction.

- External gears rotate in opposite directions
- Spur gear is simple and efficient but may be noisy at high speed
- Helical gear carries load gradually but produces axial thrust
- Bevel gears change direction between intersecting shafts
- Worm sets provide high reduction but sliding causes heat and lower efficiency
- Module, tooth count, pressure angle, face width and material identify a gear

> **Malayalam support:** Spur/helical parallel shafts-□□□□, bevel intersecting shafts-□□□□, worm skew/90° shafts-□□□□ □□□□□□□□□□□□□□□. Tooth count speed ratio □□□□□□□□□□□□□□□□□□.



Spur helical bevel and worm gears. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

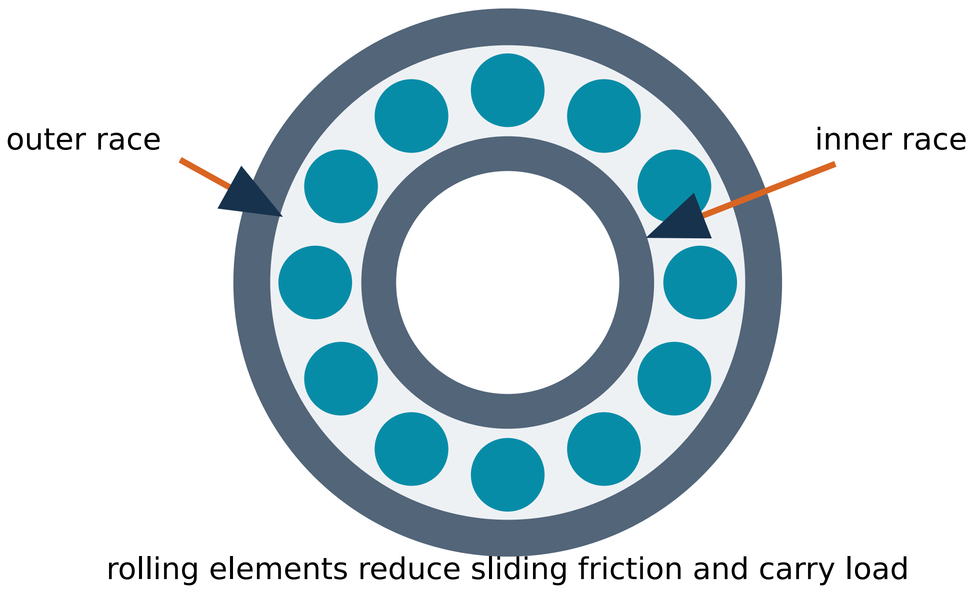
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

3.5 Journal, ball and roller bearings 3 h · Understand –

A bearing supports and locates a moving shaft while reducing friction and carrying radial, axial or combined load. Journal bearings support sliding contact through a lubricant film; rolling-element bearings place balls or rollers between inner and outer races. Bearing selection depends on load, speed, alignment, space, environment, life, lubrication and mounting.

- Ball bearings suit moderate radial/axial load and high speed
- Cylindrical rollers carry high radial load
- Taper rollers carry combined radial and axial load
- Needle rollers provide high radial capacity in small radial space
- Journal bearings can carry heavy load quietly when a stable oil film exists
- Incorrect fitting, contamination, misalignment and poor lubrication cause early failure

Malayalam support: Bearing shaft-□□ support □□□□□□ friction □□□□□□□□□□□□□□. Ball/roller bearing-□ rolling contact; journal bearing-□ lubricant film □□□□ sliding contact □□□.



Journal ball and roller bearings. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and

assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

3.6 Purpose, principle and methods of lubrication 1 h · Understand –

Lubrication introduces a suitable film between interacting surfaces to reduce friction and wear, remove heat, prevent corrosion, seal contaminants and reduce noise. Oil, grease, solid lubricants and specialised fluids are selected according to speed, load, temperature, environment and equipment design.

- Boundary lubrication: thin film with surface interaction
- Hydrodynamic lubrication: full fluid film generated by relative motion
- Grease stays in place and seals but removes heat less effectively than circulating oil
- Methods include manual oiling, grease gun, splash, ring, wick, drip, pressure-feed and mist
- Too little causes wear; too much can cause churning, heat and seal damage
- Never mix incompatible lubricants without approval

Malayalam support: Lubricant friction, wear, heat, corrosion □□□□□□□□□□□□□. Correct grade, quantity, interval, cleanliness □□□□□□ □□□□□□□□□; over-lubrication-□□□ □□□□□□□□.

oil/grease film separates surfaces, removes heat and limits wear

boundary → mixed → full-film lubrication

Lubrication regimes and methods. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

MODULE IV · CO4

Manufacturing Processes and Industrial Safety

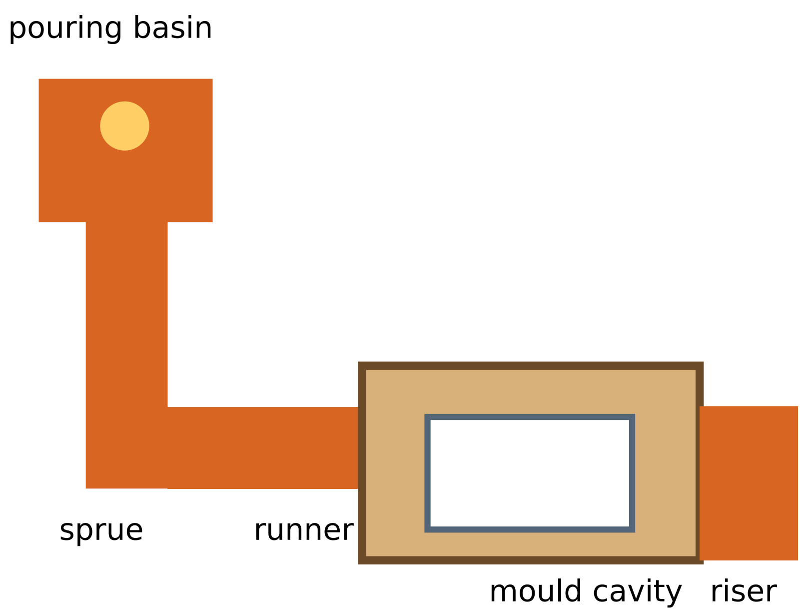
Theory 13 h | Practical 11 h | CO4 | Understand level + one Remember row

4.1 Casting principle and types 2 h · Understand –

Casting produces a component by pouring molten material into a mould cavity, allowing it to solidify and removing/finishing the casting. Sand casting is versatile for complex shapes and varied size; permanent-mould and die casting improve repeatability for suitable metals and production volume; investment casting gives fine detail.

- Pattern creates the mould cavity with allowances
- Mould contains cavity, sprue, runner, gate, riser and vents
- Core forms an internal hole or passage
- Riser feeds liquid metal to compensate for solidification shrinkage
- Common defects include blowholes, shrinkage cavity, misrun, cold shut and sand inclusion
- Molten-metal work requires trained personnel, dry tools, PPE and controlled exclusion zones

Malayalam support: Pattern □□□□□□□□□□ mould cavity □□□□□□□□□ molten metal pour □□□□□□ solidify □□□□□□□□□□□□□□ casting. Sprue, runner, gate, riser, vent, core □□□□□□□□□□□□□.



Sand casting system. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

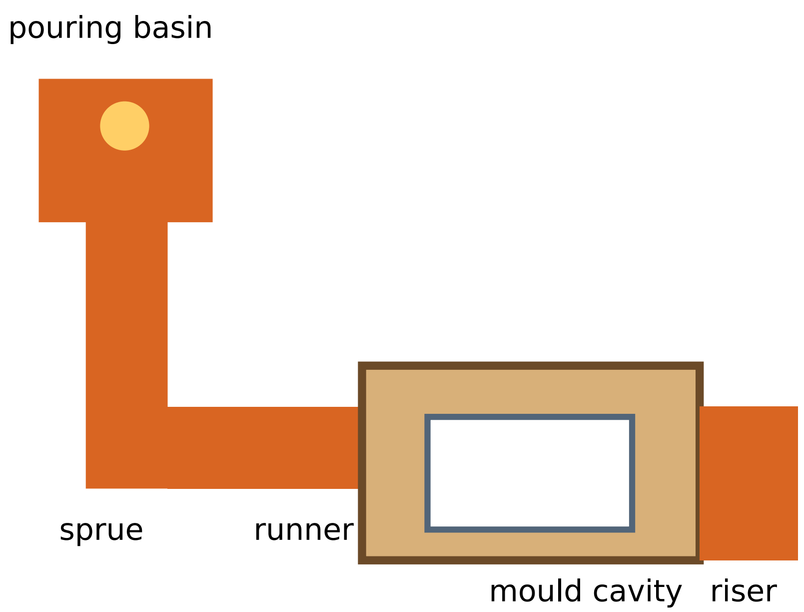
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

4.2 Components and moulding materials 2 h · Remember –

A sand mould needs refractory moulding sand with sufficient strength, permeability, plasticity and collapsibility. Silica sand is the main refractory constituent; binder gives strength, moisture activates suitable binders and additives adjust surface or breakdown behaviour. Flask, pattern, moulding tools, core and gating components must be identified correctly.

- Cope is the upper moulding flask; drag is the lower part
- Parting line separates cope and drag
- Ramming compacts sand uniformly around the pattern
- Vent wire provides gas escape paths
- Core prints locate and support a core
- Gating design controls clean, calm and adequate metal flow

Malayalam support: Moulding sand-□□ strength, permeability, plasticity, refractoriness □□□□□□□□□. Cope, drag, pattern, core, sprue, runner, gate, riser □□□□□□□□□□□□□.



Moulding tools and casting components. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

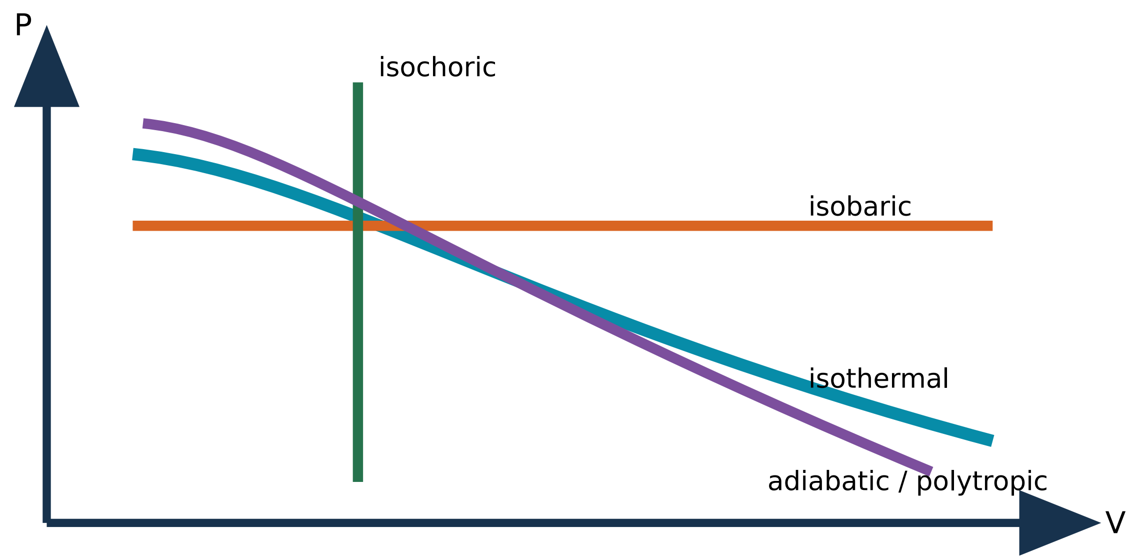
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

4.3 Forging, rolling, extrusion and drawing 2 h · Understand –

Metal forming changes shape mainly by plastic deformation without intentionally removing material. Forging compresses metal between tools; rolling reduces thickness or changes section between rolls; extrusion forces material through a die; drawing pulls wire, rod or tube through a die to reduce section.

Hot working lowers flow stress and allows large deformation	Cold working improves finish and dimensional control but increases force and work hardening
Forging improves directional grain flow and mechanical integrity	Rolling makes plates, sheets, rails and structural sections
Extrusion makes long constant profiles	Drawing makes wire, bar and tube with controlled diameter

Malayalam support: Forming-□ material remove □□□□□□□□ plastic deformation □□□ shape □□□□□□□□□□□. Forging compression, rolling rolls, extrusion die □□□ push, drawing die □□□ pull □□□□□□□□□□□.



Metal forming processes. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

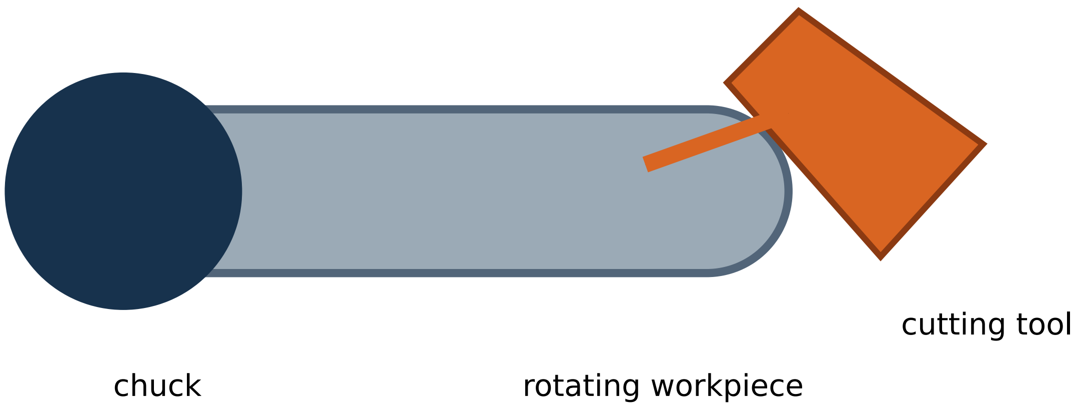
4.4 Turning, drilling, milling and grinding 2 h · Understand –

Machining removes material as chips or fine particles to obtain shape, size and finish. In turning, a rotating workpiece meets a cutting tool; drilling creates a round hole; milling uses a rotating multi-edge cutter; grinding uses abrasive grains for fine finishing and accuracy.

- Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock
- Drilling variables include drill size, speed and feed
- Milling produces flat, slot, profile and gear-related surfaces with suitable cutters
- Grinding wheel selection, guarding, ring test/mounting and dressing are safety-critical
- Cutting speed, feed and depth of cut influence time, finish, force and tool life
- Remove chips with a brush under safe conditions, never by bare hand

> **Malayalam support:** Turning-□ work rotate □□□□□□□; drilling hole □□□□□□□□□□; milling multi-point cutter □□□□□□□□□□□; grinding abrasive grains □□□□□□□□□□ fine finish □□□□□.

turning: work rotates · drilling/milling/grinding: tool rotates



Turning drilling milling and grinding. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

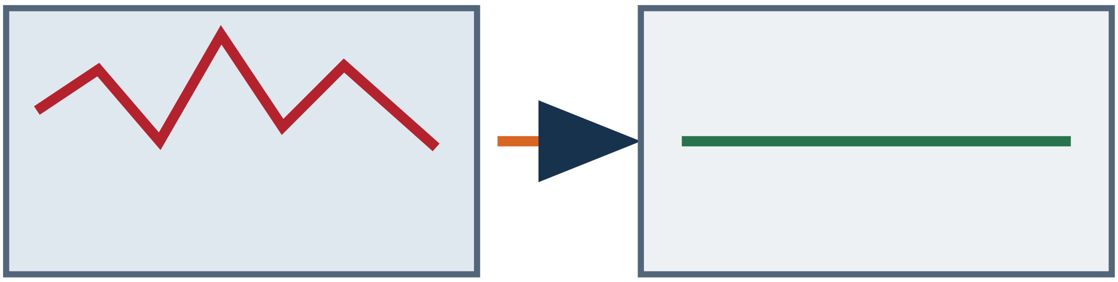
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

4.5 Surface-finishing processes 1 h · Understand −

Surface finishing improves roughness, appearance, cleanliness, dimensional accuracy, wear, friction or corrosion performance. Filing, polishing, buffing, lapping, honing, grinding, blasting and coating processes serve different purposes; the correct process depends on material, geometry and required surface function.

- Polishing removes small irregularities; buffing improves lustre
- Lapping uses fine abrasive between surfaces for very high accuracy/finish
- Honing uses bonded abrasive sticks, commonly for bores
- Blasting cleans and textures a surface using controlled media
- Coatings may protect against corrosion or wear
- Surface roughness should be specified and verified, not judged only by appearance

Malayalam support: Surface finishing roughness, appearance, accuracy, wear/corrosion performance [illegible]. Process material-[illegible] required function-[illegible] [illegible].



Surface finishing comparison. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

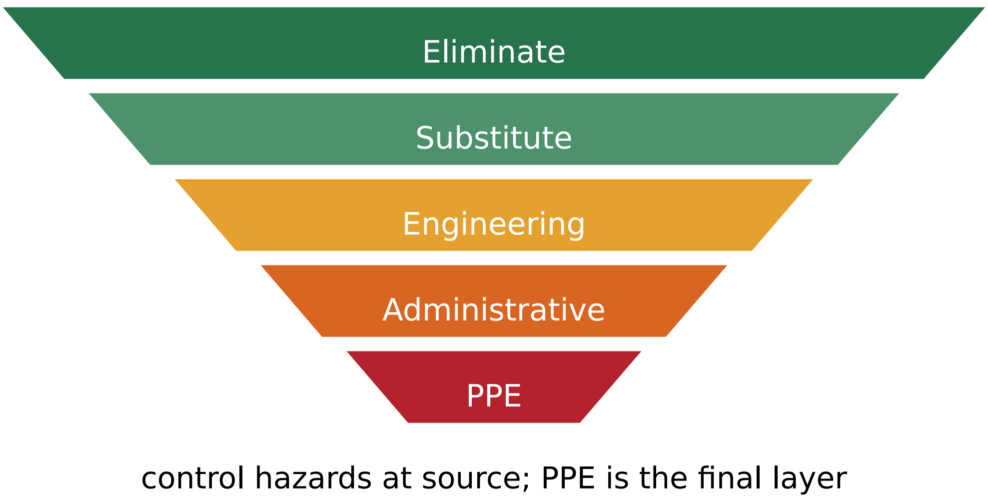
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

4.6 Importance of safety and workshop rules 1 h · Understand –

Workshop safety prevents injury, occupational illness, equipment damage, fire, environmental harm and production loss. Rules convert hazard knowledge into consistent action: authorisation, housekeeping, guarding, isolation, correct tools, clear communication, emergency readiness and reporting.

- Know the machine-specific safe operating procedure before use
- Inspect guards, tool condition, work holding and emergency stop
- Use lockout/isolation before maintenance or clearing a jam
- Keep aisles, floors and exits clean and unobstructed
- Do not distract an operator or bypass an interlock
- Report defects, near misses and incidents promptly

> **Malayalam support:** Authorisation □□□□□□□ machine operate □□□□□□□□□. Guard, work holding, emergency stop, housekeeping, isolation □□□□□□ □□□ task-□□□ □□□□□□ □□□□□□□□□□□□.



Workshop safety rules. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

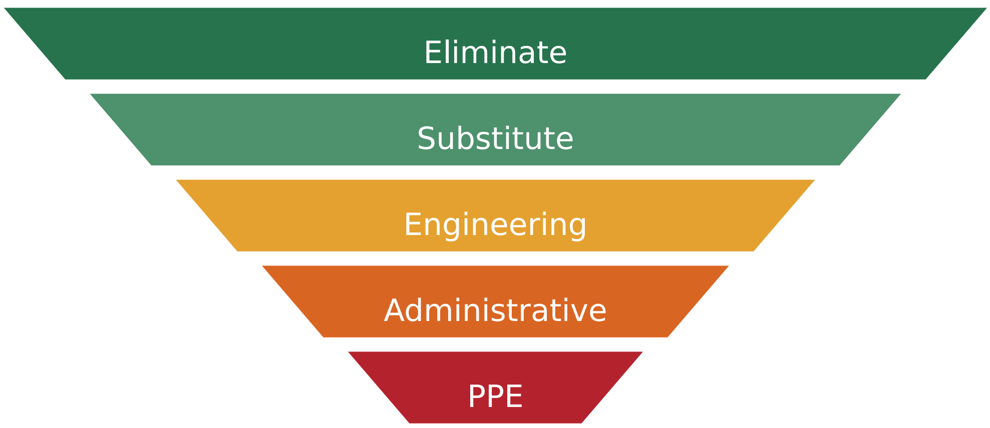
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

4.7 PPE, accident causes and prevention 2 h · Understand –

Personal protective equipment reduces exposure when hazards cannot be eliminated or fully controlled by engineering and administrative measures. Accidents commonly involve unsafe conditions and unsafe acts interacting with inadequate planning, supervision or maintenance. Prevention uses the hierarchy of controls: eliminate, substitute, engineer, administer and finally protect with PPE.

- Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection
- Welding requires rated filter protection, gloves, clothing, ventilation and fire control
- Gloves are not worn near exposed rotating machinery where entanglement is possible
- Safety footwear protects from impact, penetration and slip according to rating
- Hearing and respiratory protection require correct selection and fit
- Investigate root causes rather than blaming the injured person

Malayalam support: PPE □□□□□ control layer □□□. □□□□□ hazard eliminate/substitute □□□□□□□; guard, ventilation, procedure □□□□□□□□□□□ □□□□ □□□□□□ PPE □□□□□□□□□□□□□□□.



control hazards at source; PPE is the final layer

PPE and hierarchy of controls. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

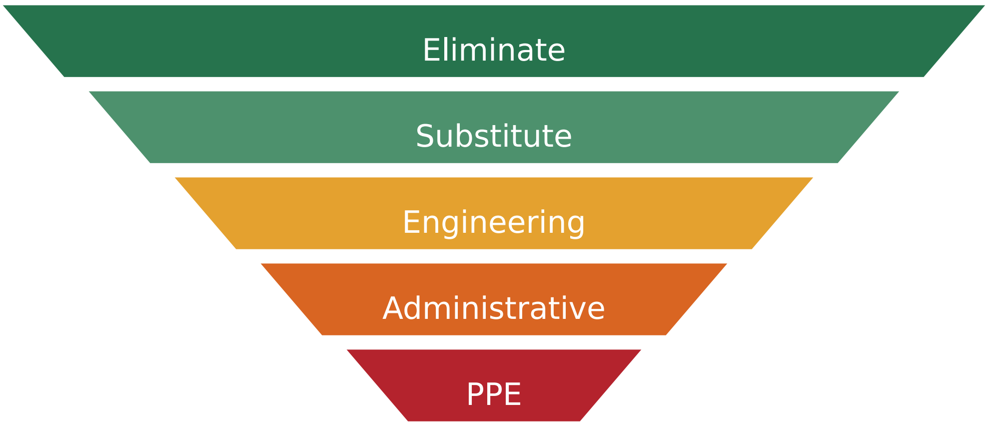
Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

4.8 Industrial safety culture and continuous improvement 1 h · Understand –

Industrial safety is a managed system involving leadership, worker participation, risk assessment, training, maintenance, permit systems, emergency preparedness, incident learning and continuous improvement. A strong safety culture makes safe production a normal value rather than a response after an accident.

- Management provides policy, resources, competent supervision and accountability
- Safety officers coordinate risk control, inspection, training and compliance
- Workers participate in hazard reporting, toolbox talks and procedure improvement
- Leading indicators include inspections and action closure; lagging indicators include injuries and damage
- Plan–Do–Check–Act supports continuous improvement
- Production pressure never justifies bypassing a critical safety control

Malayalam support: Safety culture management-□□□ workers-□□□ shared responsibility □□□. Hazard report, training, inspection, corrective action, emergency drill □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□□.



control hazards at source; PPE is the final layer

Industrial safety management cycle. Identify parts, trace motion/energy/material flow, then state one application and one safety point.

Engineering connection

Relate this concept to an actual machine, service requirement, maintenance decision or industrial safety control. Clearly state the system boundary, motion, energy path and assumptions.

Self-check

Define it, identify or draw its main parts, explain the basic working principle, compare one alternative and state an application plus limitation.

PRACTICUM

Practical identification and record centre

The official practical syllabus is expanded into safe observation records. These are not operating instructions for unsupervised work.

Faculty must isolate the equipment and demonstrate safe access. Students may identify, trace, sketch and record only within the authorised task.

Completion planner

- [ ] M1 · Activity 1: Identify tools, machines, materials, IC engine and refrigerator
- [ ] M1 · Activity 2: Identify components of petrol/diesel IC engines
- [ ] M2 · Activity 3: Identify pressure-measuring instruments
- [ ] M2 · Activity 4: Identify hydraulic pumps and their parts
- [ ] M2 · Activity 5: Identify hydraulic turbines and their parts
- [ ] M2 · Activity 6: Identify pneumatic system components
- [ ] M2 · Activity 7: Energy conservation and audit observation
- [ ] M3 · Activity 8: Identify mechanical power-transmission systems
- [ ] M3 · Activity 9: Identify belt and chain drives
- [ ] M3 · Activity 10: Identify spur, helical, bevel and worm gears
- [ ] M3 · Activity 11: Identify journal, ball and roller bearings
- [ ] M3 · Activity 12: Identify lubricants and lubrication methods
- [ ] M4 · Activity 13: Identify moulding and casting components
- [ ] M4 · Activity 14: Identify metal-forming processes
- [ ] M4 · Activity 15: Identify machining, welding and soldering equipment
- [ ] M4 · Activity 16: Identify surface finishing and safety equipment

0 of 16 activities marked complete.

Activity 1: Identify tools, machines, materials, IC engine and refrigerator

M1 · 3 h

Malayalam practical guide: Tool/material name □□□□□□ □□□□, identification feature, correct use, condition, hazard □□□□□□□□□ record □□□□□□□.

Aim

Recognise common mechanical workshop tools, engineering materials, a heat engine/IC engine and a refrigerator without operating or dismantling live equipment.

Items/components

spanner and socket set · screwdriver and pliers · torque wrench and puller · steel, cast iron, aluminium, copper, polymer and rubber samples · engine assembly · refrigerator compressor/condenser/evaporator

Mapped outcome

CO1 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Confirm isolation and faculty authorisation
2. Name each tool and match it to its safe purpose
3. Identify material samples from visible/physical features without destructive testing
4. Trace the visible energy conversion in engine and refrigerator
5. Sketch the selected equipment and label principal parts
6. Record one misuse hazard for each tool group

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Tool/component/material name, identification feature, function, condition and hazard.

Expected result statement: Tools, materials and thermal machines were correctly recognised and related to function.

Safety: Use only isolated display equipment; do not open a charged refrigeration circuit or fuel system.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 2: Identify components of petrol/diesel IC engines M1 · 3 h

Malayalam practical guide: Piston reciprocates □□□□□□□□□□□□; crankshaft rotate □□□□□□□□□□□□. SI-□ spark plug, CI-□ injector □□□□□□□□□□□□□□.

Aim

Identify principal parts of an isolated two-/four-stroke petrol or diesel engine.

Items/components

cylinder block/head · piston and rings · connecting rod and crankshaft · camshaft and valves · spark plug or injector · flywheel and lubrication/cooling parts

Mapped outcome

CO1 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Verify the engine is isolated, cool and securely supported
2. Observe an assembled section/model before separate parts
3. Locate gas path, piston path and crank rotation
4. Differentiate SI ignition part from CI injection part
5. Prepare a labelled sketch and part-function table

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Part, location, material/feature, motion and function.

Expected result statement: IC-engine parts and their role in the operating cycle were explained.

> **Safety:** Heavy components require correct lifting; sharp edges and trapped oil must be controlled.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 3: Identify pressure-measuring instruments M2 · 3 h

> **Malayalam practical guide:** Instrument disconnect ▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ pressure zero ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯.

Aim

Recognise piezometer, U-tube manometer and pressure gauges and explain their basic principle.

Items/components

piezometer · simple U-tube manometer · differential manometer · Bourdon pressure gauge · vacuum/compound gauge

Mapped outcome

CO2 · Understand

Date / equipment ID

Faculty-supervised observation sequence

1. Confirm the rig is depressurised
2. Identify pressure connection, sensing element, scale and zero
3. Read liquid levels at eye level
4. Classify the pressure indicated
5. Sketch one instrument and state unit/range

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Instrument, principle, range, unit, zero condition and application.

Expected result statement: Pressure instruments were identified and their safe reading method explained.

Safety: Never loosen a connection under pressure; use rated tubing and fittings only.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 4: Identify hydraulic pumps and their parts M2 · 2 h

Malayalam practical guide: Pump type, flow direction, prime mover, suction/delivery parts □□□□□□□□□□□□.

Aim

Differentiate centrifugal and reciprocating pumps and identify major components.

Items/components

impeller and volute casing · suction and delivery connections · shaft and seal · piston/plunger and cylinder · suction and delivery non-return valves

Mapped outcome

CO2 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Isolate electrical and hydraulic energy
2. Trace flow direction from suction to delivery
3. Identify rotating versus reciprocating elements
4. Compare continuous and pulsating delivery
5. Record priming need and one typical application

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Pump type, energy input, principal parts, flow nature and application.

Expected result statement: Centrifugal and reciprocating pumps were distinguished by construction and working.

Safety: Never run dry or remove a guard; control residual liquid and pressure.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*

3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 5: Identify hydraulic turbines and their parts M2 · 2 h

Malayalam practical guide: Pelton high head, Francis medium head, Kaplan low head/high flow □□□□ matching □□□□□□□.

Aim

Recognise Pelton, Francis and Kaplan turbine models/components.

Items/components

Pelton nozzle, spear, buckets and casing · Francis spiral casing, guide vanes, runner and draft tube · Kaplan propeller runner and guide mechanism

Mapped outcome

CO2 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Use an isolated cut-section/model
2. Identify the water entry, runner path and exit
3. Classify impulse or reaction type
4. Match each turbine to head/flow range
5. Sketch one runner and label energy direction

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Turbine type, head range, flow direction, runner feature and application.

Expected result statement: Three hydraulic turbine types and their basic working principles were explained.

Safety: No access to a running runner; isolate water and rotation before inspection.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 6: Identify pneumatic system components M2 · 2 h

Malayalam practical guide: Air path arrow □□□□□□□□□□ FRL → valve → cylinder → exhaust □□□□□□□□□□□□□□□□.

Aim

Trace a basic compressed-air path and identify preparation, control and actuator components.

Items/components

compressor/receiver · filter-regulator-lubricator · pressure gauge · directional valve · flow control · single-/double-acting cylinder · tubing and fittings

Mapped outcome

CO2 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Depressurise before touching connections
2. Follow airflow from source to actuator and exhaust
3. Identify valve ports and actuator type
4. Explain how direction and speed are controlled

5. Prepare a labelled circuit-flow sketch

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Component, symbol/ports, function, set pressure and hazard.

Expected result statement: The pneumatic energy path and component functions were correctly traced.

> **Safety:** Compressed air must never be directed at the body; hoses require secure rated fittings.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 7: Energy conservation and audit observation M2 · major-topic activity

> **Malayalam practical guide:** Boundary define ▯▯▯▯▯▯ measure → baseline → loss → action → verify sequence record ▯▯▯▯▯▯▯▯.

Aim

Observe a selected laboratory/workshop system and prepare a basic energy-use and conservation record.

Items/components

nameplate/energy meter readings where authorised · operating-hour log · leak/idle-running checklist · maintenance condition · baseline and action sheet

Mapped outcome

CO2 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Define the equipment and observation boundary
2. Record rated/observed energy and operating time without opening live panels
3. Identify idle running, leakage, throttling, poor maintenance or unnecessary load
4. Propose safe conservation actions
5. State how saving would be verified after implementation

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Energy source, operating hours, measured/estimated use, avoidable loss, action, responsible person and verification.

Expected result statement: An elementary energy-audit flow and practical conservation opportunities were documented.

> **Safety:** Electrical measurement and machine access are faculty-controlled; do not open energised enclosures.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 8: Identify mechanical power-transmission systems M3 · 3 h

Malayalam practical guide: Driver/driven, motion direction, friction/positive drive, guard □□□□□□ record □□□□□□□.

Aim

Recognise shaft, coupling, belt, rope, chain and gear transmission systems.

Items/components

shaft and coupling · flat/V belt and pulleys · rope and grooved sheave · chain and sprockets · gear pair and guards

Mapped outcome

CO3 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Confirm zero energy and guarding
2. Identify driver and driven members
3. State motion type, shaft arrangement and likely speed change
4. Classify friction or positive drive
5. Record inspection and maintenance points

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Drive, driver/driven, shaft relation, slip, ratio feature and application.

Expected result statement: Transmission systems were classified and related to service requirements.

Safety: Never rotate or tension a drive by hand unless isolated and authorised.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 9: Identify belt and chain drives M3 · 3 h

Malayalam practical guide: Belt crack/glazing, pulley alignment, chain slack/lubrication □□□□□□ □□□□□□□□□□□□.

Aim

Identify open, crossed and V-belt arrangements plus chain-drive parts and specifications.

Items/components

flat belt · V-belt section · driver/driven pulley · chain links and connecting link · driver/driven sprocket · tension/guard arrangement

Mapped outcome

CO3 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Observe only stationary isolated drives
2. Determine rotation direction
3. Inspect alignment, tension and wear indicators
4. Read available belt/chain designation
5. Compare slip, noise, lubrication and maintenance

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Type, dimensions/designation, direction, tension, wear and application.

Expected result statement: Belt and chain drive parts, operation and inspection points were identified.

Safety: Nip points and stored tension remain hazardous after power isolation.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 10: Identify spur, helical, bevel and worm gears M3 · 3 h

Malayalam practical guide: Tooth form □□□□□□□□□ shaft relation-□□ application-□□ explain □□□□□□□.

Aim

Recognise gear types from tooth direction and shaft arrangement.

Items/components

spur gear · helical gear · bevel gear · worm and worm wheel · gear shaft/bearing support

Mapped outcome

CO3 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Use loose samples or isolated demonstration units
2. Observe tooth orientation and pitch surface
3. Match gear to parallel, intersecting or skew shafts
4. Count teeth where authorised and infer speed trend
5. Record advantages, limitations and application

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Gear type, tooth form, shaft relation, direction, ratio feature and use.

Expected result statement: Four common gear types were identified and compared.

> **Safety:** Sharp teeth and pinch points require careful handling; heavy gears need lifting support.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 11: Identify journal, ball and roller bearings M3 · 2 h

> **Malayalam practical guide:** Ball/roller shape, race, cage, radial/axial load capability □□□□□□□□□□□□.

Aim

Differentiate sliding and rolling-element bearings and common bearing types.

Items/components

journal bearing/bush · deep-groove ball bearing · cylindrical roller bearing · taper roller bearing · needle bearing · inner/outer races and cage

Mapped outcome

CO3 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Clean the sample externally without spinning with compressed air
2. Identify load direction and rolling element
3. Read visible designation only if clear
4. Inspect for roughness, corrosion or damaged seal
5. Match type to application

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Bearing type, contact, load, speed suitability, lubrication and condition.

Expected result statement: Bearing types and their typical load applications were explained.

Safety: Do not overspeed a loose bearing with compressed air; use gloves only away from rotation.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 12: Identify lubricants and lubrication methods M3 · 2 h

Malayalam practical guide: Correct grade, clean tool, specified quantity/interval □□□□□□□□□□; used oil separate □□□ collect □□□□□□□.

Aim

Recognise oil, grease and common application methods without mixing products.

Items/components

lubricating oil · grease · oil can/drip/wick · grease gun · splash/ring lubrication · pressure-feed system

Mapped outcome

CO3 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Read container grade and hazard information
2. Observe colour/consistency without skin contact
3. Match method to bearing/gear arrangement
4. Identify fill, drain and level points
5. Record contamination and over-lubrication risks

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Lubricant, grade, method, application, interval and hazard.

Expected result statement: Lubricants and methods were matched to safe mechanical applications.

Safety: Avoid skin contact and spills; never mix grades or dispose oil to drain.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 13: Identify moulding and casting components M4 · 3 h

Malayalam practical guide: Cold model □□□□□□□□□□ sprue → runner → gate → cavity flow trace □□□□□□□; riser feeding explain □□□□□□□.

Aim

Identify pattern, flask, mould, core, gating and common foundry tools.

Items/components

pattern and core box · cope and drag · moulding sand · sprue, runner, gate and riser · rammer, trowel, vent wire and lifter

Mapped outcome

CO4 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Use cold demonstration material only
2. Separate pattern, mould and core functions
3. Trace proposed metal and gas paths
4. Check parting line and feeding locations
5. Prepare a labelled mould section

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Component, shape/location, function, defect prevented and safety point.

Expected result statement: Moulding/casting components and metal-flow sequence were recognised.

Safety: No molten-metal activity without full foundry controls and trained faculty.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 14: Identify metal-forming processes M4 · 3 h

Malayalam practical guide: Material removal □□□□; plastic deformation □□□. Force direction arrow □□□ sketch □□□□□□□.

Aim

Recognise forging, rolling, extrusion and drawing from equipment, force and material flow.

Items/components

hammer/press and dies · rolling mill rolls · extrusion container/die · wire/tube drawing die · formed product samples

Mapped outcome

CO4 · Understand

Date / equipment ID

Faculty-supervised observation sequence

1. Observe isolated equipment, samples or models
2. Identify whether force pushes, squeezes or pulls
3. Mark material entry and exit directions
4. Compare hot and cold working
5. Relate process to typical product

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Process, force, tooling, material flow, product and advantage.

Expected result statement: Four forming processes were identified from their working principle.

Safety: Hot metal, pinch points and stored press energy require controlled exclusion.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 15: Identify machining, welding and soldering equipment M4 · 3 h

Malayalam practical guide: Work motion, tool motion, holding method, guard, emergency stop, PPE □□□□□□ □□□□□□.

Aim

Identify lathe, drilling, milling, grinding, welding and soldering equipment and their safe zones.

Items/components

lathe parts · drill spindle/table · milling cutter/table · grinding wheel/guard · welding source/holder/return · soldering iron/stand

Mapped outcome

CO4 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Confirm equipment isolation
2. Identify work motion and tool motion
3. Locate work holding, guard and emergency stop
4. Match process to output feature
5. Demonstrate soldering only under faculty instruction with ventilation

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Machine/process, tool, work motion, output, control and PPE.

Expected result statement: Machining and joining equipment were correctly recognised with safety controls.

> **Safety:** No live machining/welding without authorisation; grinding and welding PPE are process-specific.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*

2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

Activity 16: Identify surface finishing and safety equipment M4 · 2 h

Malayalam practical guide: PPE hazard-□□□□□□□□□□ select □□□□□□□; damaged/expired item report □□□□□□ replace □□□□□□□.

Aim

Recognise finishing methods and explain the function/limitation of safety equipment and PPE.

Items/components

file and abrasive papers · polishing/buffing equipment · lapping/honing samples · blast-cleaned/coated samples · goggles, face shield, helmet, gloves, footwear, hearing/respiratory PPE · fire extinguisher and emergency stop

Mapped outcome

CO4 · Understand

Date / equipment ID

____

Faculty-supervised observation sequence

1. Compare before/after surface samples
2. Match finish to function
3. Identify PPE markings and condition
4. Match extinguisher/control to authorised training scenario
5. Record maintenance, storage and replacement needs

Record table

Item / feature	Identification clue	Function / working	Application	Safety / condition

Item / feature	Identification clue	Function / working	Application	Safety / condition

Observation focus: Item, purpose, limitation, inspection criterion and application.

Expected result statement: Finishing and safety equipment were identified and selected by hazard/function.

Safety: PPE does not permit unsafe operation; extinguisher use follows institutional emergency training.

Viva with answer cues

1. What feature identifies this item? *State shape/construction, connection or motion.*
2. What energy or motion change occurs? *Trace input, conversion and output.*
3. What is the main safety control? *Name isolation, guard, PPE or pressure/heat control as applicable.*

Faculty signature: ____________________

SUPPORTING CALCULATIONS

Formula and unit bank

The official detailed syllabus asks mainly for definitions and basic working principles. These formulas support units, examples, self-learning and engineering interpretation; they do not increase the official examination depth.

1. Density

```
ρ = m / V
```

Symbols / units: ρ density (kg/m^3), m mass (kg), V volume (m^3)

Example: A 12 kg liquid occupies 0.015 m^3: $\rho = 12/0.015 = 800$ kg/m^3.

2. Specific weight

```
γ = ρg
```

Symbols / units: γ specific weight (N/m^3), $g \approx 9.81$ m/s^2

Example: For $\rho = 800$ kg/m^3, $\gamma = 800 \times 9.81 = 7{,}848$ N/m^3.

3. Specific volume

```
v = 1 / ρ
```

Symbols / units: v specific volume (m^3/kg)

Example: For ρ = 800 kg/m^3, v = 0.00125 m^3/kg.

4. Specific gravity

```
SG = ρfluid / ρwater
```

Symbols / units: dimensionless; use consistent temperature/reference

Example: If ρ = 850 kg/m^3 and water ≈ 1000 kg/m^3, SG = 0.85.

5. Pressure

```
p = F / A
```

Symbols / units: p Pa, F N, A m^2

Example: A 1,500 N force on 0.01 m^2 gives 150,000 Pa = 150 kPa.

6. Hydrostatic pressure

```
p = ρgh
```

Symbols / units: h vertical liquid depth (m)

Example: Water at 2 m depth: p ≈ 1000 × 9.81 × 2 = 19.62 kPa gauge.

7. Pascal force multiplication

```
F₂ / F₁ = A₂ / A₁
```

Symbols / units: ideal confined-fluid relation

Example: A_2/A_1 = 20 and F_1 = 100 N gives ideal F_2 = 2,000 N.

8. Mechanical work

```
W = F s
```

Symbols / units: W joule, F newton, s displacement in force direction (m)

Example: 500 N through 3 m gives 1,500 J.

9. Power

```
P = W / t; P = Tω
```

Symbols / units: P watt, T torque N·m, ω rad/s

Example: 6,000 J in 20 s gives 300 W.

10. Efficiency

```
η = useful output / input × 100%
```

Symbols / units: use the same energy or power basis

Example: 4 kW output from 5 kW input gives η = 80%.

11. Ideal belt velocity ratio

```
N₁/N₂ = D₂/D₁
```

Symbols / units: no slip; driver 1, driven 2

Example: Driver 200 mm at 900 rpm, driven 600 mm: N_2 = 300 rpm.

12. Ideal external gear ratio

```
N₁/N₂ = T₂/T₁
```

Symbols / units: T tooth count; meshing external gears reverse direction

Example: 20-tooth driver at 1200 rpm drives 60 teeth: N_2 = 400 rpm.

13. Heat for temperature change

```
Q = mcΔT
```

Symbols / units: Q J, c J/(kg·K); supporting energy example

Example: 2 kg water, c = 4180 J/kgK, rise 10 K: Q = 83.6 kJ.

PROBLEM SOLVING

Solved examples and practice

Example 1: Classify a running air compressor−

Given / observation: Air enters and compressed air leaves while shaft work and heat cross the boundary.

Solution and conclusion: It is an open thermodynamic system because both matter and energy cross the chosen boundary.

Example 2: Distinguish petrol and diesel engine ignition−

Given / observation: Observe the combustion-initiation component.

Solution and conclusion: A petrol/SI engine normally uses a spark plug; a diesel/CI engine uses high compression and fuel injection.

Example 3: Manometer pressure difference−

Given / observation: Water column difference is 0.35 m.

Solution and conclusion: $p = \rho gh = 1000 \times 9.81 \times 0.35 = 3{,}433.5$ Pa ≈ 3.43 kPa.

Example 4: Hydraulic press force−

Given / observation: Small piston area 4 cm^2, large piston area 80 cm^2, input 120 N.

Solution and conclusion: $F_2 = 120 \times 80/4 = 2{,}400$ N ideally; actual output is lower because of losses.

Example 5: Select a turbine−

Given / observation: Site has high head and relatively low water flow.

Solution and conclusion: Pelton turbine is the basic suitable category because an impulse jet uses high head effectively.

Example 6: Belt-drive speed−

Given / observation: 200 mm driver at 900 rpm drives a 600 mm pulley without slip.

Solution and conclusion: $N_2 = N_1D_1/D_2 = 900 \times 200/600 = 300$ rpm; actual speed can be slightly lower because of slip.

Example 7: Gear direction and speed−

Given / observation: 20-tooth external driver rotates clockwise at 1200 rpm and meshes with 60 teeth.

Solution and conclusion: Driven speed = 400 rpm and direction is counter-clockwise.

Example 8: Choose a bearing−

Given / observation: A shaft carries high radial load and little axial load.

Solution and conclusion: A cylindrical roller bearing is a typical candidate; final selection also needs speed, life, fit, lubrication and environment.

Example 9: Choose a forming process−

Given / observation: A long aluminium section needs the same complex profile along its length.

Solution and conclusion: Extrusion is suitable because material is forced through a shaped die to produce a constant section.

Example 10: Apply hierarchy of controls−

Given / observation: Grinding dust exposure is high.

Solution and conclusion: First eliminate/substitute if possible, then use enclosure/extraction, procedures and training; suitable respiratory/eye PPE is the final layer, not the only control.

DRAWING PRACTICE

Complete labelled diagram library

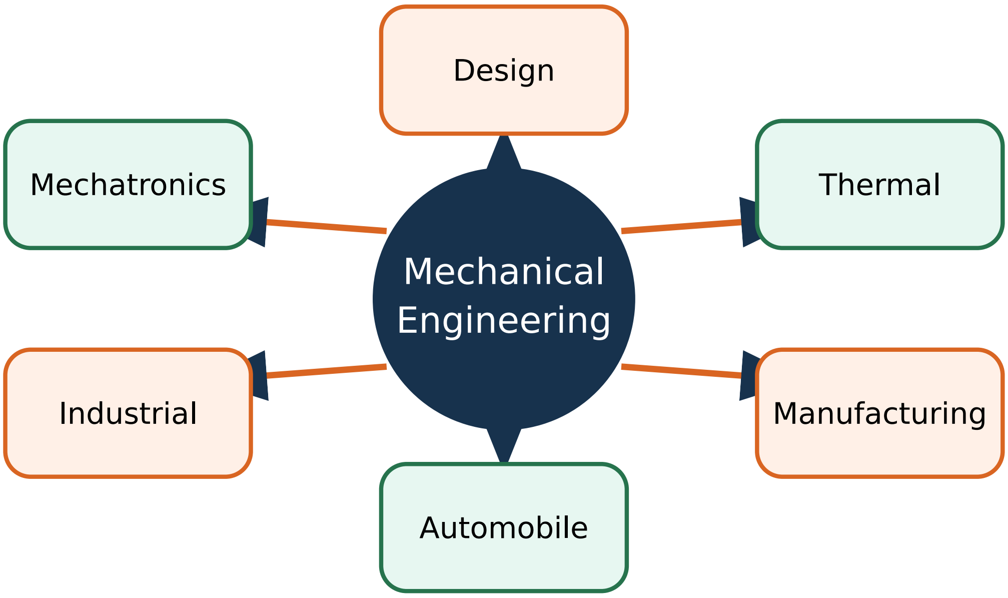


Figure 1. Mechanical Engineering branches. Reproduce the essential parts, labels and arrows in examination drawings.

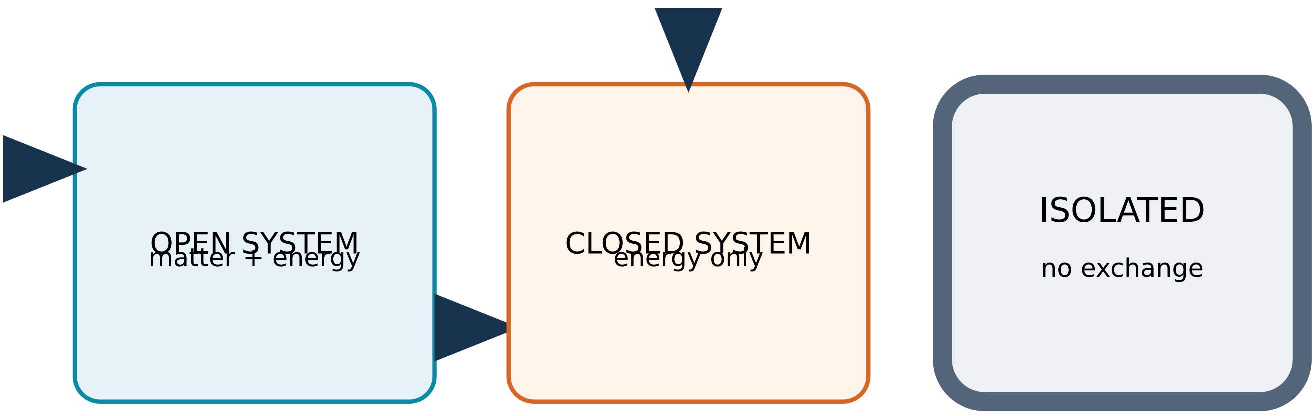


Figure 2. Open closed and isolated thermodynamic systems. Reproduce the essential parts, labels and arrows in examination drawings.

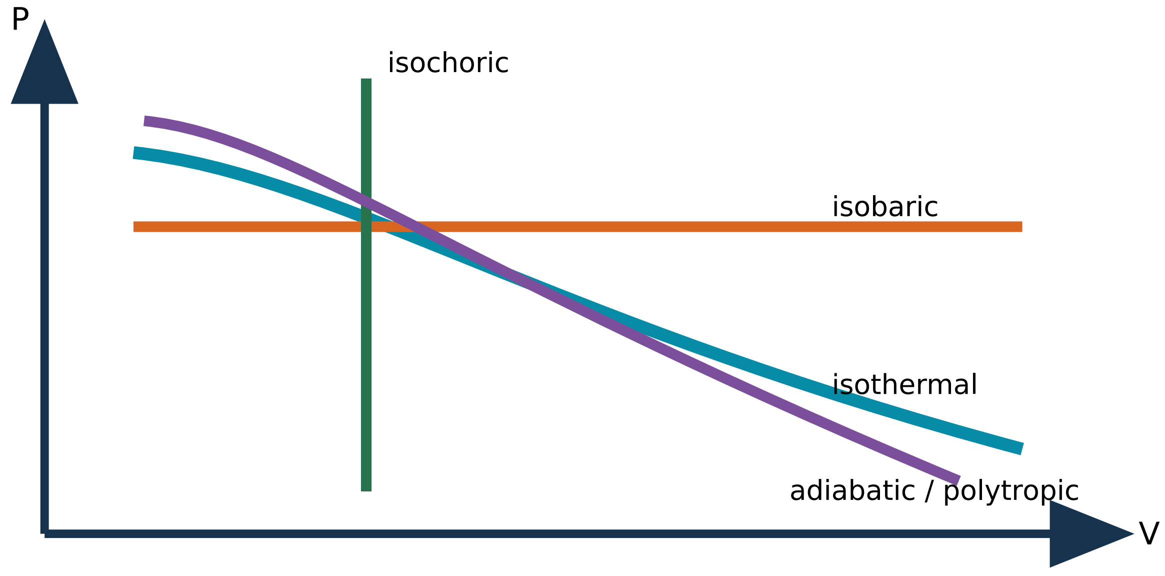


Figure 3. P-V process comparison. Reproduce the essential parts, labels and arrows in examination drawings.

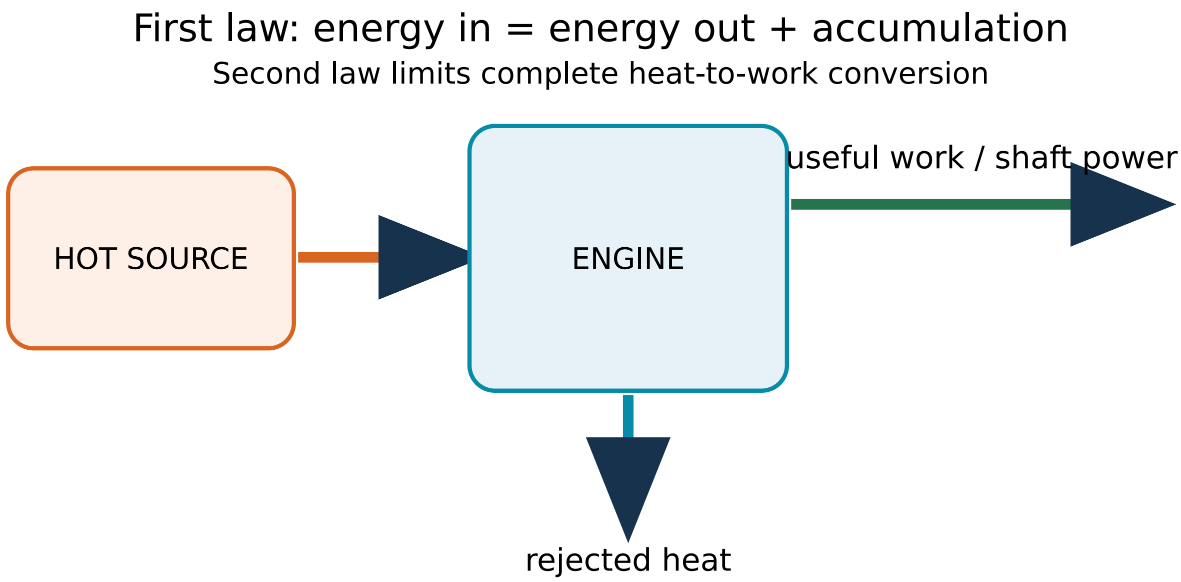


Figure 4. Thermodynamic laws and energy balance. Reproduce the essential parts, labels and arrows in examination drawings.

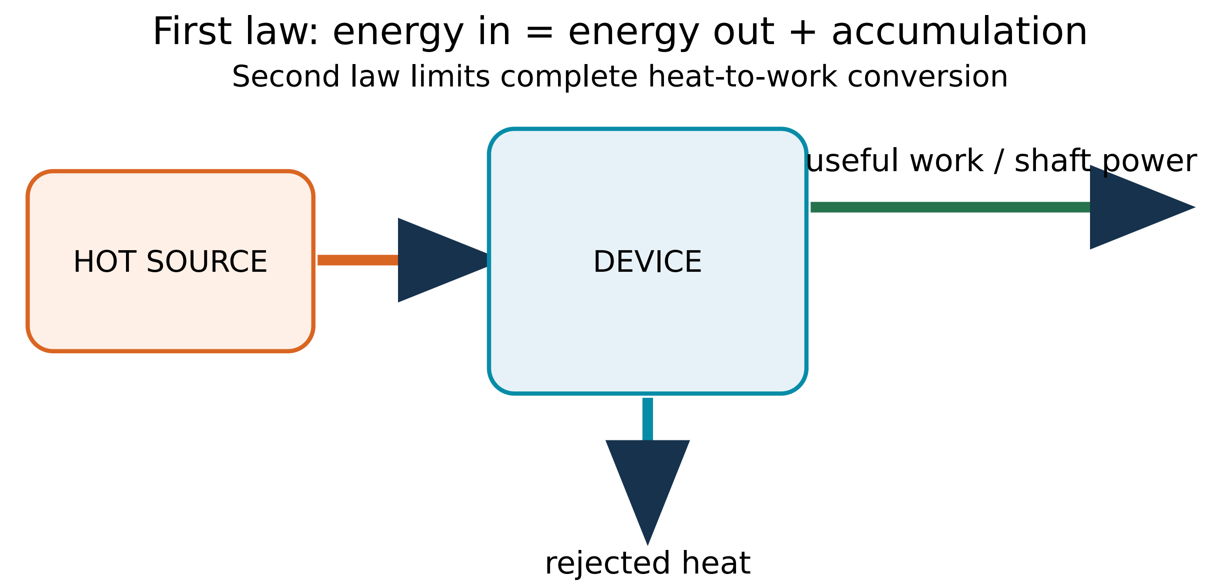


Figure 5. Heat engine and refrigerator. Reproduce the essential parts, labels and arrows in examination drawings.

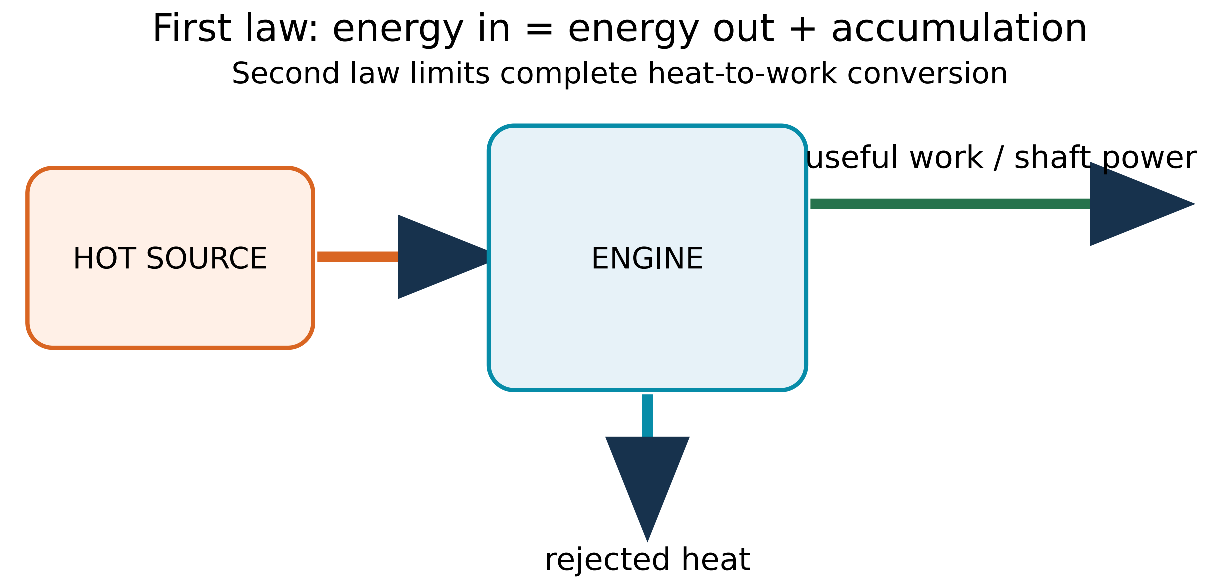


Figure 6. Boiler steam-turbine power chain. Reproduce the essential parts, labels and arrows in examination drawings.

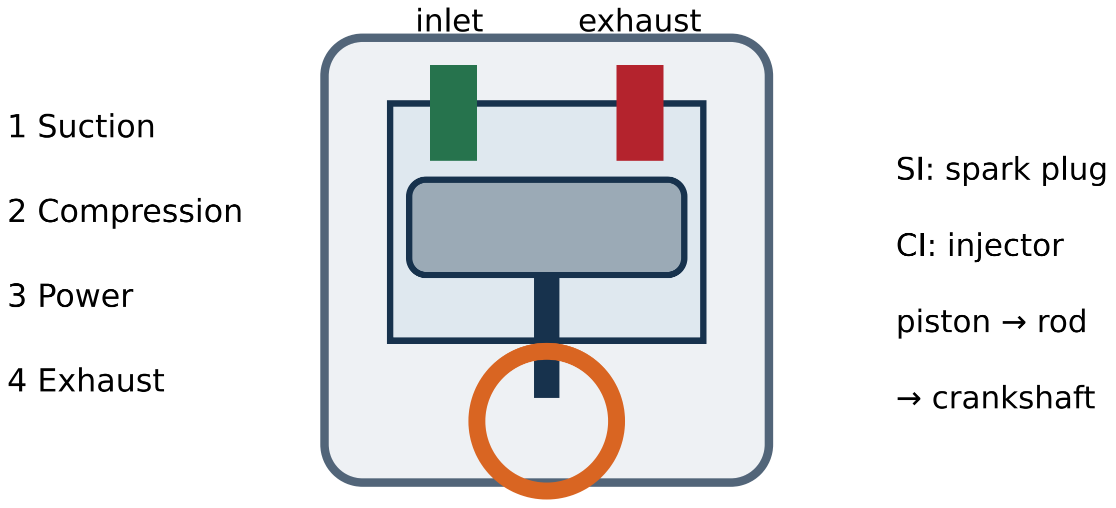


Figure 7. Four-stroke IC engine. Reproduce the essential parts, labels and arrows in examination drawings.

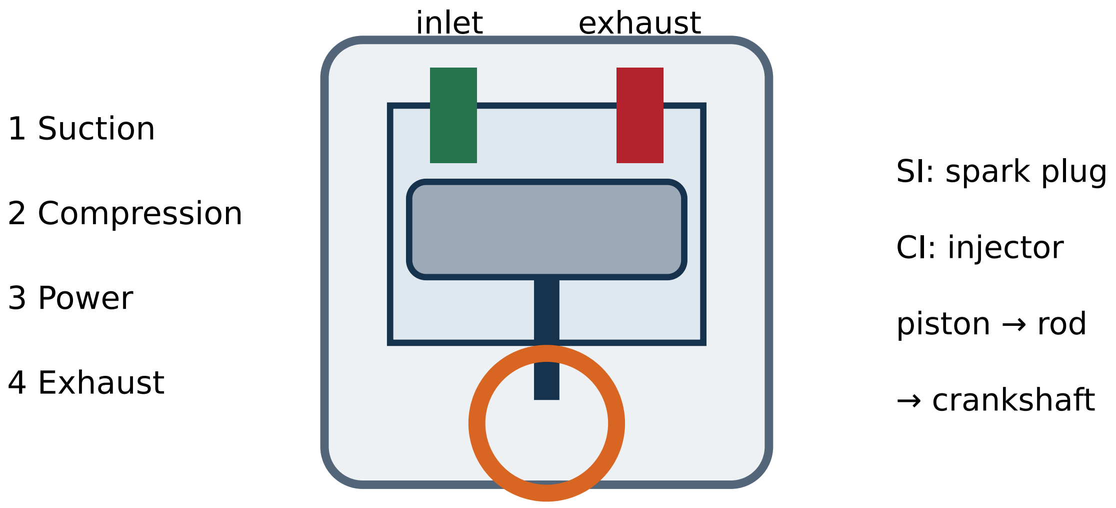


Figure 8. Petrol and diesel engine comparison. Reproduce the essential parts, labels and arrows in examination drawings.

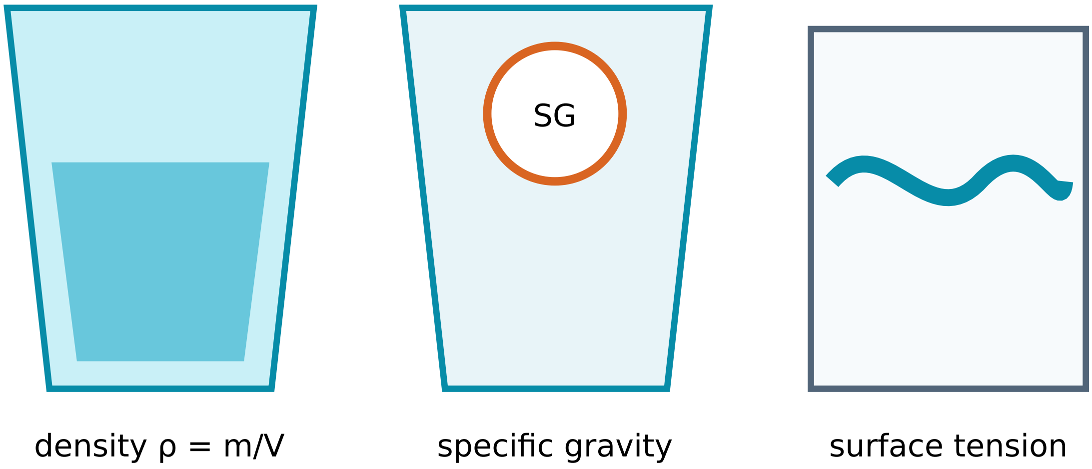


Figure 9. Fluid properties. Reproduce the essential parts, labels and arrows in examination drawings.

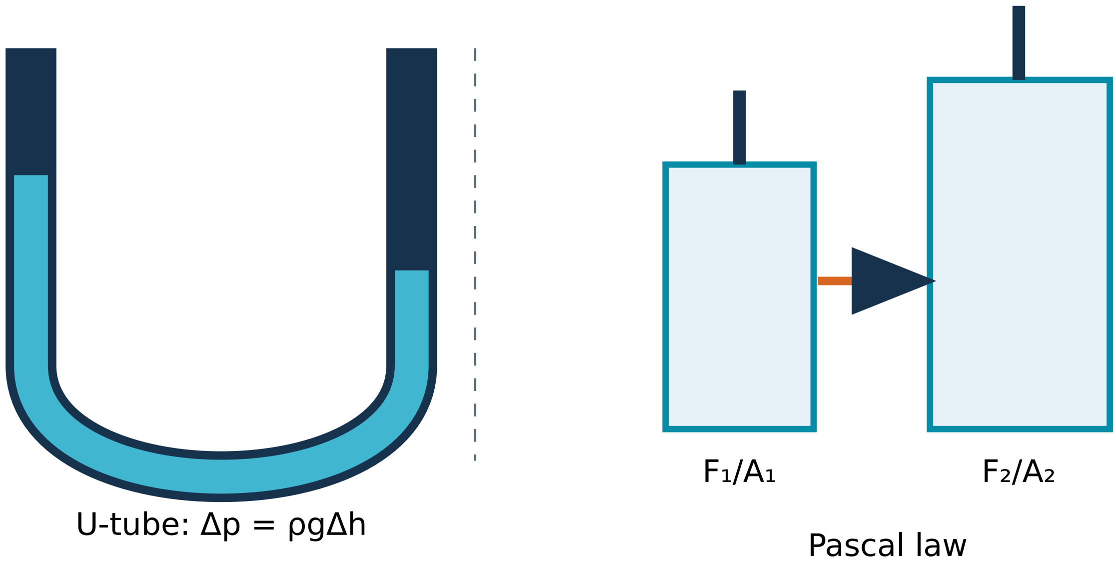


Figure 10. U-tube manometer. Reproduce the essential parts, labels and arrows in examination drawings.

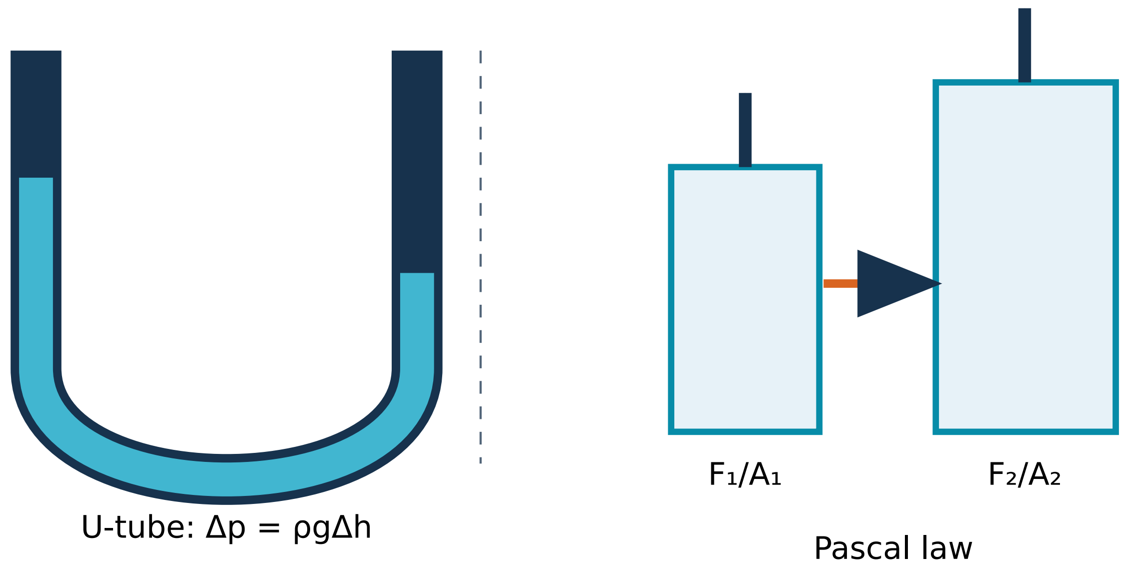


Figure 11. Hydraulic press using Pascal's law. Reproduce the essential parts, labels and arrows in examination drawings.

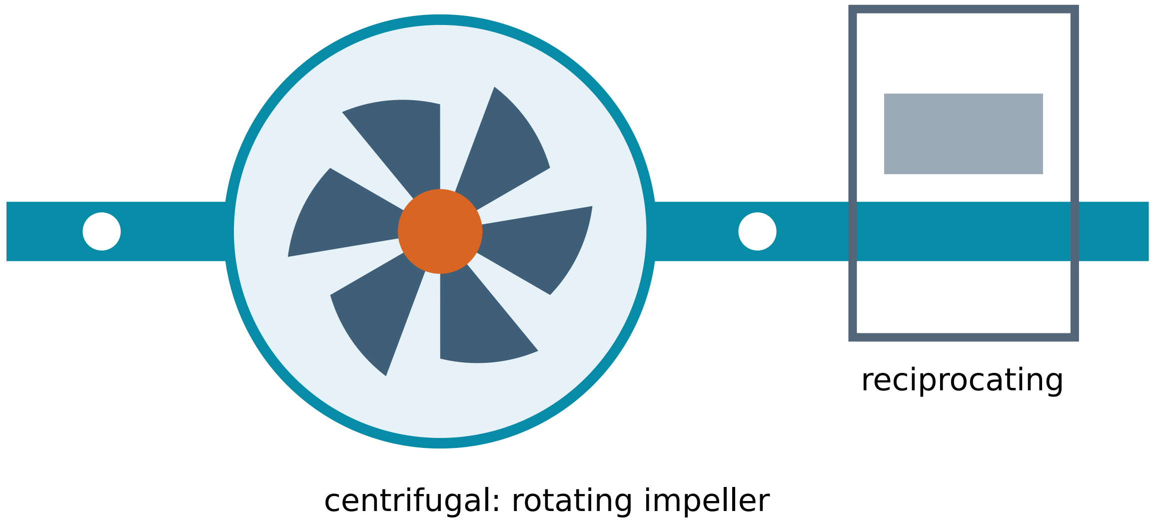


Figure 12. Centrifugal pump. Reproduce the essential parts, labels and arrows in examination drawings.

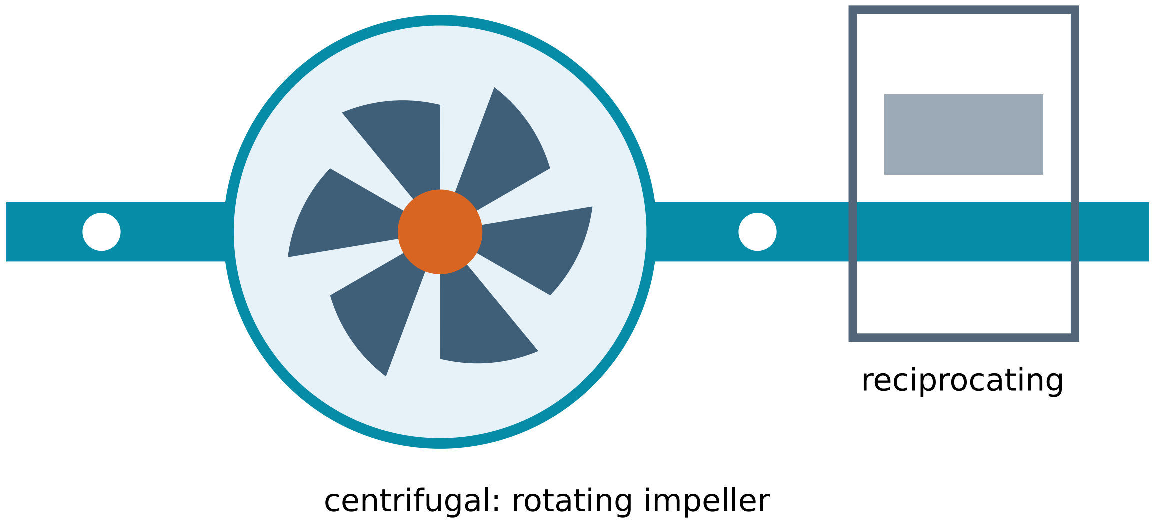


Figure 13. Reciprocating pump. Reproduce the essential parts, labels and arrows in examination drawings.

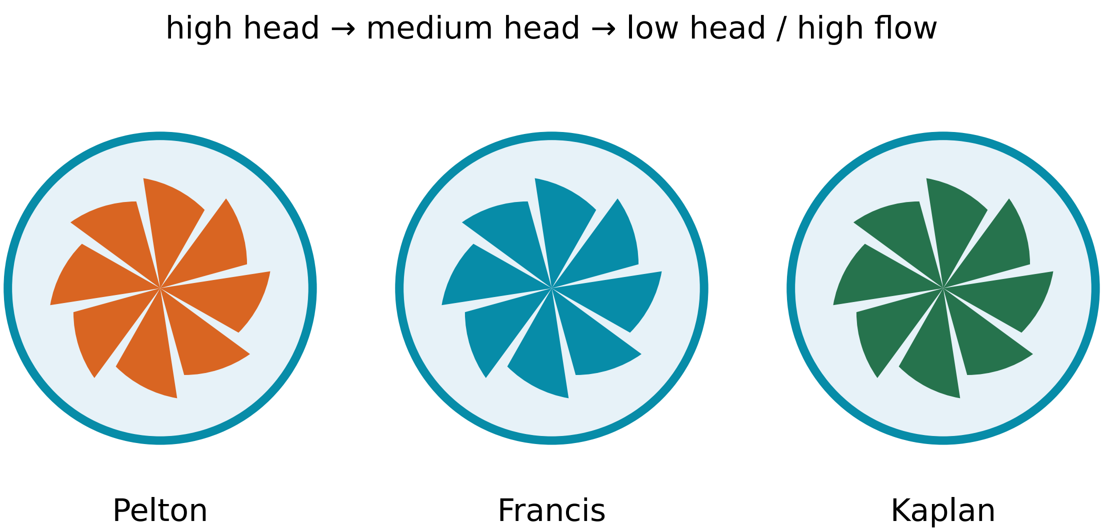


Figure 14. Pelton Francis and Kaplan turbines. Reproduce the essential parts, labels and arrows in examination drawings.

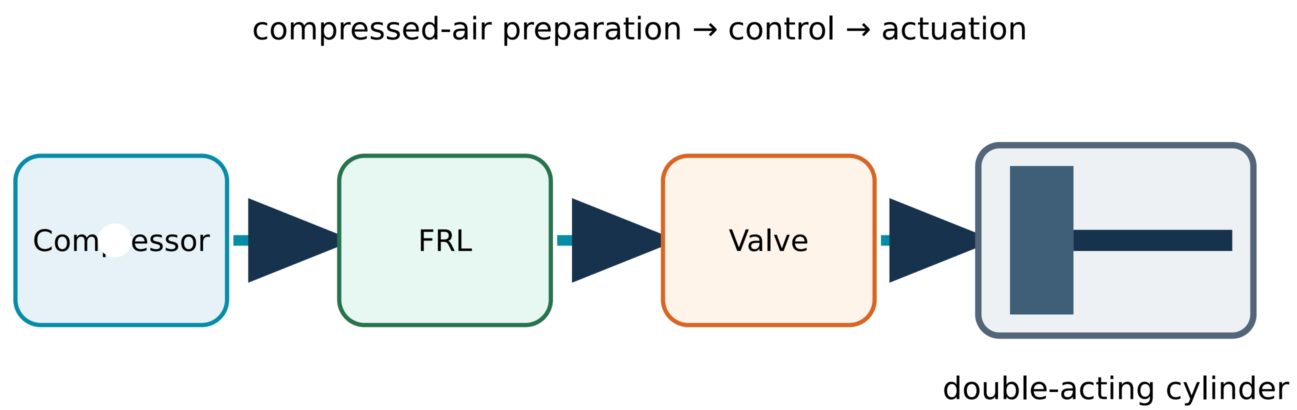


Figure 15. Pneumatic circuit. Reproduce the essential parts, labels and arrows in examination drawings.

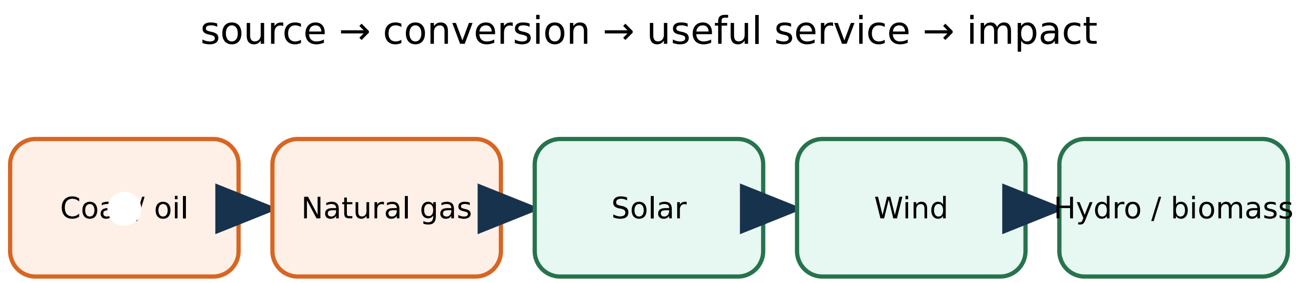


Figure 16. Conventional and renewable energy sources. Reproduce the essential parts, labels and arrows in examination drawings.

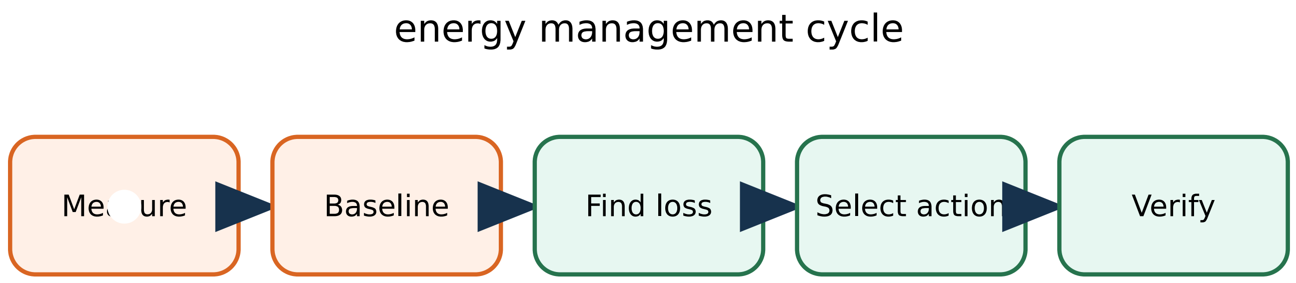


Figure 17. Energy audit flow. Reproduce the essential parts, labels and arrows in examination drawings.

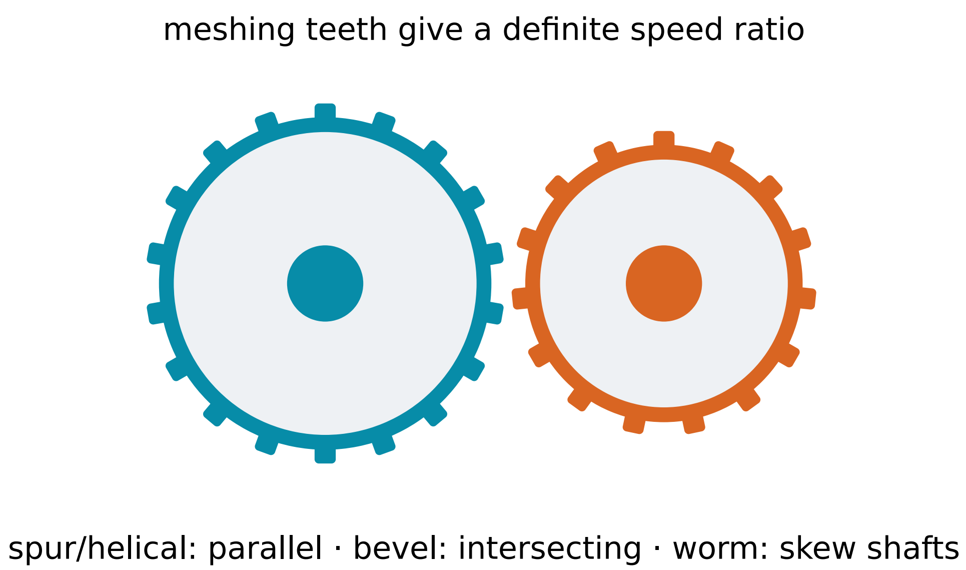


Figure 18. Mechanical power transmission. Reproduce the essential parts, labels and arrows in examination drawings.

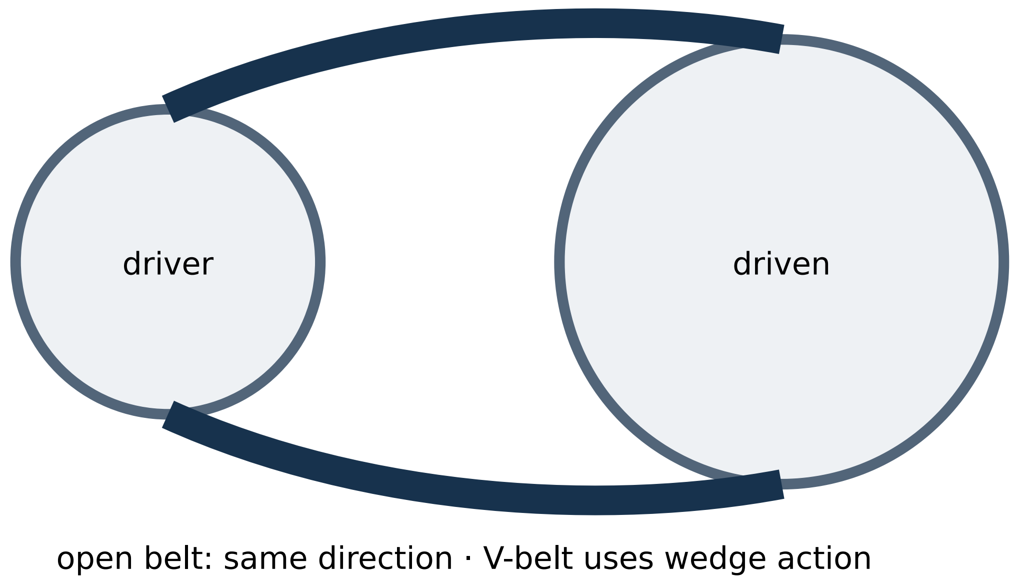


Figure 19. Open crossed and V-belt drives. Reproduce the essential parts, labels and arrows in examination drawings.

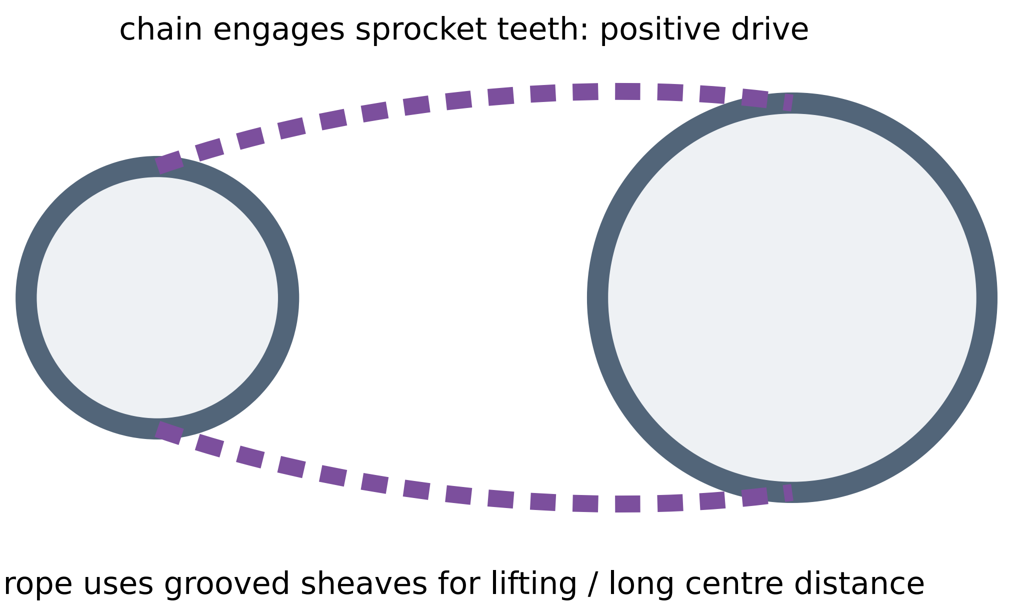


Figure 20. Rope and chain drives. Reproduce the essential parts, labels and arrows in examination drawings.

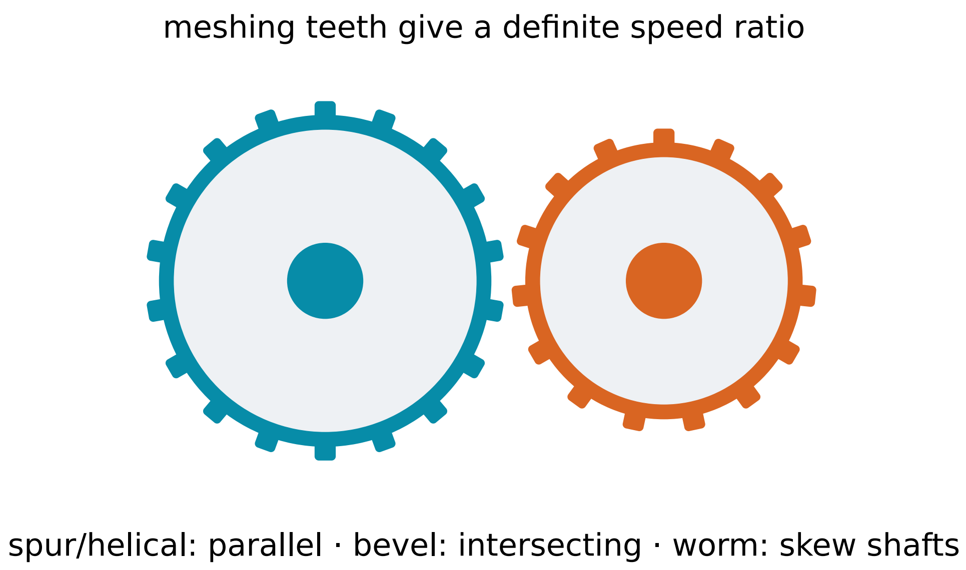


Figure 21. Spur helical bevel and worm gears. Reproduce the essential parts, labels and arrows in examination drawings.

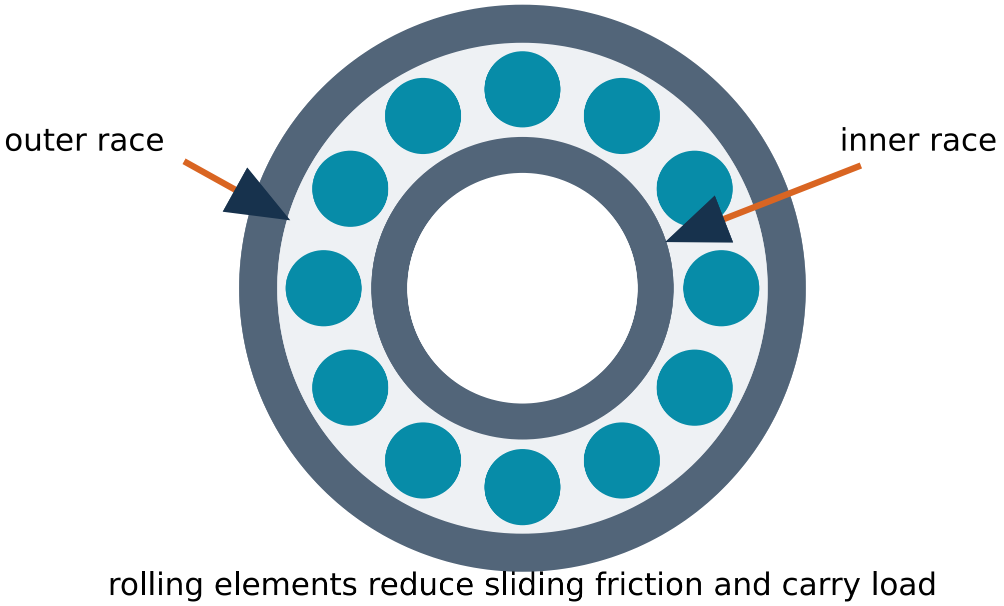


Figure 22. Journal ball and roller bearings. Reproduce the essential parts, labels and arrows in examination drawings.

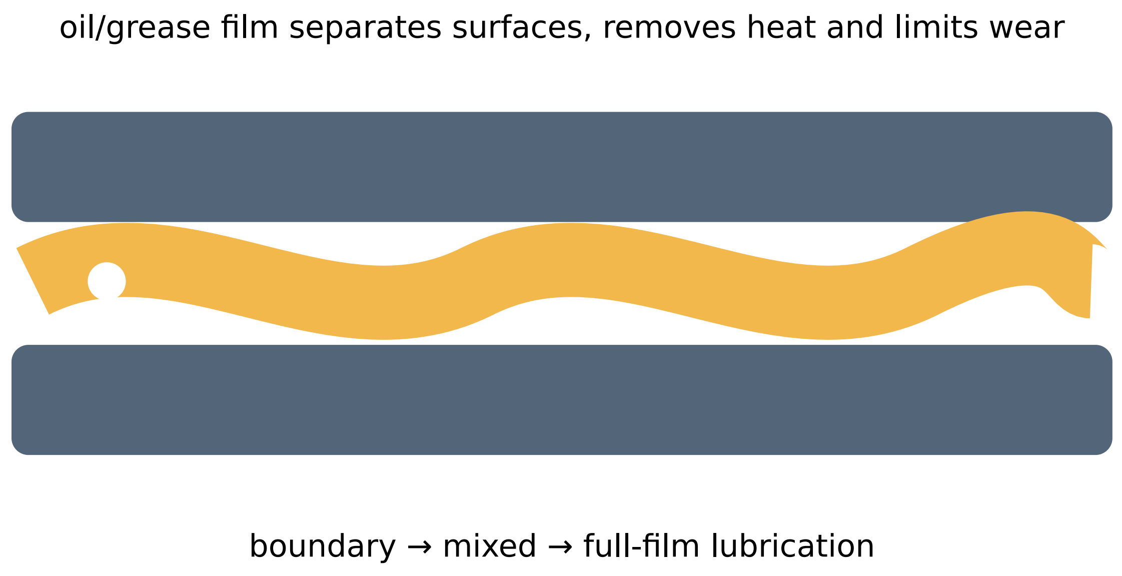


Figure 23. Lubrication regimes. Reproduce the essential parts, labels and arrows in examination drawings.

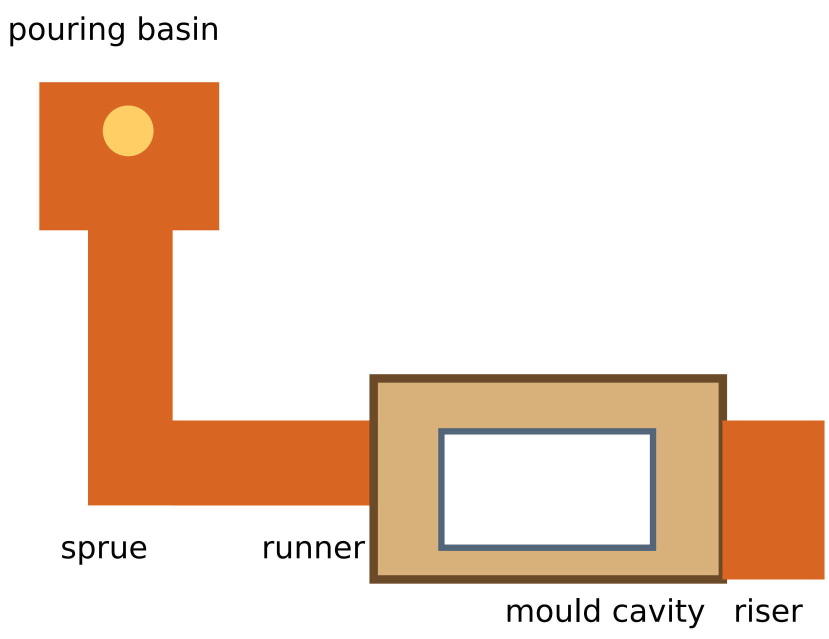


Figure 24. Sand casting system. Reproduce the essential parts, labels and arrows in examination drawings.

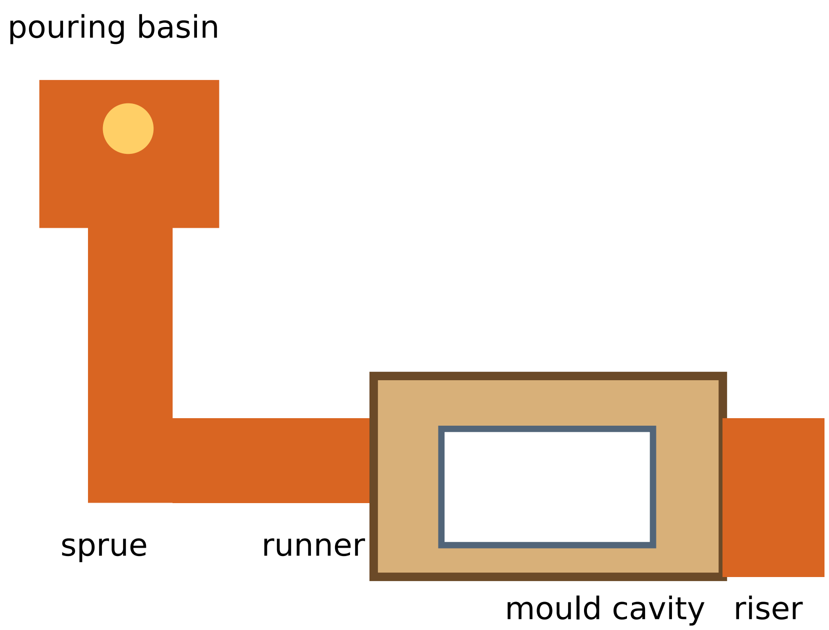


Figure 25. Moulding tools and components. Reproduce the essential parts, labels and arrows in examination drawings.

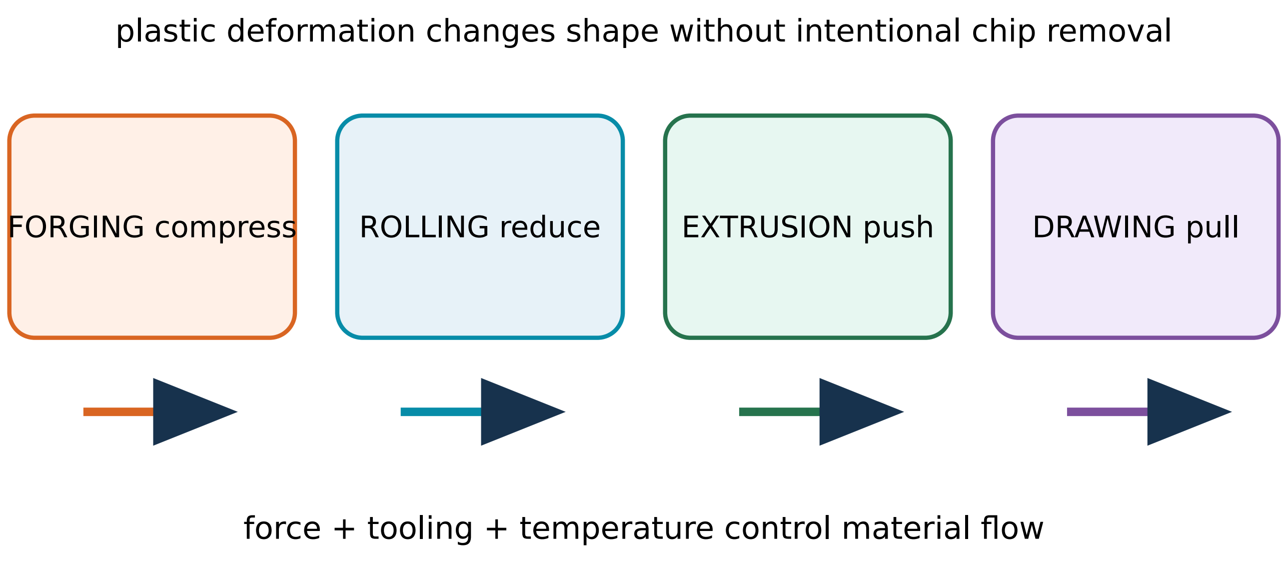


Figure 26. Forging rolling extrusion and drawing. Reproduce the essential parts, labels and arrows in examination drawings.

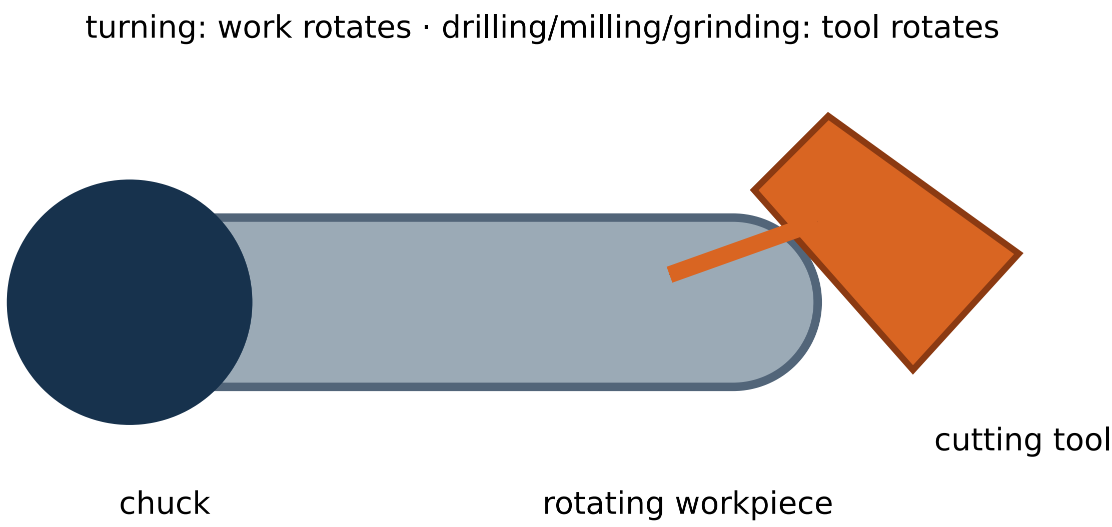


Figure 27. Turning drilling milling and grinding. Reproduce the essential parts, labels and arrows in examination drawings.

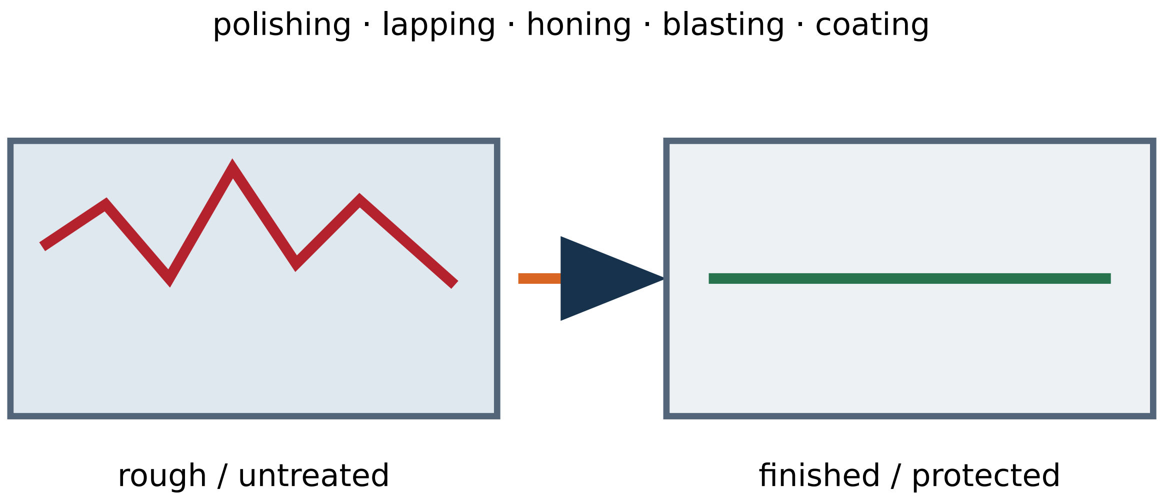


Figure 28. Surface finishing comparison. Reproduce the essential parts, labels and arrows in examination drawings.

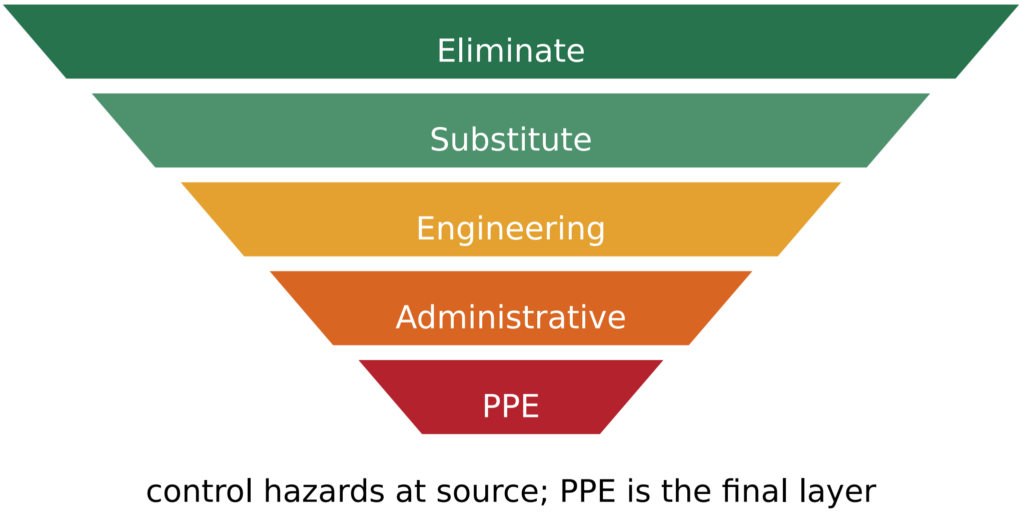


Figure 29. PPE and hierarchy of controls. Reproduce the essential parts, labels and arrows in examination drawings.

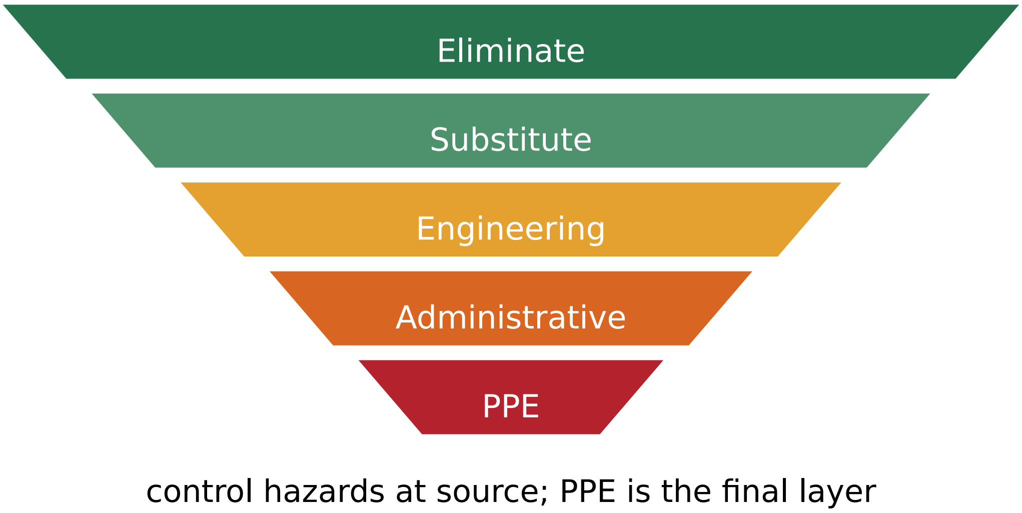


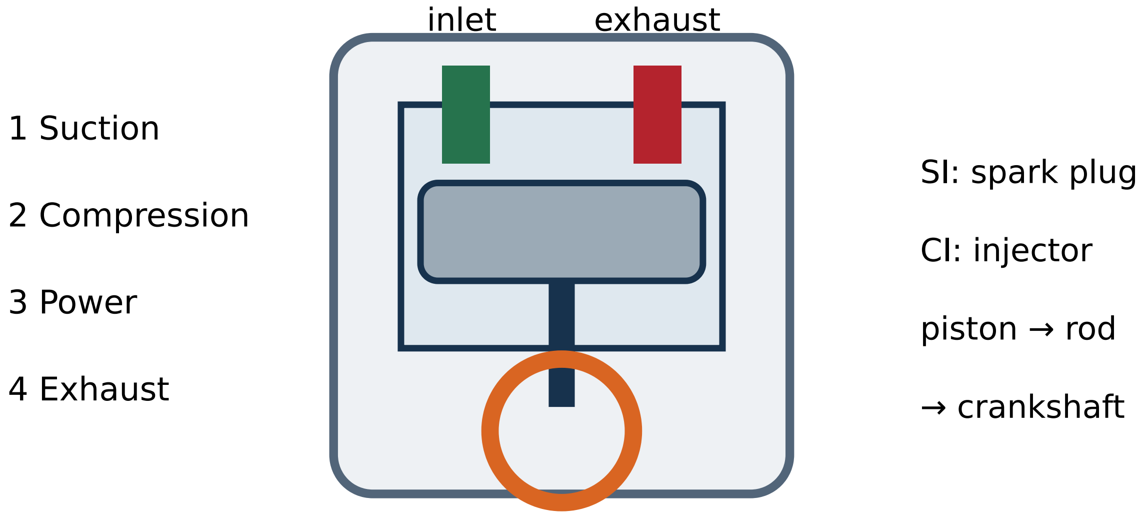
Figure 30. Industrial safety management cycle. Reproduce the essential parts, labels and arrows in examination drawings.

CONTROLLED MOTION

Mechanical working-principle animations

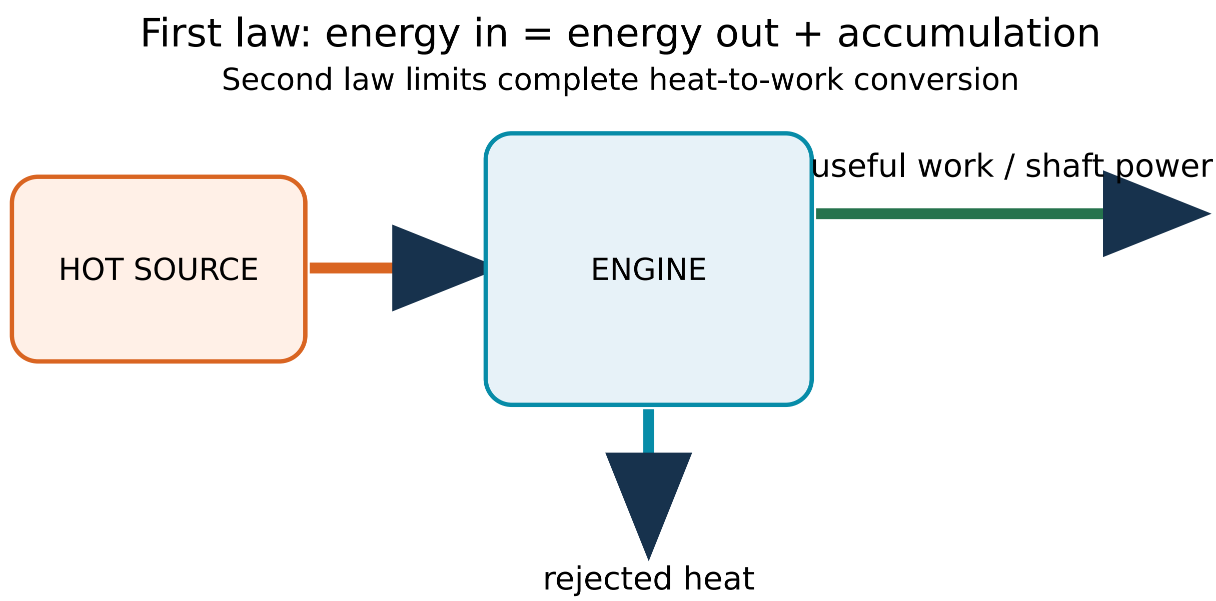
Every animation begins paused. Play, pause, step, reset or change speed; use reduced-motion mode whenever needed.

1. Four-stroke engine cycle



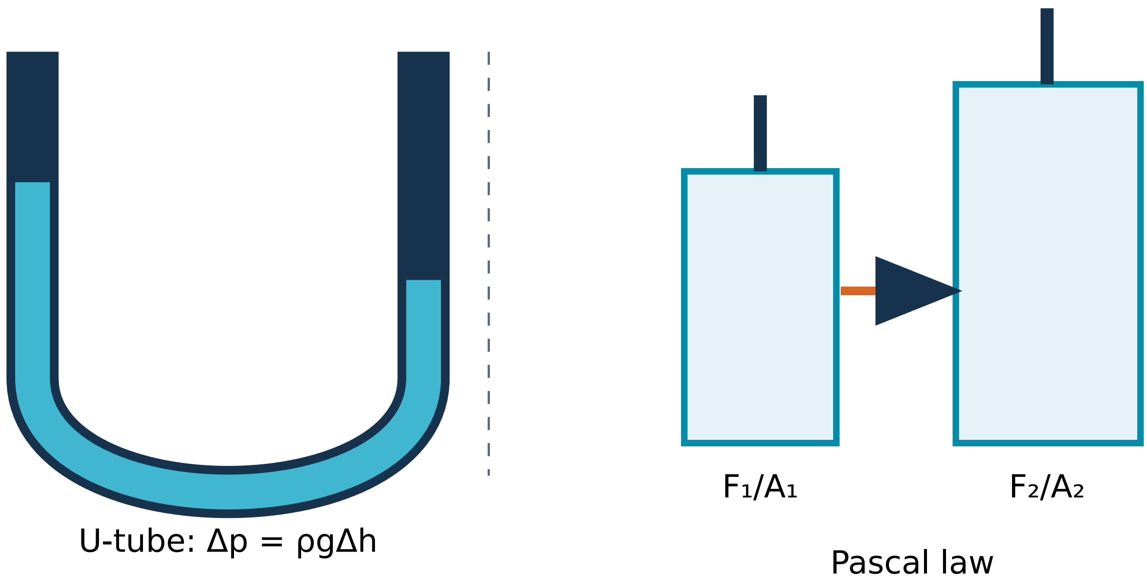
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

2. Energy path through heat engine



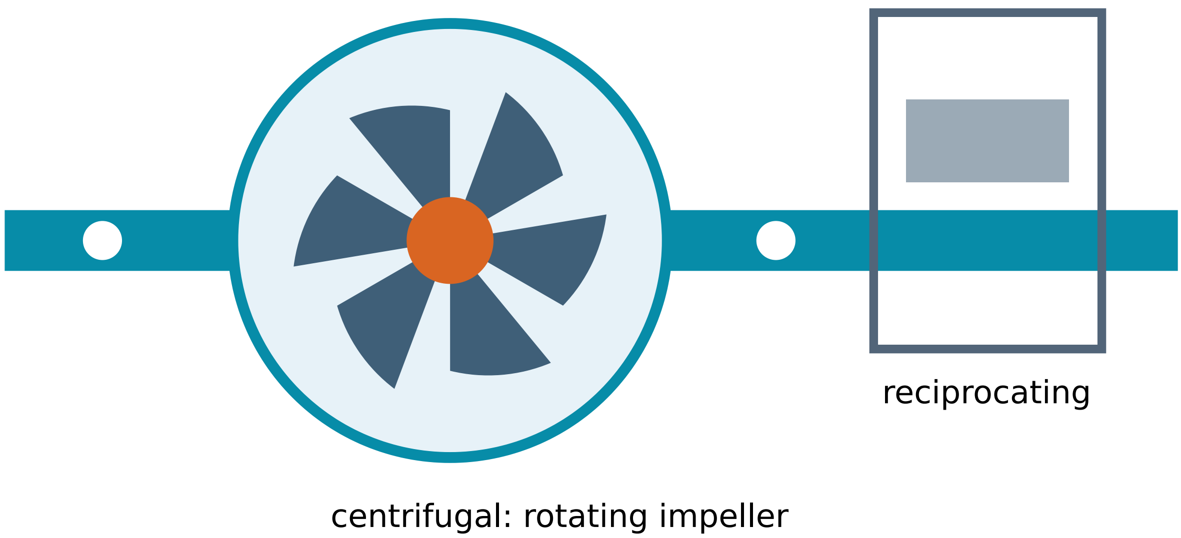
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

3. U-tube manometer response



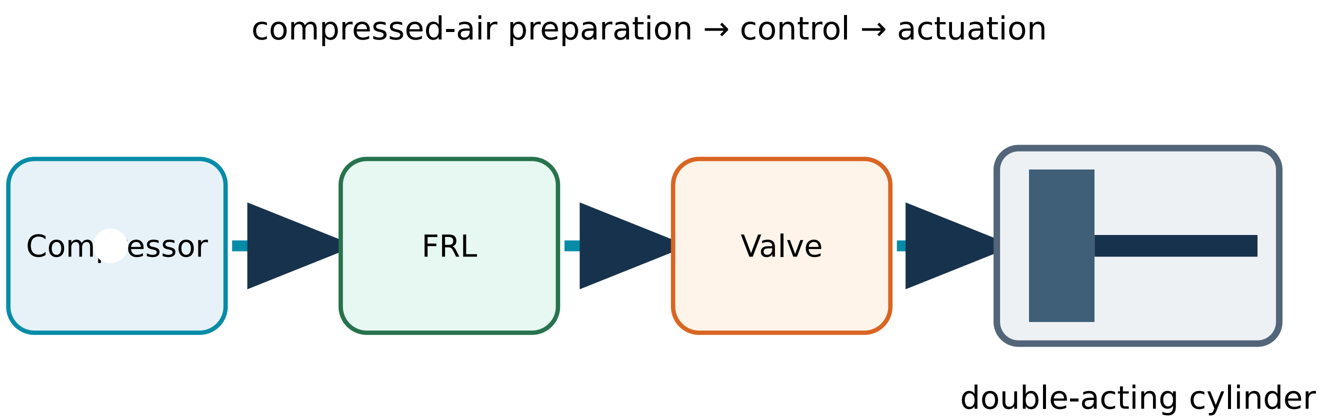
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

4. Centrifugal pump energy transfer



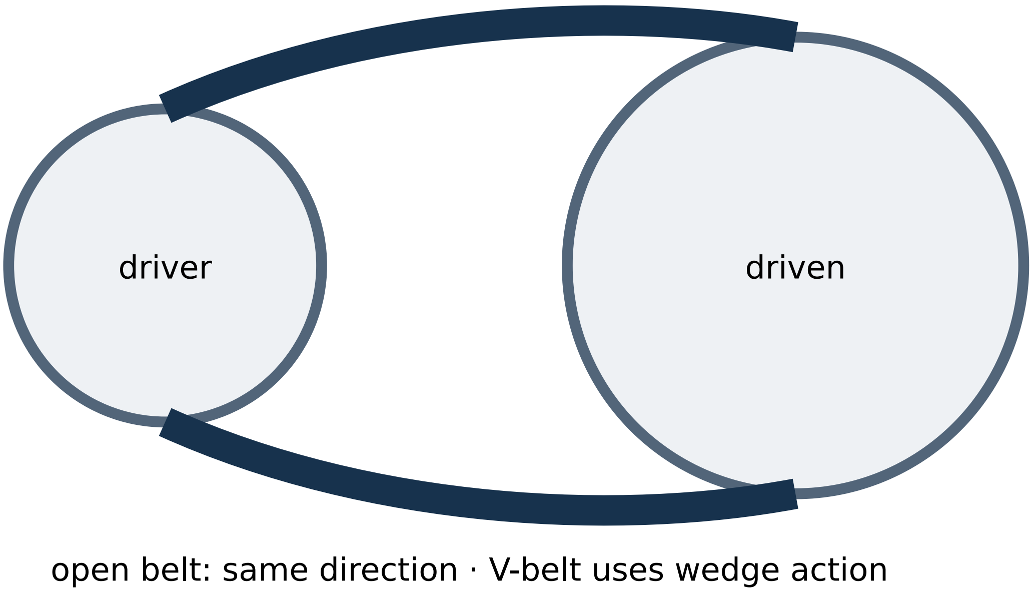
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

5. Pneumatic cylinder actuation



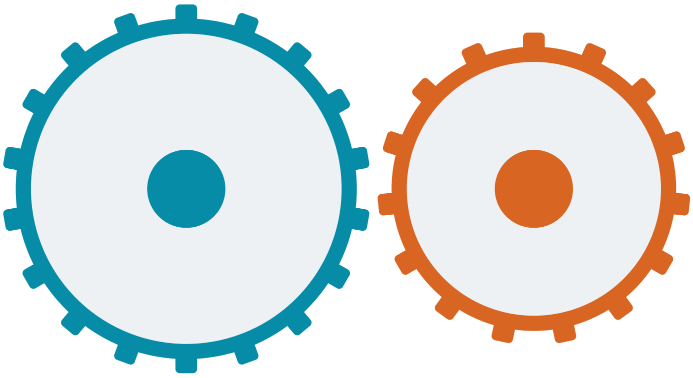
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

6. Open belt drive motion



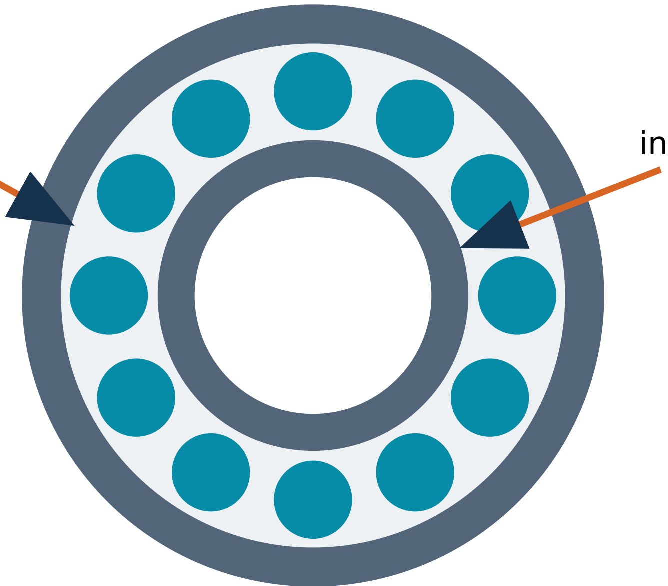
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

7. Spur gear meshing



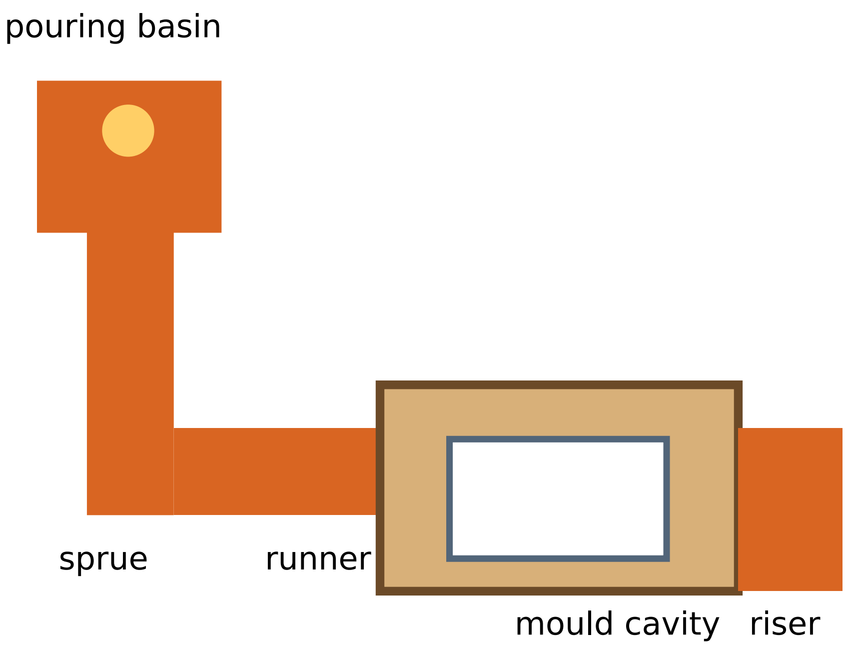
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

8. Rolling-element bearing



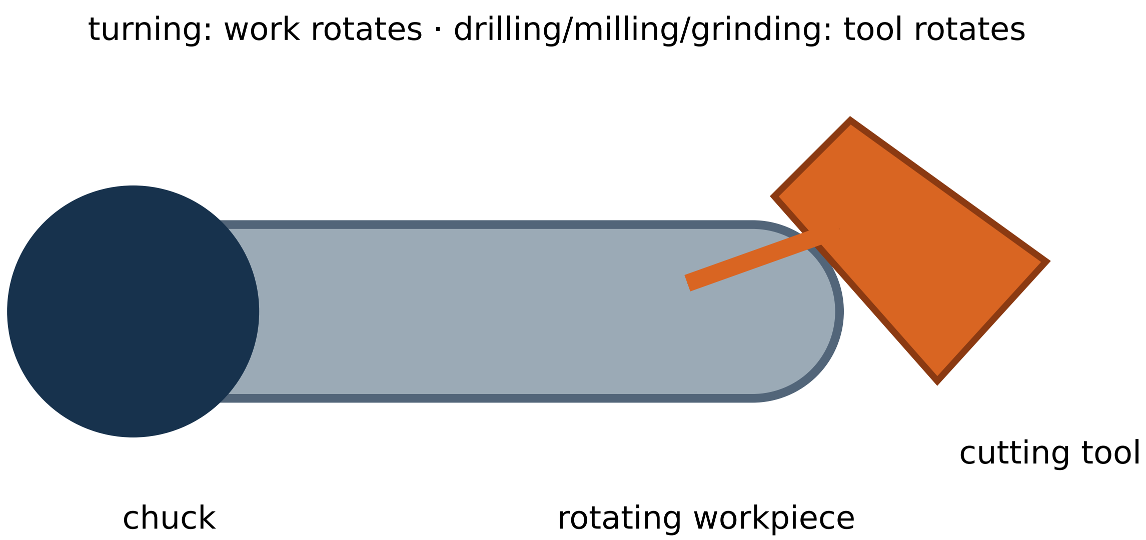
Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

9. Metal casting flow



Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

10. Lathe turning operation



Play to observe motion or flow. Pause and use Step to inspect a controlled frame.

OFFICIAL 90-HOUR PLAN

Suggested student activities for self-learning

Week 1 · Module 1 · 5 h

Official activity group: Evolution and historical development of Mechanical Engineering; contribution to sustainable development; interdisciplinary links with Mechatronics, AI and

Robotics; state, path and cycle.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 2 · Module 1 · 5 h

Official activity group: Practical energy conversion; units of work, heat and energy; mechanical and flow work; Zeroth-law applications; limitations of energy conversion; boiler safety and power-plant applications.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 3 · Module 1 · 5 h

Official activity group: IC-engine applications in automobiles and power generation; power and efficiency; SI versus CI and boiler versus turbine comparison tables.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 4 · Module 1 · 5 h

Official activity group: Study approved videos, pictures or animations of IC-engine components and automobile-workshop tools.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 5 · Module 2 · 5 h

Official activity group: Engineering importance of fluid properties, industrial pressure measurement, Pascal-law applications and hydraulic-system safety.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 6 · Module 2 · 8 h

Official activity group: Pump/turbine applications, pneumatic components, advantages and automation uses, and hydraulic-versus-pneumatic comparison.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 7 · Module 2 · 5 h

Official activity group: Coal impacts, petroleum formation/refining and products, natural-gas properties, renewable environmental/economic benefits and Indian scope.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 8 · Module 2 · 6 h

Official activity group: Observe a safe domestic hydraulic machine/pump and approved pneumatic/pressure-device resources; distinguish pumps and turbines.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 9 · Module 3 · 5 h

Official activity group: Machine motion, transmission methods, industrial systems, belt-drive advantages/limitations and rope-drive applications.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 10 · Module 3 · 5 h

Official activity group: Chain construction/application, belt-versus-chain comparison, spur-versus-helical difference, bevel functions and worm applications.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 11 · Module 3 · 5 h

Official activity group: Bearing materials/applications and purpose, importance and applications of lubrication.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 12 · Module 3 · 5 h

Official activity group: Observe a bicycle/two-wheeler drive and a domestic gear/lubrication example; capture authorised evidence and prepare a short note.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 13 · Module 4 · 9 h

Official activity group: Casting advantages/limitations/applications; forging, rolling, extrusion and drawing; welding-versus-brazing-versus-soldering comparison.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 14 · Module 4 · 10.5 h

Official activity group: Surface finishing, fire safety, emergency procedure, PPE selection/maintenance, training, management/safety-officer roles and safety culture.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

Week 15 · Module 4 · 6.5 h

Official activity group: Study approved manufacturing, machining and finishing animations/videos/documentaries and prepare a technical note.

Suggested evidence: dated comparison table, labelled diagram, authorised photograph/video note or 300–500 word technical report as suitable.

Evaluation: accuracy, relevance, originality, safety, source acknowledgement, neatness and timely submission.

EXAMINATION PREPARATION

Expected questions with answers

Questions follow the official 1-mark, 3-mark and 7-mark structure. Answers are original learning support, not copied model-paper text.

Module I · CO1

One-mark questions

1. Define or identify Scope, importance and branches of Mechanical Engineering. −

> **Answer:** Mechanical Engineering applies force, motion, energy, materials and manufacturing knowledge to design, produce, operate and maintain machines and systems.

2. Define or identify Thermodynamic systems, properties and processes.−

> **Answer:** A thermodynamic system is a chosen quantity of matter or region in space; everything outside it is the surroundings, separated by a real or imaginary boundary.

3. Define or identify Energy, work, heat and laws of thermodynamics.−

> **Answer:** Energy is the capacity to produce change.

4. Define or identify Heat engines, refrigerators, steam boilers and turbines.−

> **Answer:** A heat engine receives heat from a high-temperature source, converts part to work and rejects the remainder to a low-temperature sink.

5. Define or identify Internal combustion engines: types and working.−

> **Answer:** In an internal combustion engine, fuel releases energy inside the cylinder and gas pressure acts on a piston to rotate the crankshaft.

Three-mark questions

1. Explain three essential points about Scope, importance and branches of Mechanical Engineering.−

Answer:

1. Design converts a need into specifications, calculations, drawings and a safe product
2. Production plans processes, machines, tools, quality and cost
3. Maintenance preserves reliability through inspection, lubrication, repair and condition monitoring

2. Explain three essential points about Thermodynamic systems, properties and processes.−

Answer:

1. Isothermal process: temperature remains constant
2. Isobaric process: pressure remains constant
3. Isochoric process: volume remains constant

3. Explain three essential points about Energy, work, heat and laws of thermodynamics.−

Answer:

1. Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other
2. First law: energy is conserved; it changes form or crosses a boundary
3. Second law: heat does not spontaneously flow from cold to hot and no heat engine converts all supplied heat into work

4. Explain three essential points about Heat engines, refrigerators, steam boilers and turbines.−

Answer:

1. Heat-engine output is always less than heat supplied
2. Refrigerator components form a closed circulation system; it does not create cold
3. Boilers require pressure control, water-level control and safety devices

5. Explain three essential points about Internal combustion engines: types and working.−

Answer:

1. Four-stroke sequence: suction, compression, power and exhaust
2. Piston reciprocation becomes crankshaft rotation through connecting rod and crank
3. Inlet and exhaust valves control gas exchange in a four-stroke engine

Seven-mark questions

1. Develop a structured note on Scope, importance and branches of Mechanical Engineering with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Mechanical Engineering applies force, motion, energy, materials and manufacturing knowledge to design, produce, operate and maintain machines and systems. Mechanical engineers work in transport, power, manufacturing, building services, maintenance, safety, automation and sustainable development, often alongside electrical, electronics, civil, computer and chemical specialists.

1. Design converts a need into specifications, calculations, drawings and a safe product
2. Production plans processes, machines, tools, quality and cost
3. Maintenance preserves reliability through inspection, lubrication, repair and condition monitoring
4. Thermal Engineering deals with energy conversion, engines, refrigeration and power plants
5. Manufacturing, design, industrial, automobile, mechatronics and materials are major branches

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

2. Develop a structured note on Thermodynamic systems, properties and processes with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

A thermodynamic system is a chosen quantity of matter or region in space; everything outside it is the surroundings, separated by a real or imaginary boundary. Open systems exchange matter and energy, closed systems exchange energy but not matter, and isolated systems ideally exchange neither. A state is described by properties; a path connects states and a cycle returns to the initial state.

1. Isothermal process: temperature remains constant
2. Isobaric process: pressure remains constant
3. Isochoric process: volume remains constant
4. Adiabatic process: no heat crosses the boundary
5. Polytropic process follows PV^n = constant as a useful general relation
6. Intensive properties do not depend on mass; extensive properties do

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

3. Develop a structured note on Energy, work, heat and laws of thermodynamics with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Energy is the capacity to produce change. Work is ordered energy transfer through a force acting over a distance or another generalised displacement; heat is energy transfer caused by a temperature difference. The Zeroth law establishes thermal equilibrium and temperature measurement, the First law expresses conservation of energy, and the Second law gives direction and limits to energy conversion.

1. Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other
2. First law: energy is conserved; it changes form or crosses a boundary
3. Second law: heat does not spontaneously flow from cold to hot and no heat engine converts all supplied heat into work
4. Heat and work are boundary interactions, not stored properties
5. Use consistent SI units: joule for energy/work/heat and watt for power

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

4. Develop a structured note on Heat engines, refrigerators, steam boilers and turbines with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

A heat engine receives heat from a high-temperature source, converts part to work and rejects the remainder to a low-temperature sink. A refrigerator uses work input to remove heat from a low-temperature region and reject it at a higher temperature. A boiler converts water to steam using heat, while a steam turbine expands steam to produce shaft work.

1. Heat-engine output is always less than heat supplied
2. Refrigerator components form a closed circulation system; it does not create cold
3. Boilers require pressure control, water-level control and safety devices
4. A turbine converts fluid energy into rotating mechanical energy
5. Power-plant chain: fuel/heat source → boiler → steam turbine → generator → condenser/feed system

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

5. Develop a structured note on Internal combustion engines: types and working with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

In an internal combustion engine, fuel releases energy inside the cylinder and gas pressure acts on a piston to rotate the crankshaft. Spark-ignition petrol engines ignite a prepared air-fuel mixture using a spark plug; compression-ignition diesel engines compress air strongly and inject fuel near the end of compression. Four-stroke engines complete a cycle in two crankshaft revolutions, while two-stroke engines complete a cycle in one.

1. Four-stroke sequence: suction, compression, power and exhaust
2. Piston reciprocation becomes crankshaft rotation through connecting rod and crank
3. Inlet and exhaust valves control gas exchange in a four-stroke engine
4. Petrol/SI engine uses a spark plug; diesel/CI engine uses high compression and fuel injection
5. Two-stroke engines use ports and combine events, giving higher power-to-weight but poorer scavenging/emissions in simple designs
6. Applications include automobiles, pumps, generators and industrial drives

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

Multiple-choice questions

1. Which statement correctly belongs to Scope, importance and branches of Mechanical Engineering?

- **A. Design converts a need into specifications, calculations, drawings and a safe product**
- **B. Isothermal process: temperature remains constant**
- **C. Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**
- **D. Heat-engine output is always less than heat supplied**

−

Answer: A. Design converts a need into specifications, calculations, drawings and a safe product

2. Which statement correctly belongs to Thermodynamic systems, properties and processes?

A. **Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**
B. **Isothermal process: temperature remains constant**
C. **Heat-engine output is always less than heat supplied**
D. **Four-stroke sequence: suction, compression, power and exhaust**

–

Answer: B. Isothermal process: temperature remains constant

3. Which statement correctly belongs to Energy, work, heat and laws of thermodynamics?

A. **Heat-engine output is always less than heat supplied**
B. **Four-stroke sequence: suction, compression, power and exhaust**
C. **Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**
D. **Design converts a need into specifications, calculations, drawings and a safe product**

–

Answer: C. Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other

4. Which statement correctly belongs to Heat engines, refrigerators, steam boilers and turbines?

A. **Four-stroke sequence: suction, compression, power and exhaust**
B. **Design converts a need into specifications, calculations, drawings and a safe product**
C. **Isothermal process: temperature remains constant**
D. **Heat-engine output is always less than heat supplied**

–

Answer: D. Heat-engine output is always less than heat supplied

5. Which statement correctly belongs to Internal combustion engines: types and working?

A. **Four-stroke sequence: suction, compression, power and exhaust**
B. **Design converts a need into specifications, calculations, drawings and a safe product**
C. **Isothermal process: temperature remains constant**
D. **Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**

–

Answer: A. Four-stroke sequence: suction, compression, power and exhaust

6. Which statement correctly belongs to Scope, importance and branches of Mechanical Engineering?

A. **Isothermal process: temperature remains constant**
B. **Design converts a need into specifications, calculations, drawings and a safe product**
C. **Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**
D. **Heat-engine output is always less than heat supplied**

–

Answer: B. Design converts a need into specifications, calculations, drawings and a safe product

7. Which statement correctly belongs to Thermodynamic systems, properties and processes?

A. **Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**
B. **Heat-engine output is always less than heat supplied**
C. **Isothermal process: temperature remains constant**
D. **Four-stroke sequence: suction, compression, power and exhaust**

–

Answer: C. Isothermal process: temperature remains constant

8. Which statement correctly belongs to Energy, work, heat and laws of thermodynamics?

A. **Heat-engine output is always less than heat supplied**
B. **Four-stroke sequence: suction, compression, power and exhaust**
C. **Design converts a need into specifications, calculations, drawings and a safe product**
D. **Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other**

−

> **Answer: D.** Zeroth law: if A and B are each in thermal equilibrium with C, they are in equilibrium with each other

Practical viva questions

1. How will you safely identify and record identify tools, machines, materials, ic engine and refrigerator?−

> **Answer:** Confirm isolation and faculty authorisation; Name each tool and match it to its safe purpose; Identify material samples from visible/physical features without destructive testing. Safety: Use only isolated display equipment; do not open a charged refrigeration circuit or fuel system.

2. How will you safely identify and record identify components of petrol/diesel ic engines?−

> **Answer:** Verify the engine is isolated, cool and securely supported; Observe an assembled section/model before separate parts; Locate gas path, piston path and crank rotation. Safety: Heavy components require correct lifting; sharp edges and trapped oil must be controlled.

Module II · CO2

One-mark questions

1. Define or identify Properties of fluids.−

> **Answer:** A fluid deforms continuously under shear and takes the shape of its container.

2. Define or identify Pressure, manometers, gauges and Pascal's law.−

Answer: Pressure is normal force per unit area.

3. Define or identify Hydraulic pumps: centrifugal and reciprocating.−

Answer: A pump adds mechanical energy to a liquid so it can move or rise in pressure.

4. Define or identify Hydraulic turbines: Pelton, Francis and Kaplan.−

Answer: A hydraulic turbine removes energy from flowing or falling water and delivers rotating shaft power.

5. Define or identify Pneumatic systems and compressed-air applications.−

Answer: Pneumatics uses compressed air to transmit and control power.

6. Define or identify Conventional energy sources.−

Answer: Coal, petroleum and natural gas are concentrated fossil energy sources formed over geological time.

7. Define or identify Renewable energy: solar, wind, hydro and biomass.−

Answer: Renewable sources are replenished by natural flows.

8. Define or identify Energy conservation and management.−

Answer: Energy conservation reduces avoidable consumption while maintaining required service, safety and quality.

Three-mark questions

1. Explain three essential points about Properties of fluids.−

Answer:

1. Density $\rho = m/V$, unit kg/m^3
2. Specific weight $\gamma = \rho g$, unit N/m^3
3. Specific volume $v = 1/\rho$, unit m^3/kg

2. Explain three essential points about Pressure, manometers, gauges and Pascal's law.−

Answer:

1. Absolute pressure is measured from perfect vacuum
2. Gauge pressure is measured relative to local atmospheric pressure
3. Vacuum pressure is below atmospheric pressure

3. Explain three essential points about Hydraulic pumps: centrifugal and reciprocating.−

Answer:

1. Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover
2. Priming removes air and fills the suction/casing with liquid before starting
3. Reciprocating pump: cylinder, piston/plunger, suction valve and delivery valve

4. Explain three essential points about Hydraulic turbines: Pelton, Francis and Kaplan.−

Answer:

1. Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure
2. Francis: water enters radially/mixed and leaves axially through a draft tube
3. Kaplan: propeller-like adjustable blades suit large flow

5. Explain three essential points about Pneumatic systems and compressed-air applications.−

Answer:

1. FRL means filter, regulator and lubricator
2. Directional control valves choose actuator motion
3. Flow-control valves adjust cylinder speed

6. Explain three essential points about Conventional energy sources.−

Answer:

1. Coal is widely used for thermal power and industrial heat
2. Petroleum refining produces fuels, lubricants and petrochemical feedstocks
3. Natural gas burns more cleanly than coal per unit energy but methane leakage matters

7. Explain three essential points about Renewable energy: solar, wind, hydro and biomass.−

Answer:

1. Solar PV output varies with irradiance, temperature, orientation and shading
2. Wind power varies strongly with wind speed and requires suitable sites
3. Hydro can be controllable but may affect river ecology and communities

8. Explain three essential points about Energy conservation and management.−

Answer:

1. Avoid idle running, leakage, throttling and unnecessary heating/cooling
2. Maintain alignment, bearings, lubrication, filters and insulation
3. Match motor/pump/compressor capacity to the load

Seven-mark questions

1. Develop a structured note on Properties of fluids with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

A fluid deforms continuously under shear and takes the shape of its container. Density relates mass to volume; specific weight relates weight to volume; specific volume is volume per unit mass; specific gravity compares density with a reference, normally water for liquids. Surface tension arises from molecular attraction at a liquid surface.

1. Density $\rho = m/V$, unit kg/m^3
2. Specific weight $\gamma = \rho g$, unit N/m^3
3. Specific volume $v = 1/\rho$, unit m^3/kg
4. Specific gravity is dimensionless
5. Surface tension unit is N/m and influences droplets, capillary action and wetting

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

2. Develop a structured note on Pressure, manometers, gauges and Pascal’s law with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Pressure is normal force per unit area. A piezometer uses liquid-column height for positive liquid pressure; a U-tube manometer balances pressure against a known liquid column; mechanical pressure gauges convert pressure-induced deformation into pointer movement. Pascal’s law states that pressure applied to a confined fluid is transmitted undiminished in all directions.

1. Absolute pressure is measured from perfect vacuum
2. Gauge pressure is measured relative to local atmospheric pressure
3. Vacuum pressure is below atmospheric pressure
4. Hydrostatic pressure increases with density, gravity and depth
5. Pascal’s law enables hydraulic jack, lift, press and brake force multiplication
6. Read a manometer vertically and use the correct level difference

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

3. Develop a structured note on Hydraulic pumps: centrifugal and reciprocating with a labelled diagram, working principle, application, limitation and safety/maintenance point.–

Answer framework:

A pump adds mechanical energy to a liquid so it can move or rise in pressure. A centrifugal pump uses a rotating impeller and volute casing for continuous flow; a reciprocating pump uses a piston or plunger and non-return valves for positive displacement. Pump choice depends on flow, head, liquid, efficiency, priming and maintenance.

1. Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover
2. Priming removes air and fills the suction/casing with liquid before starting
3. Reciprocating pump: cylinder, piston/plunger, suction valve and delivery valve
4. Centrifugal pumps suit moderate head and large continuous flow
5. Reciprocating pumps suit high pressure and lower pulsating flow
6. Never run a pump dry unless the manufacturer specifically permits it

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

4. Develop a structured note on Hydraulic turbines: Pelton, Francis and Kaplan with a labelled diagram, working principle, application, limitation and safety/maintenance point.–

Answer framework:

A hydraulic turbine removes energy from flowing or falling water and delivers rotating shaft power. Pelton is an impulse turbine for high head and low flow; Francis is a mixed-flow reaction turbine for medium head; Kaplan is an axial-flow reaction turbine for low head and high flow. A casing, runner and draft tube or nozzle guide the energy conversion according to type.

1. Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure
2. Francis: water enters radially/mixed and leaves axially through a draft tube
3. Kaplan: propeller-like adjustable blades suit large flow
4. Turbine converts hydraulic energy to mechanical energy; pump performs the reverse direction of energy transfer
5. Speed control and safe water isolation are essential

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

5. Develop a structured note on Pneumatic systems and compressed-air applications with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Pneumatics uses compressed air to transmit and control power. A typical system has compressor, receiver, dryer/filter, regulator, lubricator where required, directional valve, flow control, tubing and cylinder or air motor. Pneumatic systems are clean, fast and overload-safe, but air compressibility reduces stiffness and energy efficiency.

1. FRL means filter, regulator and lubricator
2. Directional control valves choose actuator motion
3. Flow-control valves adjust cylinder speed
4. Single-acting cylinder uses air in one direction and spring/load return
5. Double-acting cylinder uses air for extension and retraction
6. Common applications: clamping, pick-and-place, packaging, doors and light automation

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

Multiple-choice questions

1. Which statement correctly belongs to Properties of fluids?

- **A. Density ρ = m/V, unit kg/m³**
- **B. Absolute pressure is measured from perfect vacuum**
- **C. Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover**
- **D. Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure**

−

Answer: A. Density ρ = m/V, unit kg/m³

2. Which statement correctly belongs to Pressure, manometers, gauges and Pascal's law?

A. **Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover**
B. **Absolute pressure is measured from perfect vacuum**
C. **Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure**
D. **FRL means filter, regulator and lubricator**

–

Answer: B. Absolute pressure is measured from perfect vacuum

3. Which statement correctly belongs to Hydraulic pumps: centrifugal and reciprocating?

A. **Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure**
B. **FRL means filter, regulator and lubricator**
C. **Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover**
D. **Coal is widely used for thermal power and industrial heat**

–

Answer: C. Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover

4. Which statement correctly belongs to Hydraulic turbines: Pelton, Francis and Kaplan?

A. **FRL means filter, regulator and lubricator**
B. **Coal is widely used for thermal power and industrial heat**
C. **Solar PV output varies with irradiance, temperature, orientation and shading**
D. **Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure**

–

Answer: D. Pelton: jet strikes double-cup buckets; runner operates approximately at atmospheric pressure

5. Which statement correctly belongs to Pneumatic systems and compressed-air applications?

A. **FRL means filter, regulator and lubricator**
B. **Coal is widely used for thermal power and industrial heat**
C. **Solar PV output varies with irradiance, temperature, orientation and shading**
D. **Avoid idle running, leakage, throttling and unnecessary heating/cooling**

–

Answer: A. FRL means filter, regulator and lubricator

6. Which statement correctly belongs to Conventional energy sources?

A. **Solar PV output varies with irradiance, temperature, orientation and shading**
B. **Coal is widely used for thermal power and industrial heat**
C. **Avoid idle running, leakage, throttling and unnecessary heating/cooling**
D. **Density $\rho = m/V$, unit kg/m^3**

–

Answer: B. Coal is widely used for thermal power and industrial heat

7. Which statement correctly belongs to Renewable energy: solar, wind, hydro and biomass?

A. **Avoid idle running, leakage, throttling and unnecessary heating/cooling**
B. **Density $\rho = m/V$, unit kg/m^3**
C. **Solar PV output varies with irradiance, temperature, orientation and shading**
D. **Absolute pressure is measured from perfect vacuum**

–

Answer: C. Solar PV output varies with irradiance, temperature, orientation and shading

8. Which statement correctly belongs to Energy conservation and management?

A. **Density ρ = m/V, unit kg/m³**
B. **Absolute pressure is measured from perfect vacuum**
C. **Centrifugal pump: suction pipe, impeller, casing, delivery pipe and prime mover**
D. **Avoid idle running, leakage, throttling and unnecessary heating/cooling**

–

Answer: D. Avoid idle running, leakage, throttling and unnecessary heating/cooling

Practical viva questions

1. How will you safely identify and record identify pressure-measuring instruments?–

Answer: Confirm the rig is depressurised; Identify pressure connection, sensing element, scale and zero; Read liquid levels at eye level. Safety: Never loosen a connection under pressure; use rated tubing and fittings only.

2. How will you safely identify and record identify hydraulic pumps and their parts?–

Answer: Isolate electrical and hydraulic energy; Trace flow direction from suction to delivery; Identify rotating versus reciprocating elements. Safety: Never run dry or remove a guard; control residual liquid and pressure.

3. How will you safely identify and record identify hydraulic turbines and their parts?–

Answer: Use an isolated cut-section/model; Identify the water entry, runner path and exit; Classify impulse or reaction type. Safety: No access to a running runner; isolate water and rotation before inspection.

4. How will you safely identify and record identify pneumatic system components?–

Answer: Depressurise before touching connections; Follow airflow from source to actuator and exhaust; Identify valve ports and actuator type. Safety: Compressed air must never be directed at the body; hoses require secure rated fittings.

5. How will you safely identify and record energy conservation and audit observation?−

Answer: Define the equipment and observation boundary; Record rated/observed energy and operating time without opening live panels; Identify idle running, leakage, throttling, poor maintenance or unnecessary load. Safety: Electrical measurement and machine access are faculty-controlled; do not open energised enclosures.

Module III · CO3

One-mark questions

1. Define or identify Motion and mechanical power transmission.−

Answer: Machines transmit power from a prime mover to a driven member while changing speed, torque, direction, shaft position or motion type.

2. Define or identify Open, crossed and V-belt drives.−

Answer: A belt drive transmits power by friction between a flexible belt and pulleys.

3. Define or identify Rope drives and chain drives.−

Answer: Rope drives use fibrous or wire ropes with grooved pulleys for long-distance or lifting duties.

4. Define or identify Spur, helical, bevel and worm gears.−

Answer: Gears transmit power through meshing teeth and provide a definite speed ratio.

5. Define or identify Journal, ball and roller bearings.−

Answer: A bearing supports and locates a moving shaft while reducing friction and carrying radial, axial or combined load.

6. Define or identify Purpose, principle and methods of lubrication.−

Answer: Lubrication introduces a suitable film between interacting surfaces to reduce friction and wear, remove heat, prevent corrosion, seal contaminants and reduce noise.

Three-mark questions

1. Explain three essential points about Motion and mechanical power transmission.−

Answer:

1. Shafts transmit torque; couplings join shafts
2. Belt and rope drives use friction and suit longer centre distances
3. Chain and gear drives are positive drives with no normal slip

2. Explain three essential points about Open, crossed and V-belt drives.−

Answer:

1. Driver is connected to the power source; driven pulley receives power
2. Tight side carries greater tension than slack side
3. Slip and creep cause actual velocity ratio to differ from the ideal

3. Explain three essential points about Rope drives and chain drives.−

Answer:

1. Rope drives suit large centre distance and heavy duty according to rope type
2. Chain drive has driver sprocket, driven sprocket and chain links
3. Chain polygonal action creates small speed fluctuation

4. Explain three essential points about Spur, helical, bevel and worm gears.−

Answer:

1. External gears rotate in opposite directions
2. Spur gear is simple and efficient but may be noisy at high speed
3. Helical gear carries load gradually but produces axial thrust

5. Explain three essential points about Journal, ball and roller bearings.−

Answer:

1. Ball bearings suit moderate radial/axial load and high speed
2. Cylindrical rollers carry high radial load
3. Taper rollers carry combined radial and axial load

6. Explain three essential points about Purpose, principle and methods of lubrication.−

Answer:

1. Boundary lubrication: thin film with surface interaction
2. Hydrodynamic lubrication: full fluid film generated by relative motion
3. Grease stays in place and seals but removes heat less effectively than circulating oil

Seven-mark questions

1. Develop a structured note on Motion and mechanical power transmission with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Machines transmit power from a prime mover to a driven member while changing speed, torque, direction, shaft position or motion type. Motion may be rotary, reciprocating, oscillating or linear. Selection considers power, speed ratio, centre distance, slip, shock load, noise, maintenance, alignment and safety guarding.

1. Shafts transmit torque; couplings join shafts
2. Belt and rope drives use friction and suit longer centre distances
3. Chain and gear drives are positive drives with no normal slip
4. Gears provide precise ratios and compact transmission
5. Guards prevent access to nip points, rotating shafts and thrown parts

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

2. Develop a structured note on Open, crossed and V-belt drives with a labelled diagram, working principle, application, limitation and safety/maintenance point. –

Answer framework:

A belt drive transmits power by friction between a flexible belt and pulleys. An open belt drive rotates parallel shafts in the same direction; a crossed belt reverses the driven direction and increases wrap but adds rubbing; a V-belt wedges into a grooved pulley for compact higher-capacity transmission.

1. Driver is connected to the power source; driven pulley receives power
2. Tight side carries greater tension than slack side
3. Slip and creep cause actual velocity ratio to differ from the ideal
4. Correct tension and alignment prevent heat, wear and bearing overload
5. Advantages: simple, quiet, shock-absorbing and inexpensive
6. Applications: fans, blowers, conveyors and machine tools

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

3. Develop a structured note on Rope drives and chain drives with a labelled diagram, working principle, application, limitation and safety/maintenance point. –

Answer framework:

Rope drives use fibrous or wire ropes with grooved pulleys for long-distance or lifting duties. Chain drives use an articulated metal chain engaging sprocket teeth, giving a positive velocity ratio without slip. Chain drives need lubrication, alignment and controlled slack; ropes require inspection for wear, broken strands and correct groove contact.

1. Rope drives suit large centre distance and heavy duty according to rope type
2. Chain drive has driver sprocket, driven sprocket and chain links
3. Chain polygonal action creates small speed fluctuation
4. Chain drives suit bicycles, motorcycles, conveyors and timing systems
5. Never operate an exposed chain or rope drive; inspect under isolation

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

4. Develop a structured note on Spur, helical, bevel and worm gears with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Gears transmit power through meshing teeth and provide a definite speed ratio. Spur gears join parallel shafts with straight teeth; helical gears use inclined teeth for smoother engagement; bevel gears intersect shaft axes; worm gears connect skew shafts, commonly near 90°, and can provide a large reduction.

1. External gears rotate in opposite directions
2. Spur gear is simple and efficient but may be noisy at high speed
3. Helical gear carries load gradually but produces axial thrust
4. Bevel gears change direction between intersecting shafts
5. Worm sets provide high reduction but sliding causes heat and lower efficiency
6. Module, tooth count, pressure angle, face width and material identify a gear

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

5. Develop a structured note on Journal, ball and roller bearings with a labelled diagram, working principle, application, limitation and safety/maintenance point. −

Answer framework:

A bearing supports and locates a moving shaft while reducing friction and carrying radial, axial or combined load. Journal bearings support sliding contact through a lubricant film; rolling-element bearings place balls or rollers between inner and outer races. Bearing selection depends on load, speed, alignment, space, environment, life, lubrication and mounting.

1. Ball bearings suit moderate radial/axial load and high speed
2. Cylindrical rollers carry high radial load
3. Taper rollers carry combined radial and axial load
4. Needle rollers provide high radial capacity in small radial space
5. Journal bearings can carry heavy load quietly when a stable oil film exists
6. Incorrect fitting, contamination, misalignment and poor lubrication cause early failure

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

Multiple-choice questions

1. Which statement correctly belongs to Motion and mechanical power transmission?

A. **Shafts transmit torque; couplings join shafts**
B. **Driver is connected to the power source; driven pulley receives power**
C. **Rope drives suit large centre distance and heavy duty according to rope type**
D. **External gears rotate in opposite directions**

–

Answer: A. Shafts transmit torque; couplings join shafts

2. Which statement correctly belongs to Open, crossed and V-belt drives?

A. **Rope drives suit large centre distance and heavy duty according to rope type**
B. **Driver is connected to the power source; driven pulley receives power**
C. **External gears rotate in opposite directions**
D. **Ball bearings suit moderate radial/axial load and high speed**

–

Answer: B. Driver is connected to the power source; driven pulley receives power

3. Which statement correctly belongs to Rope drives and chain drives?

A. **External gears rotate in opposite directions**
B. **Ball bearings suit moderate radial/axial load and high speed**
C. **Rope drives suit large centre distance and heavy duty according to rope type**
D. **Boundary lubrication: thin film with surface interaction**

–

Answer: C. Rope drives suit large centre distance and heavy duty according to rope type

4. Which statement correctly belongs to Spur, helical, bevel and worm gears?

A. **Ball bearings suit moderate radial/axial load and high speed**
B. **Boundary lubrication: thin film with surface interaction**
C. **Shafts transmit torque; couplings join shafts**
D. **External gears rotate in opposite directions**

–

Answer: D. External gears rotate in opposite directions

5. Which statement correctly belongs to Journal, ball and roller bearings?

A. **Ball bearings suit moderate radial/axial load and high speed**
B. **Boundary lubrication: thin film with surface interaction**
C. **Shafts transmit torque; couplings join shafts**
D. **Driver is connected to the power source; driven pulley receives power**

–

Answer: A. Ball bearings suit moderate radial/axial load and high speed

6. Which statement correctly belongs to Purpose, principle and methods of lubrication?

A. **Shafts transmit torque; couplings join shafts**
B. **Boundary lubrication: thin film with surface interaction**
C. **Driver is connected to the power source; driven pulley receives power**
D. **Rope drives suit large centre distance and heavy duty according to rope type**

–

Answer: B. Boundary lubrication: thin film with surface interaction

7. Which statement correctly belongs to Motion and mechanical power transmission?

A. **Driver is connected to the power source; driven pulley receives power**
B. **Rope drives suit large centre distance and heavy duty according to rope type**
C. **Shafts transmit torque; couplings join shafts**
D. **External gears rotate in opposite directions**

−

Answer: C. Shafts transmit torque; couplings join shafts

8. Which statement correctly belongs to Open, crossed and V-belt drives?

A. **Rope drives suit large centre distance and heavy duty according to rope type**
B. **External gears rotate in opposite directions**
C. **Ball bearings suit moderate radial/axial load and high speed**
D. **Driver is connected to the power source; driven pulley receives power**

−

Answer: D. Driver is connected to the power source; driven pulley receives power

Practical viva questions

1. How will you safely identify and record identify mechanical power-transmission systems?−

Answer: Confirm zero energy and guarding; Identify driver and driven members; State motion type, shaft arrangement and likely speed change. Safety: Never rotate or tension a drive by hand unless isolated and authorised.

2. How will you safely identify and record identify belt and chain drives?−

Answer: Observe only stationary isolated drives; Determine rotation direction; Inspect alignment, tension and wear indicators. Safety: Nip points and stored tension remain hazardous after power isolation.

3. How will you safely identify and record identify spur, helical, bevel and worm gears?−

Answer: Use loose samples or isolated demonstration units; Observe tooth orientation and pitch surface; Match gear to parallel, intersecting or skew shafts. Safety: Sharp teeth and pinch points require careful handling; heavy gears need lifting support.

4. How will you safely identify and record identify journal, ball and roller bearings?−

Answer: Clean the sample externally without spinning with compressed air; Identify load direction and rolling element; Read visible designation only if clear. Safety: Do not overspeed a loose bearing with compressed air; use gloves only away from rotation.

5. How will you safely identify and record identify lubricants and lubrication methods?−

Answer: Read container grade and hazard information; Observe colour/consistency without skin contact; Match method to bearing/gear arrangement. Safety: Avoid skin contact and spills; never mix grades or dispose oil to drain.

Module IV · CO4

One-mark questions

1. Define or identify Casting principle and types.−

Answer: Casting produces a component by pouring molten material into a mould cavity, allowing it to solidify and removing/finishing the casting.

2. Define or identify Components and moulding materials.−

Answer: A sand mould needs refractory moulding sand with sufficient strength, permeability, plasticity and collapsibility.

3. Define or identify Forging, rolling, extrusion and drawing.−

Answer: Metal forming changes shape mainly by plastic deformation without intentionally removing material.

4. Define or identify Turning, drilling, milling and grinding.−

Answer: Machining removes material as chips or fine particles to obtain shape, size and finish.

5. Define or identify Surface-finishing processes.−

Answer: Surface finishing improves roughness, appearance, cleanliness, dimensional accuracy, wear, friction or corrosion performance.

6. Define or identify Importance of safety and workshop rules.−

Answer: Workshop safety prevents injury, occupational illness, equipment damage, fire, environmental harm and production loss.

7. Define or identify PPE, accident causes and prevention.−

Answer: Personal protective equipment reduces exposure when hazards cannot be eliminated or fully controlled by engineering and administrative measures.

8. Define or identify Industrial safety culture and continuous improvement.−

Answer: Industrial safety is a managed system involving leadership, worker participation, risk assessment, training, maintenance, permit systems, emergency preparedness, incident learning and continuous improvement.

Three-mark questions

1. Explain three essential points about Casting principle and types.−

Answer:

1. Pattern creates the mould cavity with allowances
2. Mould contains cavity, sprue, runner, gate, riser and vents
3. Core forms an internal hole or passage

2. Explain three essential points about Components and moulding materials.−

Answer:

1. Cope is the upper moulding flask; drag is the lower part
2. Parting line separates cope and drag
3. Ramming compacts sand uniformly around the pattern

3. Explain three essential points about Forging, rolling, extrusion and drawing.−

Answer:

1. Hot working lowers flow stress and allows large deformation
2. Cold working improves finish and dimensional control but increases force and work hardening
3. Forging improves directional grain flow and mechanical integrity

4. Explain three essential points about Turning, drilling, milling and grinding.−

Answer:

1. Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock
2. Drilling variables include drill size, speed and feed
3. Milling produces flat, slot, profile and gear-related surfaces with suitable cutters

5. Explain three essential points about Surface-finishing processes.−

Answer:

1. Polishing removes small irregularities; buffing improves lustre
2. Lapping uses fine abrasive between surfaces for very high accuracy/finish
3. Honing uses bonded abrasive sticks, commonly for bores

6. Explain three essential points about Importance of safety and workshop rules. −

Answer:

1. Know the machine-specific safe operating procedure before use
2. Inspect guards, tool condition, work holding and emergency stop
3. Use lockout/isolation before maintenance or clearing a jam

7. Explain three essential points about PPE, accident causes and prevention.−

Answer:

1. Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection
2. Welding requires rated filter protection, gloves, clothing, ventilation and fire control
3. Gloves are not worn near exposed rotating machinery where entanglement is possible

8. Explain three essential points about Industrial safety culture and continuous improvement.−

Answer:

1. Management provides policy, resources, competent supervision and accountability
2. Safety officers coordinate risk control, inspection, training and compliance
3. Workers participate in hazard reporting, toolbox talks and procedure improvement

Seven-mark questions

1. Develop a structured note on Casting principle and types with a labelled diagram, working principle, application, limitation and safety/maintenance point. –

Answer framework:

Casting produces a component by pouring molten material into a mould cavity, allowing it to solidify and removing/finishing the casting. Sand casting is versatile for complex shapes and varied size; permanent-mould and die casting improve repeatability for suitable metals and production volume; investment casting gives fine detail.

1. Pattern creates the mould cavity with allowances
2. Mould contains cavity, sprue, runner, gate, riser and vents
3. Core forms an internal hole or passage
4. Riser feeds liquid metal to compensate for solidification shrinkage
5. Common defects include blowholes, shrinkage cavity, misrun, cold shut and sand inclusion
6. Molten-metal work requires trained personnel, dry tools, PPE and controlled exclusion zones

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

2. Develop a structured note on Components and moulding materials with a labelled diagram, working principle, application, limitation and safety/maintenance point.–

Answer framework:

A sand mould needs refractory moulding sand with sufficient strength, permeability, plasticity and collapsibility. Silica sand is the main refractory constituent; binder gives strength, moisture activates suitable binders and additives adjust surface or breakdown behaviour. Flask, pattern, moulding tools, core and gating components must be identified correctly.

1. Cope is the upper moulding flask; drag is the lower part
2. Parting line separates cope and drag
3. Ramming compacts sand uniformly around the pattern
4. Vent wire provides gas escape paths
5. Core prints locate and support a core
6. Gating design controls clean, calm and adequate metal flow

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

3. Develop a structured note on Forging, rolling, extrusion and drawing with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Metal forming changes shape mainly by plastic deformation without intentionally removing material. Forging compresses metal between tools; rolling reduces thickness or changes section between rolls; extrusion forces material through a die; drawing pulls wire, rod or tube through a die to reduce section.

1. Hot working lowers flow stress and allows large deformation
2. Cold working improves finish and dimensional control but increases force and work hardening
3. Forging improves directional grain flow and mechanical integrity
4. Rolling makes plates, sheets, rails and structural sections
5. Extrusion makes long constant profiles
6. Drawing makes wire, bar and tube with controlled diameter

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

4. Develop a structured note on Turning, drilling, milling and grinding with a labelled diagram, working principle, application, limitation and safety/maintenance point.−

Answer framework:

Machining removes material as chips or fine particles to obtain shape, size and finish. In turning, a rotating workpiece meets a cutting tool; drilling creates a round hole; milling uses a rotating multi-edge cutter; grinding uses abrasive grains for fine finishing and accuracy.

1. Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock
2. Drilling variables include drill size, speed and feed
3. Milling produces flat, slot, profile and gear-related surfaces with suitable cutters
4. Grinding wheel selection, guarding, ring test/mounting and dressing are safety-critical
5. Cutting speed, feed and depth of cut influence time, finish, force and tool life
6. Remove chips with a brush under safe conditions, never by bare hand

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

5. Develop a structured note on Surface-finishing processes with a labelled diagram, working principle, application, limitation and safety/maintenance point.

–

Answer framework:

Surface finishing improves roughness, appearance, cleanliness, dimensional accuracy, wear, friction or corrosion performance. Filing, polishing, buffing, lapping, honing, grinding, blasting and coating processes serve different purposes; the correct process depends on material, geometry and required surface function.

1. Polishing removes small irregularities; buffing improves lustre
2. Lapping uses fine abrasive between surfaces for very high accuracy/finish
3. Honing uses bonded abrasive sticks, commonly for bores
4. Blasting cleans and textures a surface using controlled media
5. Coatings may protect against corrosion or wear
6. Surface roughness should be specified and verified, not judged only by appearance

Draw the handbook visual with correct labels and arrows; conclude with an application and the most relevant safety or maintenance control.

Multiple-choice questions

1. Which statement correctly belongs to Casting principle and types?

- **A. Pattern creates the mould cavity with allowances**
- **B. Cope is the upper moulding flask; drag is the lower part**
- **C. Hot working lowers flow stress and allows large deformation**
- **D. Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock**

–

Answer: A. Pattern creates the mould cavity with allowances

2. Which statement correctly belongs to Components and moulding materials?

A. **Hot working lowers flow stress and allows large deformation**
B. **Cope is the upper moulding flask; drag is the lower part**
C. **Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock**
D. **Polishing removes small irregularities; buffing improves lustre**

–

Answer: B. Cope is the upper moulding flask; drag is the lower part

3. Which statement correctly belongs to Forging, rolling, extrusion and drawing?

A. **Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock**
B. **Polishing removes small irregularities; buffing improves lustre**
C. **Hot working lowers flow stress and allows large deformation**
D. **Know the machine-specific safe operating procedure before use**

–

Answer: C. Hot working lowers flow stress and allows large deformation

4. Which statement correctly belongs to Turning, drilling, milling and grinding?

A. **Polishing removes small irregularities; buffing improves lustre**
B. **Know the machine-specific safe operating procedure before use**
C. **Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection**
D. **Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock**

–

Answer: D. Lathe main parts: bed, headstock, spindle/chuck, carriage, tool post and tailstock

5. Which statement correctly belongs to Surface-finishing processes?

A. **Polishing removes small irregularities; buffing improves lustre**
B. **Know the machine-specific safe operating procedure before use**
C. **Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection**
D. **Management provides policy, resources, competent supervision and accountability**

–

Answer: A. Polishing removes small irregularities; buffing improves lustre

6. Which statement correctly belongs to Importance of safety and workshop rules?

A. **Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection**
B. **Know the machine-specific safe operating procedure before use**
C. **Management provides policy, resources, competent supervision and accountability**
D. **Pattern creates the mould cavity with allowances**

–

Answer: B. Know the machine-specific safe operating procedure before use

7. Which statement correctly belongs to PPE, accident causes and prevention?

A. **Management provides policy, resources, competent supervision and accountability**
B. **Pattern creates the mould cavity with allowances**
C. **Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection**
D. **Cope is the upper moulding flask; drag is the lower part**

–

Answer: C. Safety goggles protect eyes; face shields supplement rather than replace suitable eye protection

8. Which statement correctly belongs to Industrial safety culture and continuous improvement?

A. **Pattern creates the mould cavity with allowances**
B. **Cope is the upper moulding flask; drag is the lower part**
C. **Hot working lowers flow stress and allows large deformation**
D. **Management provides policy, resources, competent supervision and accountability**

−

Answer: D. Management provides policy, resources, competent supervision and accountability

Practical viva questions

1. How will you safely identify and record identify moulding and casting components?−

Answer: Use cold demonstration material only; Separate pattern, mould and core functions; Trace proposed metal and gas paths. Safety: No molten-metal activity without full foundry controls and trained faculty.

2. How will you safely identify and record identify metal-forming processes?−

Answer: Observe isolated equipment, samples or models; Identify whether force pushes, squeezes or pulls; Mark material entry and exit directions. Safety: Hot metal, pinch points and stored press energy require controlled exclusion.

3. How will you safely identify and record identify machining, welding and soldering equipment?−

Answer: Confirm equipment isolation; Identify work motion and tool motion; Locate work holding, guard and emergency stop. Safety: No live machining/welding without authorisation; grinding and welding PPE are process-specific.

4. How will you safely identify and record identify surface finishing and safety equipment?−

Answer: Compare before/after surface samples; Match finish to function; Identify PPE markings and condition. Safety: PPE does not permit unsafe operation; extinguisher use follows institutional emergency training.

OFFICIAL EVALUATION

Assessment pattern and practice model paper

Teaching and assessment table

Component	CIE	ESE	Total
Theory	20	60	80
Practical	20	40	60
Combined	**40**	**100**	**140**

Series theory examination: 2 hours, any two modules, total 50 marks. End-semester theory: 2.5 hours, 60 marks. End-semester lab: 3 hours, 40 marks.

Continuous practical evaluation — 50

- Preparation and punctuality — 10
- Experimental setup and procedure — 10
- Observation and data recording — 5
- Analysis and result interpretation — 10
- Viva/concept understanding — 10
- Workmanship, discipline and teamwork — 5

Practical test — 40

- Write-up/procedure — 10
- Setup and execution — 10
- Observation/result/output — 8
- Viva voce — 8
- Record — 4

Series Lab Examinations

Series Lab Exam I: Module II detailed row, 3 hours, mapped to CO2, Understand level. The methodology table lists the first two-module series exam as 2 hours/50 raw marks.

Series Lab Exam II: Module IV detailed row, 3 hours, mapped to CO4, Understand level. The methodology table lists the second two-module series exam as 2 hours/50 raw marks.

> Series exam-□ identification, setup/flowchart, observation, result, viva, safety □□□□□□ module-wise revise □□□□□□□.

Detailed practicum conversion

Component	Raw	Converted
Attendance	10	5
Average self-learning activities	Not separately printed	5

Component	Raw	Converted
Continuous evaluation	50	10
Practical test — first half	40	10
Practical test — second half	40	10
Series Exam I	50	10
Series Exam II	50	10
ESE written theory	60	60
ESE lab	40	40

End-semester theory pattern — 60

- Part A: 2 × 1 mark from each module = 8
- Part B: 3 questions per module; answer 2 × 3 = 6 per module, total 24
- Part C: one 7-mark set with choice from each module = 28

Original practice model paper

Duration: 2.5 hours · **Maximum:** 60 marks

Part A — answer all (8 × 1 = 8)

1. Define Scope, importance and branches of Mechanical Engineering.
2. Define Thermodynamic systems, properties and processes.
3. Define Properties of fluids.
4. Define Pressure, manometers, gauges and Pascal’s law.
5. Define Motion and mechanical power transmission.
6. Define Open, crossed and V-belt drives.
7. Define Casting principle and types.
8. Define Components and moulding materials.

Part B — answer two from each module (8 × 3 = 24)

Module 1: (1) Explain Scope, importance and branches of Mechanical Engineering. (2) Explain Thermodynamic systems, properties and processes. (3) Explain Energy, work, heat and laws of thermodynamics.

Module 2: (1) Explain Properties of fluids. (2) Explain Pressure, manometers, gauges and Pascal’s law. (3) Explain Hydraulic pumps: centrifugal and reciprocating.

Module 3: (1) Explain Motion and mechanical power transmission. (2) Explain Open, crossed and V-belt drives. (3) Explain Rope drives and chain drives.

Module 4: (1) Explain Casting principle and types. (2) Explain Components and moulding materials. (3) Explain Forging, rolling, extrusion and drawing.

Part C — answer one alternative from each module (4 × 7 = 28)

Module 1: (a) Explain Heat engines, refrigerators, steam boilers and turbines with labelled figure and application. **OR** (b) Explain Internal combustion engines: types and working with principle, comparison and safety/maintenance points.

Module 2: (a) Explain Renewable energy: solar, wind, hydro and biomass with labelled figure and application. **OR** (b) Explain Energy conservation and management with principle, comparison and safety/maintenance points.

Module 3: (a) Explain Journal, ball and roller bearings with labelled figure and application. **OR** (b) Explain Purpose, principle and methods of lubrication with principle, comparison and safety/maintenance points.

Module 4: (a) Explain PPE, accident causes and prevention with labelled figure and application. **OR** (b) Explain Industrial safety culture and continuous improvement with principle, comparison and safety/maintenance points.

LAST-MINUTE REVIEW

Quick revision and answer checklist

Module I anchors

System–surroundings–boundary; open/closed/isolated; isothermal/isobaric/isochoric/adiabatic; Zeroth/First/Second laws; heat engine versus refrigerator; four-stroke sequence; SI versus CI.

Module II anchors

ρ, γ, v and SG; pressure types; manometer; Pascal law; centrifugal versus reciprocating pump; Pelton/Francis/Kaplan; FRL-valve-cylinder; renewable limits; audit cycle.

Module III anchors

Driver/driven; friction versus positive drive; open/cross/V-belt; chain and sprocket; spur/helical/bevel/worm; journal versus rolling bearing; lubrication purpose and methods.

Module IV anchors

Pattern–mould–core; sprue–runner–gate–riser; forging/rolling/extrusion/drawing; turning/drilling/milling/grinding; finishing; hierarchy of controls; PPE; PDCA safety culture.

Diagram checklist

Use a title, correct outline, principal parts, motion/flow arrows, input/output labels and a caption. Avoid decorative detail that hides the working principle.

Answer checklist

Start with definition, state principle, list or compare, use correct units, add labelled diagram/application, state limitation and finish with safety or maintenance.

OFFICIAL LEARNING RESOURCES

Textbooks, references and equipment

Official textbooks

1. *Basics of Mechanical Engineering* — J. Benjamin, Pentex Publications.
2. *Basic Mechanical Engineering* — Raghuwanshi & Choubey, Laxmi Publications.
3. *Basic Mechanical Engineering* — V. P. Arora, Tata McGraw-Hill Education.

Official references

1. *Engineering Thermodynamics (Basic Concepts)* — P. K. Nag, Tata McGraw-Hill Education.
2. *Fluid Mechanics and Hydraulic Machines (Introductory Level)* — R. K. Rajput, S. Chand & Company Ltd.
3. *Workshop Technology (Vol. I & II)* — B. S. Raghuwanshi, Dhanpat Rai & Co.
4. *Mechanical Engineering — Principles and Applications* — P. N. Rao, Tata McGraw-Hill Education.

Official web resources: NIL.

Major equipment for 30 students

1. One or more disassembled two-/four-stroke petrol/diesel engine/component sets.
2. One or more U-tube manometers, pressure gauges and hydraulic pump/turbine laboratory units.
3. One or more chain-drive, gear-drive and ball/roller-bearing demonstration sets.
4. One or more lathe, milling, foundry and welding equipment groups, with institution-approved specifications.

Official software requirement: NIL.

> Equipment operate □□□□□□□□□□□□□□□□ □□□□□□□ faculty demonstration, machine guard, isolation, PPE □□□□□□□ □□□□□□□□□□□□□.

POLY PMNA · Course 1021 · Diploma Curriculum Revision 2026

Original bilingual educational handbook based on the supplied official syllabus. Always follow institution laboratory rules and faculty instructions.